

Data Entity Builder Manual

SW11-611

**Implementation
Engineering Operations - 1**

Data Entity Builder Manual

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Honeywell Process Solutions
1860 W. Rose Garden Lane
Phoenix, AZ 85027 USA
1-800 822-7673

About This Publication

This publication provides instructions for the use of the Data Entity Builder functions in the Universal Personality.

This publication supports **TotalPlant** Solution (TPS) system network Releases 682 and greater. TPS is the evolution of TDC 3000^X

IS THIS THE RIGHT PUBLICATION?

This publication describes the use of the Data Entity Builder (DEB) to build data points and reserved entities. Instructions are provided for entering data in the DEB's Parameter Entry Displays (PEDs) and for executing DEB commands.

Use this publication when you select one of the following picks (targets) on the Engineering Main Menu:

HIWAY GATEWAY	HM HISTORY GROUPS
APPLICATION MODULE	AREA DATA BASE
COMPUTING MODULE	BUILDER COMMANDS
NETWORK INTERFACE MODULE	

WHAT SHOULD I BE FAMILIAR WITH BEFORE USING THIS PUBLICATION?

The content of these publications:

- *Implementation Overview* in the Implementation/Startup & Reconfiguration - 1 binder
- *Configuration Data Collection Guide*, in the Implementation/Startup & Reconfiguration - 2 binder
- *System Startup Guide - Cartridge Drive*, in the Implementation/Startup & Reconfiguration - 3 binder
- *System Startup Guide - Zip Drive*, in the Implementation/Startup & Reconfiguration - 3 binder
- *System Startup Guide – CD-ROM*, in the Implementation/Startup & Reconfiguration - 3 binder

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United States and Canada

Contact: Honeywell Solution Support Center
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Calls are answered by dispatcher between 6:00 am and 4:00 pm Mountain Standard Time. Emergency calls outside normal working hours are received by an answering service and returned within one hour.
Fascimile: 1-973-455-5000
Mail: Honeywell TAC, MS L17
1860 W. Garden Lane
Phoenix, AZ, 85027 USA

Europe, Middle East, and Africa (EMEA)

Contact: Honeywell TAC-EMEA
Phone: +32-2-728-2345
Fascimile: +32-2-728-2696
e: TAC-BE02
Mail: Hermes Plaza
Hermeslaan, 1H
B-1831 Diegem, Belgium

Pacific

Contact: Honeywell Global TAC – Pacific
Phone: 1300-364-822 (toll free within Australia)
+61-8-9362-9559 (outside Australia)
Fascimile: +61-8-9362-9564
e: Honeywell Limited Australia
Mail: 5 Kitchener Way
Burswood 6100, Western Australia
Email: GTAC@honeywell.com

India

Contact: Honeywell Global TAC – India
Phone: +91-20- 6603-9400
Fascimile: +91-20- 6603-9800
e: Honeywell Automation India Ltd
Mail: 56 and 57, Hadapsar Industrial Estate
Hadapsar, Pune –411 013, India
Email: Global-TAC-India@honeywell.com

Korea

Contact: Honeywell Global TAC – Korea
Phone: +82-80-782-2255 (toll free within Korea)
Fascimile: +82-2-792-9015
e: Honeywell Co., Ltd
Mail: 4F, Sangam IT Tower B4-4 Block
1590, DMC Sangam-dong, Mapo-gu,
Seoul, 121-835, Korea
Email: Global-TAC-Korea@honeywell.com

People's Republic of China

Contact: Honeywell Global TAC – China
Phone: +86- 21-2219-6888
800-820-0237
400-820-0386
Mail: Honeywell (China) Co., Ltd
33/F, Tower A, City Center, 100 Zunyi Rd.
Shanghai 200051, People's Republic of China
Email: Global-TAC-China@honeywell.com

Singapore

Contact: Honeywell Global TAC – South East Asia
Phone: +65-6580-3500
Fascimile: +65-6580-3501
+65-6445-3033
Mail: Honeywell Private Limited
Honeywell Building
17, Changi Business Park Central 1
Singapore 486073
Email: GTAC-SEA@honeywell.com

Taiwan

Contact: Honeywell Global TAC – Taiwan
Phone: +886-7-536-2567
Fascimile: +886-7-536-2039
e: Honeywell Taiwan Ltd.
Mail: 17F-1, No. 260, Jhongshan 2nd Road.
Cianjhen District
Kaohsiung, Taiwan, ROC
Email: Global-TAC-Taiwan@honeywell.com

Japan

Contact: Honeywell Global TAC – Japan
Phone: +81-3-6730-7160
Fascimile: +81-3-6730-7228
e: Honeywell Japan Inc.
Mail: New Pier Takeshiba, South Tower Building,
20th Floor, 1-16-1 Kaigan, Minato-ku,
Tokyo 105-0022, Japan
Email: Global-TAC-JapanJA25@honeywell.com

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SECTION 1 - INTRODUCTION

This section describes this publication and its intended use. It also provides references to related publications.

This publication provides instructions for the use of the Data Entity Builder (DEB). The Data Entity Builder is a function that is available when the Universal Personality is loaded into a Universal Station. The Data Entity Builder accomplishes the following activities on the Engineering Main Menu:

Hiway Gateway	HM History Groups
Application Module	Area Data Base
Computing Module	Builder Commands
Network Interface Module	

1.1 WHO THIS PUBLICATION IS INTENDED FOR

The procedures in this publication are intended for process engineers, control system engineers, and application engineers, or anyone who does similar work. It is possible for a technician or clerk to enter data from filled-in configuration forms; therefore, the data entry instructions are written so that minimal technical expertise is needed. There is more to system configuration than data entry, however, and the remaining procedures in this publication require a process engineer's technical knowledge.

1.2 HOW TO USE THIS PUBLICATION

Before consulting this publication, you should read the *Configuration Data Collection Guide*, which gives you an overview of the entire configuration process. By now you may have used the *System Startup Guide*, which provides step-by-step instructions for starting up your system, including the activities mentioned above and others.

You should refer to this publication for:

- guidance in the effective use of the Data Entity Builder
- detailed descriptions of all DEB functions
- descriptions of useful features and functions that are not called for in the *System Startup Guide*
- the meaning of prompters and error messages
- help in error recovery.

The index in the final pages of this publication may help you to quickly find the information you need.

1.3 RELATED PUBLICATIONS

1.3.1 Configuration Forms and Instructions

While it's possible to enter data without configuration forms, you'll probably need filled-in forms to build entities for a new system or to build more than a few entities for an existing system. The table below shows how the DEB activities on the Engineering Main Menu relate to the forms and the publications that provide information for filling out the forms.

HIWAY GATEWAY

HG PROCESS POINT BUILD TYPE	<i>Data Hiway, Box/Slot, and Data Point Form Instructions and Hiway Gateway Parameter Reference Dictionary in the Implementation/Hiway Gateway - 1 binder</i>
--------------------------------	---

HG CONFIGURATION

HIWAY CONFIGURATION	<i>Data Hiway, Box/Slot, and Data Point Form Instructions and Network Form Instructions in the Implementation/Startup and Reconfiguration - 1 binder</i>
------------------------	--

	<i>PLC Gateway Forms and PLC Gateway Parameter Reference Dictionary in the Implementation/PLC Gateway binder</i>
--	--

BOX/SLOT CONFIGURATION	<i>Data Hiway, Box/Slot, and Data Point Form Instructions and Network Form Instructions in the Implementation/Startup and Reconfiguration - 1 binder</i>
---------------------------	--

HG LIBRARY CONFIGURATION	<i>Hiway Gateway Library Forms in the Implementation/Configuration Forms binder</i>
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APPLICATION MODULE	<i>Application Module Form Instructions and Application Module Parameter Reference Dictionary in the Implementation/Application Module - 2 binder</i>
--------------------	---

COMPUTING MODULE	<i>Computer Gateway Form Instructions in the Implementation/Computer Gateway binder Computer Gateway Parameter Reference Dictionary in the Implementation/Computer Gateway binder</i>
------------------	--

NETWORK INTERFACE MODULE	<i>Process Manager Configuration Forms</i> in the Implementation/Process Manager – 1 binder
	<i>PM Family Parameter Reference Dictionary</i> in the Implementation/Process Manager - 2 binder
	<i>Advanced Process Manager Configuration Forms</i> in the Implementation/Advanced Process Manager – 1 binder
	<i>PM Family Parameter Reference Dictionary</i> in the Implementation/Advanced Process Manager - 2 binder
	<i>High-Performance Process Manager Configuration Forms</i> in the Implementation/HPM - 4 binder
	<i>PM Family Parameter Reference Dictionary</i> in the Implementation/ High-Performance Process Manager - 2 binder
HM HISTORY GROUPS	<i>Logic Manager Forms and Logic Manager Parameter Reference Dictionary</i> in the Implementation/Logic Manager binder
	<i>Safety Manager Module Configuration Forms and Safety Manager Parameter Reference Dictionary</i> in the Implementation/Triconex Safety Manager binder
	<i>History Module History Group Forms</i> in the Implementation/Configuration Forms binder
AREA DATA BASE	<i>HM History Group Form Instructions</i> in the Implementation/Engineering Operations - 1 binder
	<i>Area Forms</i> in the Implementation/Configuration Forms binder
BUILDER COMMANDS	<i>Area Form Instructions</i> in the Implementation/Engineering Operations - 1 binder
	See Section 7 in this manual

1.3.2 System Startup Guide and Engineer's Reference Manual

The following publications are very useful for the configuration and startup of the system:

- *System Startup Guides (Cartridge, CD-ROM, Zip Drive)*, in the Implementation /Startup & Reconfiguration – 3 binder. We recommend that you use these guides the first time you use any of the activities on the Engineering Main Menu. It provides step-by-step procedures for all available configuration activities. The tasks in the *System Startup Guides* that involve the DEB-based activities are as follows:

HIWAY GATEWAY	
HG PROCESS POINT BUILD TYPE	Task 21
HG CONFIGURATION	
HIWAY CONFIGURATION	Task 19
BOX/SLOT CONFIGURATION	Task 20
HG LIBRARY CONFIGURATION	Task 25
APPLICATION MODULE	Task 21
COMPUTING MODULE	Task 21
NETWORK INTERFACE MODULE	Task 32
HM HISTORY GROUPS	Task 27
AREA DATA BASE	Task 29

- *Engineer's Reference Manual*, in the Implementation/Startup & Reconfiguration - 2 binder. This publication provides guidance and tips that are helpful in starting up your system. Guidance is provided for file management, configuring the Data Hiway and process-connected boxes, point-building tips, and more. Listed below are the Sections in the *Engineer's Reference Manual* pertaining to DEB-based activities:

HM HISTORY GROUPS	Section 7
AREA DATA BASE	Section 13
CONFIGURING FOR PERFORMANCE	Section 23

Section 14 of the *Engineer's Reference Manual* provides important information about data binding in the TDC 3000^X system. DEB users should review 14.4—Binding of Entity Data in the Data Entity Builder and 14.7—Binding of Data in CL Programs.

In the *Implementation/Hiway Gateway - 1* binder in the *HG Implementation Guidelines* manual the following is found:

HIWAY GATEWAY	
HG CONFIGURATION	
HIWAY CONFIGURATION	Section 1
BOX/SLOT CONFIGURATION	Section 1

1.3.3 Text Editor Operations

Text Editor Operation, in the Implementation/Engineering Operations - 3 binder provides detailed instructions for the use of the Text Editor. The Text Editor is used to build or edit selection-list files and exception-build source files for use with certain DEB commands.

1.3.4 Utilities Operation

Command Processor Operation, in the Implementation/Startup & Reconfiguration - 1 binder, provides detailed instructions for the use of the Universal Personality's Command Processor functions. The Command Processor functions are used in file management (cataloging, listing, copying, and deleting files) and to modify volume pathnames.

1.4 REMOVABLE MEDIA

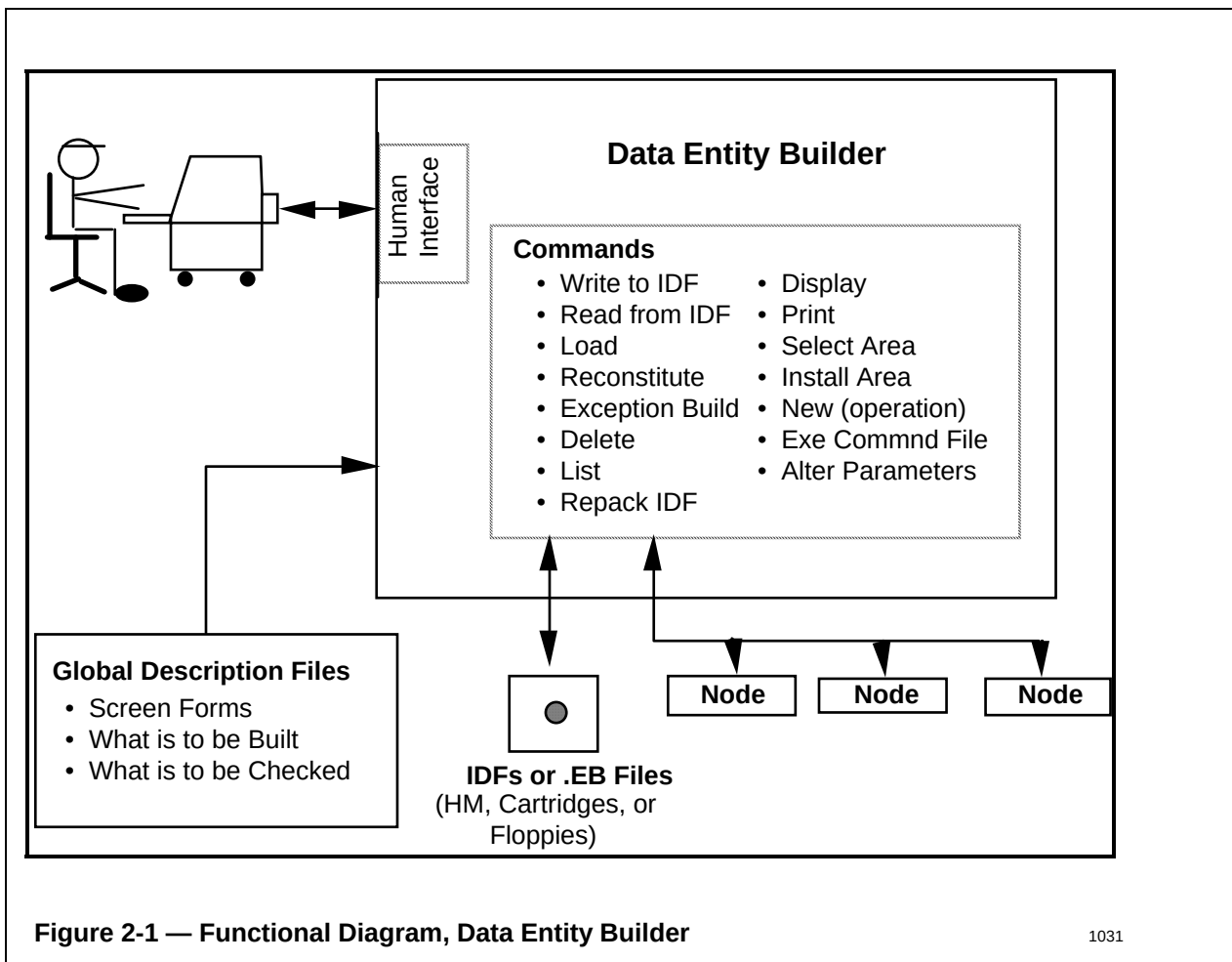
The term “removable media” as used in this manual means Zip or cartridge disks.

SECTION 2 - A USER'S GUIDE TO THE DATA ENTITY BUILDER

This section provides an introduction to the Data Entity Builder.

2.1 INTRODUCTION TO THE DATA ENTITY BUILDER (DEB)

When you select any of the DEB-based activities on the Engineering Main Menu, you are invoking the DEB software. Figure 2-1 is a diagram of the DEB's major functions. You need to know only enough about the DEB to understand how it leads you through the entity building process and to clarify some of the information that appears on the displays, particularly the Command Displays.



An entity is a named collection of data in the configured database that contains a set of parameters. These parameters define the functions of the entity and contain the data related to the entity. There are two classes of entities, as follows:

- Data Points define the functions and contain the data for a process entity. You name each data point as you build it, by keying in the value for parameter NAME. For example, you might name an HG regulatory point FC101.
- Reserved Entities define the functions and contain the data for a system entity. For example, \$HY02B05 is a system entity that you might build during HG Box/Slot Configuration. It defines Box 05 on Data Hiway 02. You do not give names to reserved entities—the system does. All system entity names begin with "\$," so to avoid confusion, data point names cannot begin with "\$". Appendix C contains a list of Reserved Entity names typically referenced by Engineers.

The DEB operates independently from the type of entity that is being built. The Global Description Files (GDFs) allow it to do this. The GDFs provide the screen forms for the Parameter Entry Displays; they define to the DEB what data is to be entered and built, and how the syntax and definition are to be checked.

There is a GDF for each type of entity. The GDFs are files in Honeywell-supplied volumes on removable media (cartridges or floppy disks) and you can store them on a History Module. The names of the volumes that contain the GDFs are as follows:

HIWAY GATEWAY	
HG PROCESS POINT BUILD TYPE	&HGG
HG CONFIGURATION	
HIWAY CONFIGURATION	&HGG ¹
BOX/SLOT CONFIGURATION	&HGG
HG LIBRARY CONFIGURATION	&HGG
APPLICATION MODULE	&AMG
COMPUTING MODULE	&AMG
NETWORK INTERFACE MODULE	&NMG and &NM2 ²
HM HISTORY GROUPS	&AMG
AREA DATA BASE	&ARG

Notes: (1) Also applies to PLC Gateway

(2) Also applies to PM, APM, HPM, and Logic Manager

2.2 SUMMARY OF DEB FUNCTIONS AND FILES

As you will see when you become familiar with the DEB, it has several functions and options, and some of these involve various types of files on removable media (floppy disks or cartridges) or on a History Module. While the DEB may seem complicated, with only a little practice such as you can have by using the *System Startup Guide*, you should find it easy to use.

Figure 2-2 is a pictorial summary of the DEB functions, options, files, and their relationships. After you have had some experience with the DEB functions, this illustration should serve as a quick way to recall the functions and relationships, so that you might not have to reread the detailed descriptions.

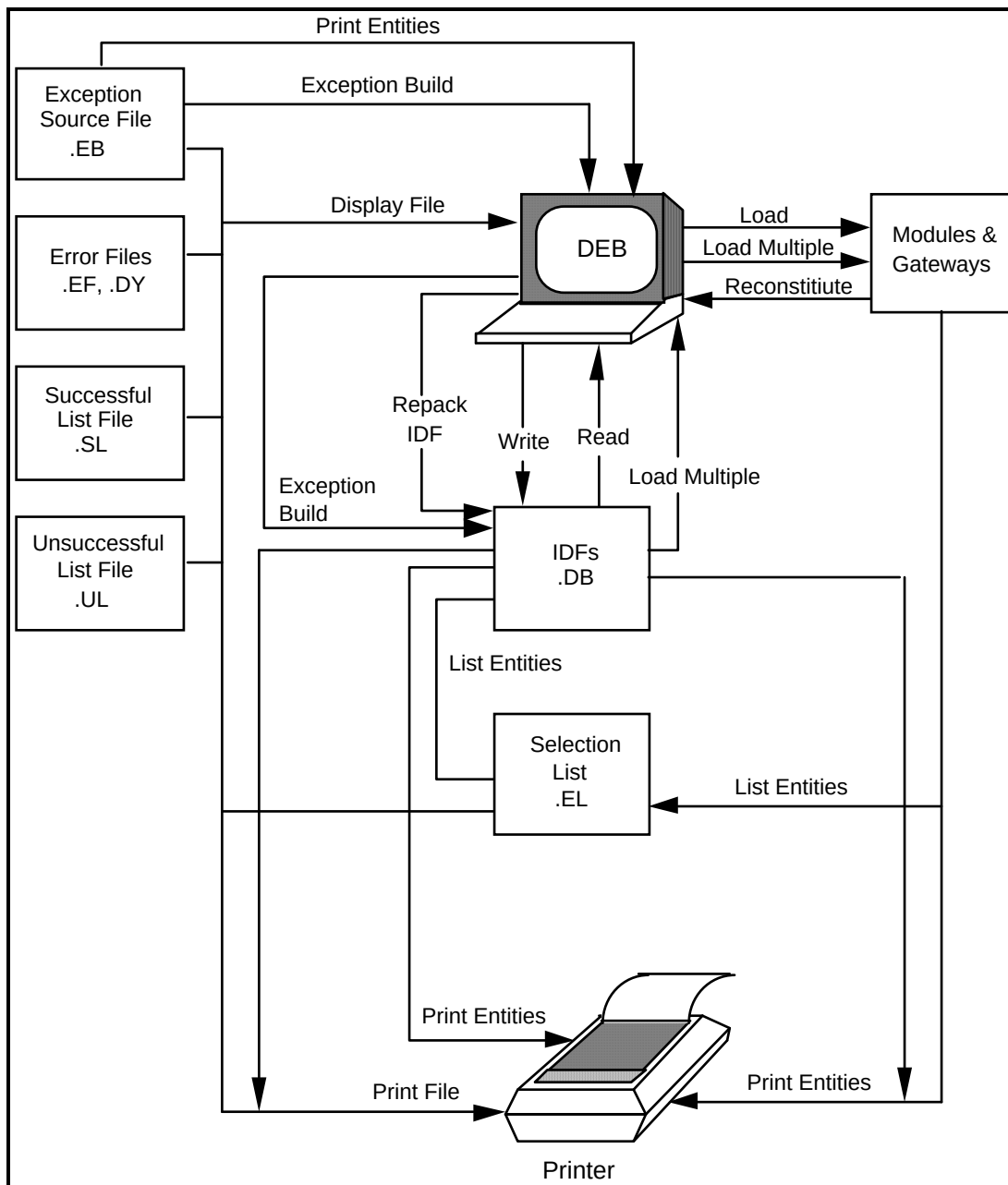


Figure 2-2 — Summary of DEB Functions and Files

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SECTION 3 - PREREQUISITES FOR A DEB SESSION

This section defines what you have to do before building entities.

3.1 CONFIGURATION DATA COLLECTION

An easy, trouble free DEB session requires that the data that you enter is correct. This means that the data must faithfully meet the intent of your system or process design, and that it must be compatible with any process or system data that has already been built and installed in the system.

Obviously, correct data requires careful design. If your system is new or nearly new, we recommend that you carefully document your design or any design changes, before you attempt any activity from the Engineering Main Menu. The configuration forms for the types of entities that the DEB builds are a very effective means to document your design, in a form that is ready for entry in the DEB's Parameter Entry Displays (PEDs). Refer to 1.3.1 for the applicable forms and forms instructions.

The *Engineer's Reference Manual* has many guidelines and suggestions that can help you to develop correct data for entry through the DEB.

3.2 COMPLETE THE FOLLOWING SYSTEM STARTUP TASKS

While you can enter data for any of the DEB-based activities and then dispose of that data at any time, for the data to have meaning, and in some cases for it to be loaded properly in its destination, the following tasks in the *System Startup Guide* must be completed.

You could for example, enter data for HG data points in the Parameter Entry Displays and then store that data in an Intermediate Data File (IDF) or exception-build (.EB) source file to be loaded in the HG sometime later. For a first time startup, however, it is better to perform each of the tasks in the order they occur in the *System Startup Guide*. The order of the startup tasks is as follows:

- Task 1 — Reset Nodes
- Task 2 — Load the Universal Personality
- Task 3 — Set Time and Date
- Task 4 — Format Removable Media
- Task 5 — Copy the System Startup Removable Media
- Task 6 — Initialize the NCF
- Task 7 — Configure Unit, Area, and Console Names
- Task 8 — Configure LCN Nodes
- Task 9 — Configure System Wide Values
- Task 10 — Configure HM Volumes
- Task 11 — Install the NCF

- Task 12 — Reload the Universal Personality
- Task 13 — (not used)
- Task 14 — Load the HM with the HM Init. Personality
- Task 15 — Initialize the HM
- Task 16 — Copy Software from Removable Media to the HM
- Task 17 — (not used)
- Task 18 — Load On-Process Personalities
- Task 19 — Configure Hiways—HIWAY CONFIGURATION
- Task 20 — Configure Boxes and Slots—BOX/SLOT CONFIGURATION
- Task 21 — Configure Points—HG PROCESS POINT BUILD TYPE
APPLICATION MODULE
COMPUTING MODULE
- Task 22 — Configure Logic Blocks
- Task 23 — Build and Compile Pictures
- Task 24 — Build and Compile Buttons
- Task 25 — Build and Load HG Libraries—HG LIBRARY CONFIGURATION
- Task 26 — Build and Compile Free-Format Logs
- Task 27 — Build and Load HM History Groups—HM HISTORY GROUPS
- Task 28 — Build, Compile, and Load CL Programs
- Task 29 — Build, Select, and Install Area Database—AREA DATA BASE
- Task 30 — Load the Operator Personality in Universal Stations
- Task 31 — Change Network Configuration
- Task 32 — Configure NIM, PM, APM, HPM —NETWORK INTERFACE MODULE and
associated UCN Modules

3.3 VOLUMES/DIRECTORIES FOR IDFS ESTABLISHED WITH SUFFICIENT CAPACITY

In Task 4 of the *System Startup Guide*, we recommend that you format removable media for use as user volumes. In Task 10, one or more user volumes are configured for an HM. You can store your IDFs or .EB files on user volumes that have been set up on either a removable medium or on an HM, but you will find it easier to use the HM.

Either way, you must configure one or more volumes with sufficient capacity to store all of the entities you intend to build. You can use Task 4 (removable media) or Task 10 (HM) of the *System Startup Guide* to format and configure your user volumes. Task 10 involves changing the NCF, which requires installing the new NCF and restarting all nodes with it, so sufficient capacity should be provided in the first place. For a detailed explanation of how to format removable media, refer to the *Command Processor Operation* manual.

To facilitate DEB operations, you should plan the IDFs/.EB files you intend to use for entity loading. Entities that refer to each other should be stored in the same IDF/.EB file. With IDFs, you must use the Load Multiple command. With .EB files, you must load the file twice. This means that all such entities in a control strategy must be in the same IDF/.EB file. It is also good practice to store only entities of the same type, such as regulatory data points, in one IDF/.EB file.

We recommend that you copy any entities you have stored on HMs to removable media so that you can recover them in case the data on the HM is lost. The Copy Volume command can be used to do this. See the *Command Processor Operation* manual.

3.4 UNIVERSAL STATION GUIDELINES

The Universal Station (US) used for entity building must be running the Universal Personality and must have an Engineer's Keyboard. It should have a printer connected to it to print records of your work.

If you are using removable media (cartridges or floppies), rather than an HM, the US should have two removable media drives connected to it. Where the drives are connected directly to the station, the transfers are faster than if they have to travel over the Local Control Network to another US.

The key switch on the US must be in the ENGR position to execute any DEB command that changes the database. It is easiest to check it when you start a DEB session and leave it in ENGR until you are finished.

3.5 VOLUME/DIRECTORY HANDLING GUIDELINES

The entity building activities selected through the Engineering Main Menu requires the following GDF volumes (or directories).

HIGHWAY GATEWAY	
HG PROCESS POINT BUILD TYPE	&HGG
HG CONFIGURATION	
HIWAY CONFIGURATION	&HGG
BOX/SLOT CONFIGURATION	&HGG
HG LIBRARY CONFIGURATION	&HGG
APPLICATION MODULE	&AMG
COMPUTING MODULE	&AMG
NETWORK INTERFACE MODULE	&NMG and &NM2
HM HISTORY GROUPS	&AMG
AREA DATA BASE	&ARG

NOTE

In this manual, the term "volume" refers to a volume or directory on a removable medium (a cartridge or a floppy disk) or to a volume or directory on an HM. When the information used by the DEB is loaded from a removable medium into an HM, that information is loaded into directories on the HM that have the same names as the volume or directory names on the removable medium.

Each removable medium contains just one volume and up to 63 directories.

To load the Universal Personality and to get the Engineering Main Menu on the screen, use startup Task 2, or refer to the *Engineer's Digest* (pocket guide), which is provided in the Implementation/Startup and Reconfiguration - 2 binder.

When you select one of the picks listed above, you must have the &OP1 volume and the appropriate Global Descriptor File (GDF) volume available on an HM, on a cartridge (&OP1 and the GDFs are provided by Honeywell on a single cartridge). Use the Modify Default Volume Pathnames display to set up the device IDs (the first three characters in each pathname) to NET or to those of your removable media drives.

If you use an HM, your user volumes are normally available on the HM because you have configured them in System Startup Task 10. The Universal Personality, the &OV1 volume, and the GDFs are normally on an HM too, because they are loaded with all of the system software in System Startup Task 16.

Commands executed in building AREA DATA BASE entities often read from or write to the area database directory. Its name is in this form:

&Dnn

where "nn" is the area number, which can range from 01 to 10.

For area database entities, if you use floppies, you need to temporarily replace one of the floppies with the &Dnn floppy when you execute a Load, Load Multiple, Reconstitute, Reconstitute Multiple, Select, or Install Area command. You can temporarily replace the GDF floppy when &Dnn is needed, for all of these commands except Reconstitute Multiple. Reconstitute Multiple needs the &ARG GDF volume, the &Dnn volume and a user volume with an IDF, so if floppies are used, you will need to temporarily exchange one floppy for another as the operation proceeds (prompters ask for the floppies as they are needed).

You need one &Dnn volume for each area. For removable media, these are created in System Startup Task 4, and for HMs they are configured in System Startup Task 10.

To summarize, the following volumes are required for DEB activities:

- &OP1—Universal Personality Overlay No. 1, which contains the Data Entity Builder and other software.

- The appropriate GDF volume, as defined above.
- &Dnn—Area Database volumes

Plus one not yet mentioned,

- CDSG—CL Custom GDF volume, required when building custom GDF parameters. See the *Control Language Data Entry* publications.

3.6 TAG NAME FORMAT OPTION

The formatting option is the result of a standard written by ISA (International Standards Association). This ISA standard is a guideline for engineers to use when defining their databases. The ISA standard includes symbol and identification codes to be used for designing instruments and instrumentation systems. As a result of this standard, a tag will contain fields delimited with an under bar. Formatting, for display, of a tagname attempts to separate the tagname at the delimiter instead of in the middle of the field. The formatting option only applies to tagnames that must be displayed on two lines.

The Tag Name format option is a second option available for the user that selects the longer tagname option on the SYSTEM WIDE VALUES display.

	<u>OPTION</u>
TAG NAME SIZE	Short Long
FORMATTING ON LONG TAG NAME?	Yes No

The result of selecting the option is only noticed where a tagname field must wrap to a second line in order to display the entire tagname. Fields that do not wrap are not affected by this option. When the formatting option is selected, the tagname is separated at a delimiter. If one of the fields should extend beyond 8 or 11 characters, depending on the field size, then it is placed on the second line providing there is room for the complete tagname. The delimiter for fields is an under bar (_).

The following examples show how tagnames are formatted for an 8-character field.

8 CHAR TAG NAME	FORMAT OPTION APPLIED	FORMAT OPTION NOT APPLIED
1234567890123456	12345678	12345678
AA NNN A	AA NNN A	AA NNN A

12 CHAR TAG NAME	FORMAT OPTION APPLIED	FORMAT OPTION NOT APPLIED
1234567890123456	12345678	12345678
AA NNN ANNNN	AA NNN	AA NNN A
	ANNNN	NNNN

12 CHAR TAG NAME	FORMAT OPTION APPLIED	FORMAT OPTION NOT APPLIED
1234567890123456	12345678	12345678
AA NNNNN ANN	AA NNNNN	AA NNNNN
	ANN	ANN

16 CHAR TAG NAME	FORMAT OPTION APPLIED	FORMAT OPTION NOT APPLIED
1234567890123456	12345678	12345678
AA NNN ANNNN AAA	AA NNN A	AA NNN A
	NNNN AAA	NNNN AAA

The 11-character field is used when it is necessary to include the PIN ID with the tagname. The rules apply here as they did for the 8-character field names. The following examples show how tagnames are formatted for an 8-character field.

8 CHAR TAG NAME	FORMAT OPTION APPLIED	FORMAT OPTION NOT APPLIED
1234567890123456789	12345678901	12345678901
X1\AA NN NN	X1\AA NN NN	X1\AA NN NN

12 CHAR TAG NAME	FORMAT OPTION APPLIED	FORMAT OPTION NOT APPLIED
1234567890123456789	12345678901	12345678901
X1\AA NN AANNNN	X1\AA NN	X1\AA NN AA
	AANNNN	NNNN

12 CHAR TAG NAME	FORMAT OPTION APPLIED	FORMAT OPTION NOT APPLIED
1234567890123456789	12345678901	12345678901
X1\AA NNNNN AAN	X1\AA NNNNN	X1\AA NNNNN
	AAN	AAN

16 CHAR TAG NAME	FORMAT OPTION APPLIED	FORMAT OPTION NOT APPLIED
1234567890123456789	12345678901	12345678901
X1\AA NN AANNNN AAA	X1\AA NN	X1\AA NN AA
	AANNNN AAA	NNNN AAA

SECTION 4 - TYPICAL SESSIONS WITH THE DEB

4.1 PURPOSE OF THIS SECTION

Typical DEB sessions are described in this section for two reasons:

- To provide a quick overview of typical sessions for new users.
- To define the order in which we recommend that you accomplish the major steps.

For detailed instructions for these major steps, refer to Sections 6 and 7.

Figure 4-1 shows the typical flow of the activities in a DEB session.

4.2 BUILDING DATA POINTS THROUGH PARAMETER ENTRY DISPLAYS

NOTE

If you need to build only a few points of one type (i.e., three or four numeric points) you can use the procedure outlined in this section. You should use exception building (4.5) if you need to build a larger number of points of the same type.

The following procedure describes how to use the PED and a few DEB commands to build, save, list and load process type points:

1. Check that the prerequisites defined in Section 3 have been met.
2. On the Engineering Main Menu, select HIWAY GATEWAY, APPLICATION MODULE, COMPUTING MODULE, or NETWORK INTERFACE MODULE.
 - 2a. If you select NETWORK INTERFACE MODULE, on the next display, select PROCESS POINT BUILDING.

DD MM YY HH:MM			
ENGINEERING MAIN MENU - SELECT DESIRED			
Network		Function	
UNIT NAMES	HIWAY GATEWAY	PICTURE EDITOR	
AREA NAMES	LOGIC BLOCKS	FREE FORMAT	
CONSOLE NAMES	APPLICATION	BUTTON	
LCN NODES	COMPUTING	HM HISTORY	
SYSTEM WIDE			
VOLUME	NETWORK INTERFACE MODULE		
		DOCUMENTATION	
	BUILDER	AREA DATA	
Support Functions and Utility Functions			
COMMAND PROCESSOR	SYSTEM MENU	SYSTEM STATUS	CONSOLE STATUS
SUPPORT UTILITIES	SMCC/ MAINTENANCE	R500 (C) Honeywell Inc.,	

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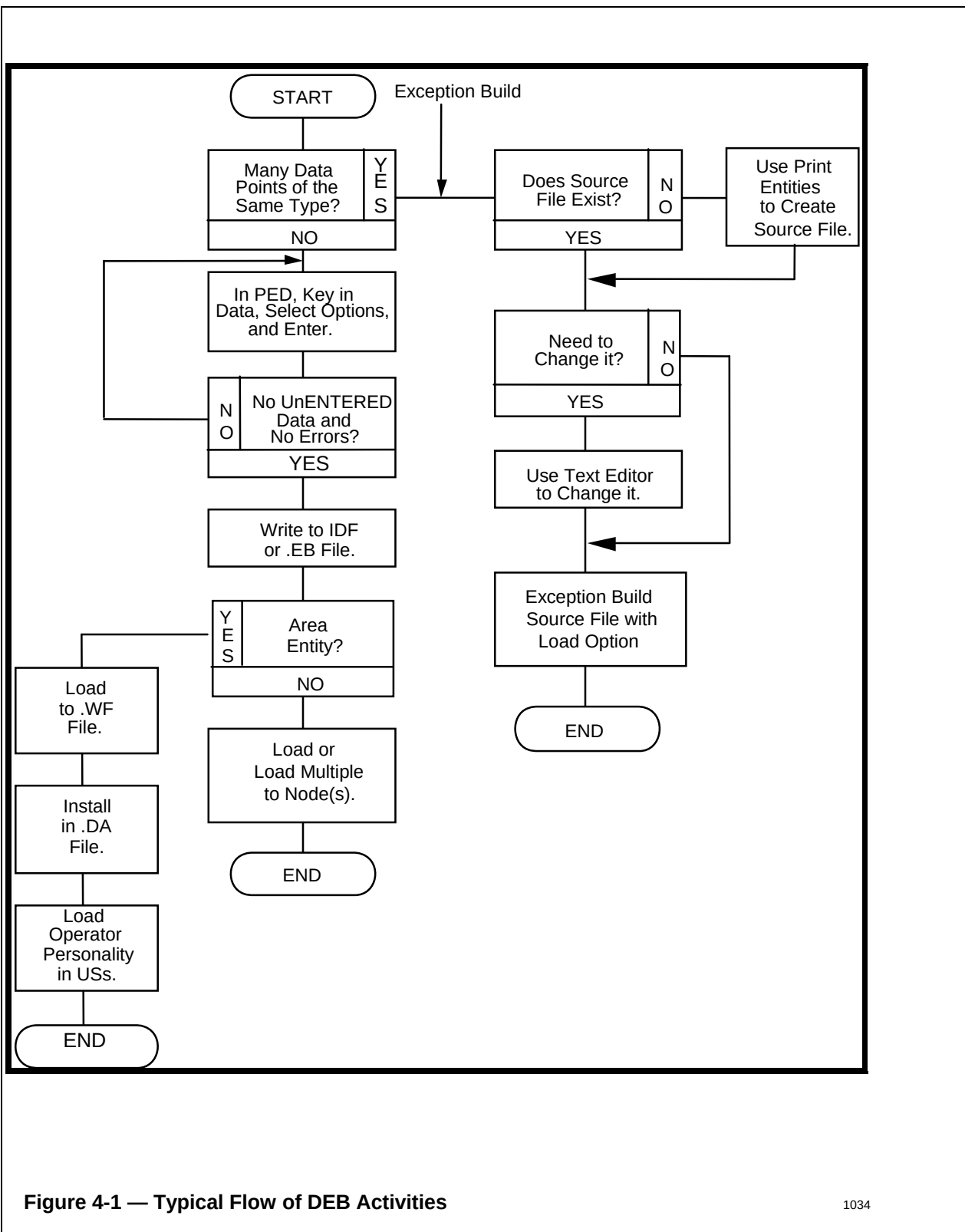


Figure 4-1 — Typical Flow of DEB Activities

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- 2b. The AM, HG, NIM or CM Build-Type Select Menu appears.
3. Select the appropriate point type.
- 3a. The first page of the Parameter Entry Display (PED) appears.

DD MM YY HH:MM 01

MENU

HG BUILD TYPE AND CONFIGURATION MENU

HG PROCESS POINT BUILD TYPE

ANALOG INPUT ANALOG OUTPUT ANALOG COMPOSITE

DIGITAL INPUT DIGITAL OUTPUT DIGITAL COMPOSITE

REGULATORY COUNTER TIMER

FLAG NUMERIC PROCESS MODULE

LOGIC BLOCK CONTROL COUNTER BATCH ASSIGN

HG HIWAY CONFIGURATION

HIWAY CONFIG BOX/SLOT CONFIG

HG LIBRARY CONFIGURATION

HG LIBRARY

F1=PED F3= F5=OVERWRITE F7=RECON F9=WLK BACK F11= F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD

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4. Key data into the ports and make selections as required. In most cases, you should get this information from your filled-in configuration forms. The data you key in is white.

DD MM YY HH:MM 01

PARAM. ENTRY DISPLAY - POINT: UNIT: UNENTERED PAGE 01 OF 04

HG - POINT ASSIGNMENT REGULATORY

TAG NAME (NAME)

UNIT ID (UNIT)

POINT DESCRIPTOR (PTDESC)

POINT KEYWORD (KEYWORD)

PRIMARY MODULE POINT ID (PRIMMOD)

HIWAY NUMBER (HWYNUM) 01

BOX TYPE (PNTBOXTY) CB EC MC

BOX NUMBER (BOX NUM) 00

POINT DISCLOSURE (PTDISCL) BRIEF FULL

SLOT NUMBER (SLOTNUM) 01

LOAD DESTINATION (LOADDEST) HG HG_HIWAY

F1=PED F3= F5=OVERWRITE F7=RECON F9=WLK BACK F11= F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD

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NOTE: With TPN R683, there is an option to display the priority and the selective cutout parameters for the NIM points in PED, for which the trip point parameters are set to NaN. By default, the PED does not display these parameters, if the corresponding trip point parameters are set to NaN. To display these parameters in the PED, you must change the value of the following option in the NCF.

System Wide Values > Console data: PRINT NIM PRIORITIES TO .EB FILE WHEN TP IS NAN.

See the *Network Form Instructions* for more information.

5. Press ENTER. If no errors were detected, this page of the PED is redisplayed with the data in light blue.

NOTE: If you selected a pick outlined in yellow or if you keyed data into a port outlined in yellow, new picks or ports are added to the PED or new pages are added to the PED.

6. If necessary, key in new data or make new selections on this page of the PED and press ENTER. If no errors were detected, this page of the PED is redisplayed with the data in light blue.
7. Press PAGE FWD. The next page of this PED appears.

DD MM YY HH:MM 01

PARAM ENTRY DISPLAY - POINT:FC100 UNIT01

ALGORITHM

(ALGIDDAC)

DAS	PIDNORM	PIDRATIO	AUTRATIO
AUTOBIAS	PIDCMA	PIDCM	PIDSPC
PIDERSOG	PIDERSQI	PIDGAP	LEADLAG
DVERHISL	DVERLOSL	SUMRWMAN	MULTWMAN
AUTOMAN	SWITCH	CBSUMR	MULT
PIDPL50	PIDPL50C	SORTPROD	SUMSORT
DIVDR	SORT	HISEL	LOSEL

F1=PED F3= F5=OVERWRITE F7=RECON F9=WLK BACK F11=

F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD

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8. Repeat Steps 4 through 6 until the status on line 2 of the PED looks like this example:

PED>>>>>-POINT: FC100 UNIT:nn PAGE 008 OF 008

Status Field Should be blank. If not, _____

ERRS (Red): DEB detected error(s) or missing data

UNENTERED (White): UnENTERed data

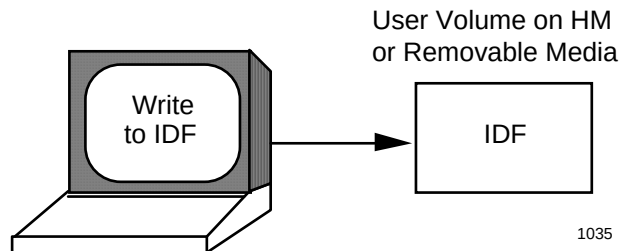
This is page 8 and it is the last page in this PED

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With TPN R683, the following DEB functions display the NIM priority parameters and the related parameters when the trip points are set to NaN, if the NCF option PRINT NIM PRIORITIES TO .EB FILE WHEN TP IS NAN is set to YES.

- Write to IDF
- Read to PED
- Load / Load Multiple
- Reconstitute / Reconstitute Multiple
- Exception Build
- Print Entities

9. Use the Write to IDF command (7.2.1) to store the point in an IDF for future use. When the point is stored, the first page of the PED reappears. For details on this and other ways to store entities, section 9.6.



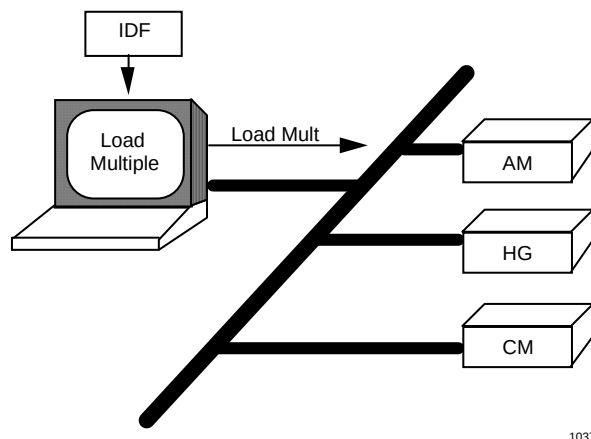
- 10a. To build another point of this type, repeat Steps 4 through 9.
- 10b. To build the first point of another type for this node, hold CTL and press F9, then repeat Steps 3 through 9.
- 10c. To build a point for another node type, hold CTL and press MENU, then repeat steps 2 through 9.

11. Use the List Entities in IDF command (7.2.10) to create a list of the points you have built. In Step 13, you will use this list file to print a list of the entities you have built.

COMMAND DISPLAY						DD MM YY HH:MM 01 PAGE 01 OF 01	
WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA		
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA		
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE			
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW		
REFERENCE PATH NAME pathname for IDF pathname for SELECTION LIST BUILD TYPES of Entities to List: ALL AM HG CIU NIM							
F1=PED F2=RECALL DISP F3= F4= F5=OVERWRITE F6= F7=RECON F8=PED STATUS F9=WLK BACK F10=WRITE F11= F12=LOAD							

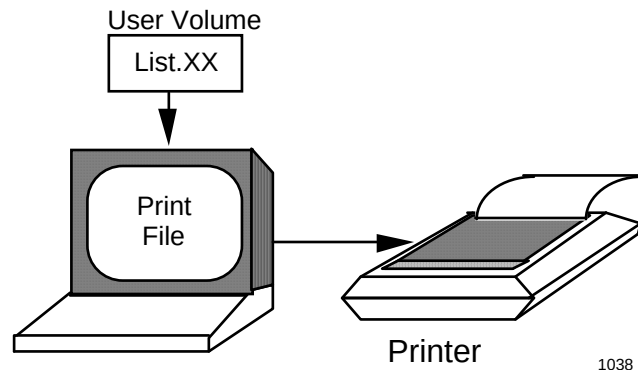
5

12. Use a Load command (7.2.3) (Load Multiple (7.2.4) if the points refer to each other) to load the points in the destination modules or gateways.



NOTE: The order in which the points are loaded is important. See the second note under 7.2.4.

13. Use the Print File command (7.2.11) to print the list of points that you built.



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14. At a US with the Operator personality, execute a SAVE DATA command to request a demand (manual) checkpoint from the node(s) in which you loaded the new entities. If those nodes are NIMs or HGs, checkpointing should be enabled (see the *Process Operations Manual* for instructions).

4.3 BUILDING AREA DATABASE ENTITIES

The process of building an area database is generally the same as the set of procedures described in 4.2 Building Data Points through Parameter Entry Displays. The difference is that after you print a list of the entities you must install them using the Select Area and Install Area commands.

This is the order of the recommended major tasks:

- Key data into the PEDs and make appropriate selections.
- When all pages have been ENTERed, and the PED status line indicates no errors and no unENTERed data; store the entity in an IDF for safekeeping and future use.
- Repeat the first two tasks for other entities of the same type.
- For the HIWAY GATEWAY, APPLICATION MODULE, COMPUTING MODULE, and AREA DATA BASE activities, return to the last menu (hold CTL and press F9), select the next type of entity, and repeat the preceding tasks.
- Load the entity (or entities) in its (their) destination(s).
- Print a list of the entities in your IDF to provide a written record.

Next, for area database entities only, the entities are installed in volume &Dnn, file AREAnn, using the Install Area command, where "nn" is the area number.

4.3.1 Aborting the Area Database Workfile (Areann.WA)

With R640, an abort function allows the user to delete the area database workfile (AREAnn.WA). The abort function is implemented using the F3 key.

The area database abort function operates in almost the same manner as the NCF abort function, which deletes the NCF workfile (NCF.WF).

If you have selected an area and are building an area database file, using the F3 key to delete the area database workfile causes the following message to appear:

```
WARNING: AREA WORK FILE FOR THE SELECTED AREA WILL BE DELETED TO  
COMPLETE THE ABORT OPERATION PRESS ENTER, OTHERWISE PRESS CANCEL.
```

Pressing ENTER at this time causes the area database workfile to be deleted, and causes the following message to be displayed:

```
AREA WORKFILE HAS BEEN DELETED
```

If CANCEL was pressed instead of ENTER, the area database builder returns you to the .WA workfile, thereby allowing you to make more entries in the file.

If the ABORT function is initiated and an area database was not previously selected, the following message is displayed:

WARNING: AREA NOT SELECTED. TO COMPLETE THE ABORT OPERATION PRESS ENTER, OTHERWISE PRESS CANCEL.

If you press ENTER, the workfile is not deleted because the workfile did not exist. If you press CANCEL, the abort operation is canceled.

If the selected area work file is protected, the following message appears:

```
FILE PROTECT OR ACCESS PRIVILEGE VIOLATION (21)  
NET>&Dnn>AREAnn.WA
```

where nn is 1 of 10 areas that the user has selected through SELECT AREA target.

If the selected area database file has been opened by another user, the following message appears:

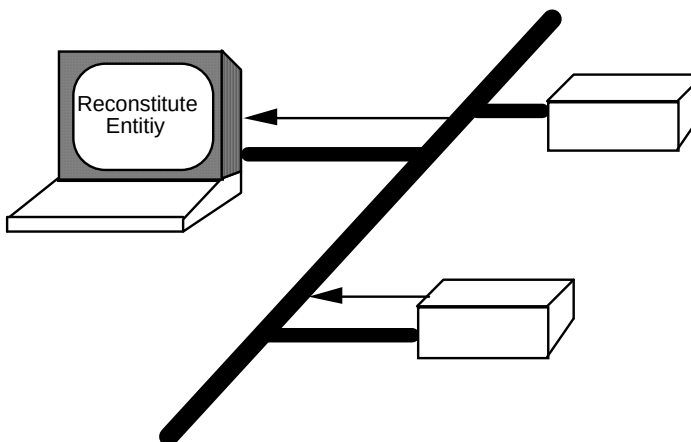
```
FILE IS OPEN BY ANOTHER USER (34) NET>&Dnn>AREAnn.WA
```

where nn is 1 of 10 areas that the user has selected through SELECT AREA target.

4.4 MAKING CHANGES TO AN EXISTING ENTITY

The Reconstitute command is used to make changes to an existing entity. Do not try to make simple changes by deleting an entity and rebuilding it. This can cause very difficult problems where entities refer to each other. See 9.4, Fixing Duplicate Entities and Deleting Entities for more information.

1. Use the Reconstitute command (7.2.5) to retrieve the entity and place it in a PED set.



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- 1a. The first page of the PED appears.

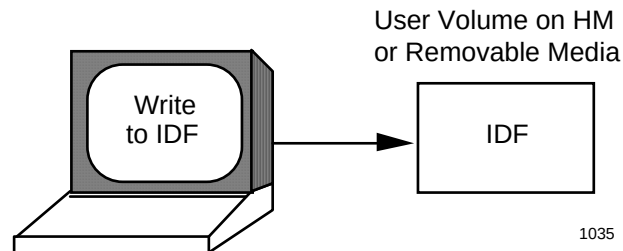
PARAM. ENTRY DISPLAY - POINT:		UNIT:		UNENTERED PAGE 01 OF 04	
HG - POINT ASSIGNMENT					
REGULATORY					
TAG NAME	(NAME)				
UNIT ID	(UNIT)				
POINT DESCRIPTOR	(PTDESC)				
POINT KEYWORD	(KEYWORD)				
PRIMARY MODULE POINT ID	(PRIMMOD)				
HIWAY NUMBER	(HWYNUM)	01			
BOX TYPE	(PNTBOXTY)	☒ CB ☐ EC ☐ MC			
BOX NUMBER	(BOX NUM)	00			
POINT DISCLOSURE	(PTDISCL)	☒ BRIEF ☐ FULL			
SLOT NUMBER	(SLOTNUM)	01			
LOAD DESTINATION	(LOADDEST)	☒ HG ☐ HG_HIWAY			
F1=PED F3= F5=OVERWRITE F7=RECON F9=WLK BACK F11= F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD					

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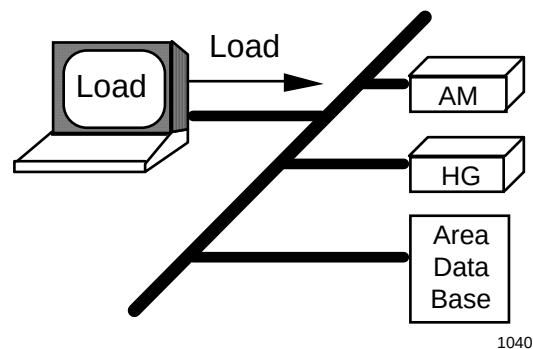
NOTE: The selections and values that appear on this display are the "live" values from the module or gateway where the entity resides, and these values are not necessarily the values that were configured. For area entities, the values are reconstituted from the area work file.

2. Key in changed data and make new selections, as required. If there are no errors, press ENTER. The PED reappears with the data in light blue.

- When there are no unENTERed data and no errors, use the Write to IDF command (7.2.1) to store the entity in an IDF for future use. For details on this and other ways to store entities, see 9.6.



- Use the Load command (7.2.3) to load the entity in its module or gateway. Area Database entities are loaded in the work file for the area.



- If this is an Area Database entity, you must also use the Install Area command (7.2.13) to store the area database entities in the area database data file, which is used in loading the operator personality in Universal Stations assigned to the area.

COMMAND DISPLAY						DD MM YY HH:MM 01	
						PAGE 01 OF 01	
WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA		
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA		
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE			
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW		

INSTALL AREA DATA BASE WORKFILE AS AREA DATA BASE FILE

REFERENCE PATH NAME

Give path OR select area number and give logical device id for data base.

AREA NUMBER

current AREA DATA BASE pathname

pathname for WORKFILE

F1=PED	F3=	F5=OVERWRITE	F7=RECON	F9=WLK BACK	F11=
F2=RECALL DISP	F4=	F6=	F8=PED STATUS	F10=WRITE	F12=LOAD

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4.5 EXCEPTION BUILDING

We recommend exception building for most data point building. Individual points should be built as described in section 4.2, only to create the initial point of each type, or where very few points of a type (typically one or two) need to be built.

Exception building consists of the building of new data points that are copies of the initial point of a type, except that certain parameter values, which you specify, are different from the values in the initial point. For example, you might need to build 50 PID points to reside in an AM and all of these points are similar, except for the input and output connections and some of the tuning constants.

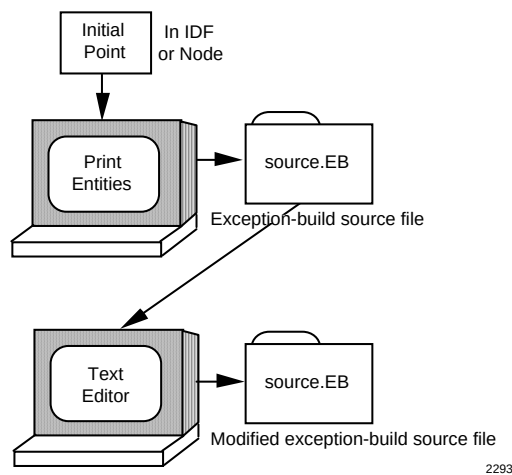
A convenient way to create an Exception Build (.EB) source file is to use the Print Entities command (7.2.11) to copy an existing point to a source file. If you enter a file name with a .EB suffix, the data is copied into the file in the proper form for exception building. You can then use the Text Editor to change parameters in the file and use it for exception building.

With TPN R683, there is an option to print the priority and the selective cutout parameters for the NIM points in the .EB file, for which the trip point parameters are set to NaN. By default, the EB files do not print these parameters if the corresponding trip point parameters are set to NaN. To print these parameters in the .EB files, you must change the value of the following option in the NCF.

System Wide Values > Console data: PRINT NIM PRIORITIES TO .EB FILE WHEN TP IS NAN. See the *Network Form Instructions* for more information.

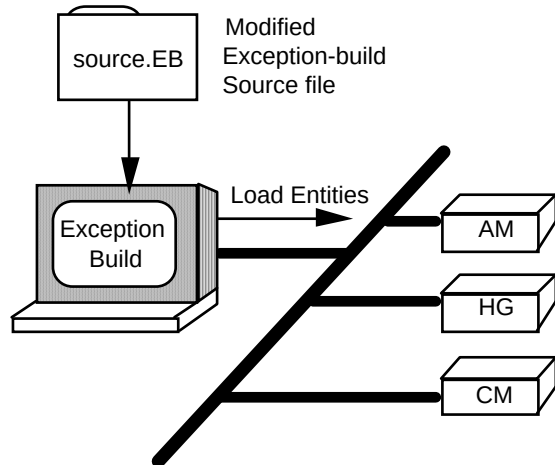
To find a point to copy as a source file, you can use the List Entities command (7.2.10) to query the system (modules, gateways, or IDFs) for existing points that are similar.

1. Either build an initial point of the type you wish to Exception Build, or use the List Entities command (7.2.10) to find one in a module or gateway or IDF.
2. Use the Print Entities command (7.2.7.2) to create an Exception Build (.EB) source file from the initial point. Use the Text Editor to edit that file as needed to specify the new points you need and the new parameter values for each new point.



3. To build and load new points in one operation, go to Step 4. To build new points and save them in an IDF, go to Step 6.

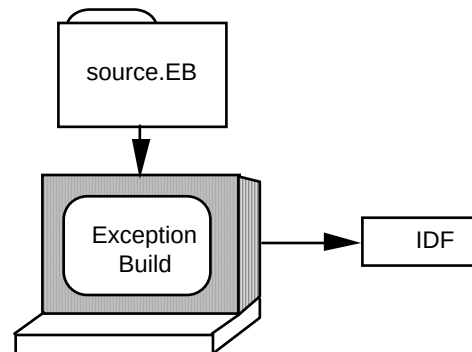
4. Use the Exception Build command with Load Entities pick selected (7.2.7.3) to build and load new points in the destination modules or gateways.



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5. Repeat Steps 1, 2, and 4 to build and load any other types of data points required.

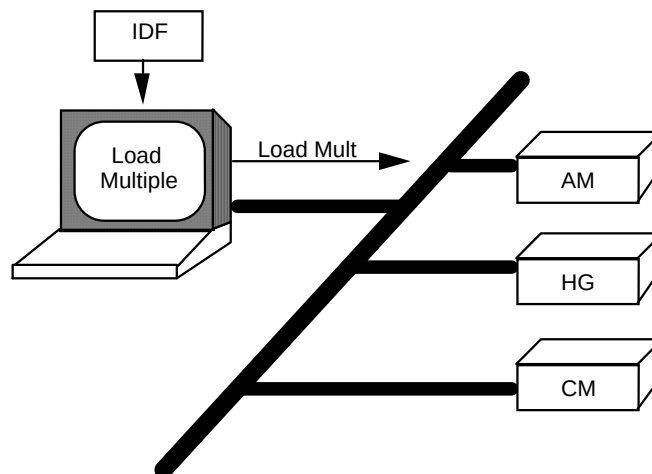
6. Use the Exception Build command (7.2.7.4) to build and store new points in an IDF. Do not select the Load Entities pick.



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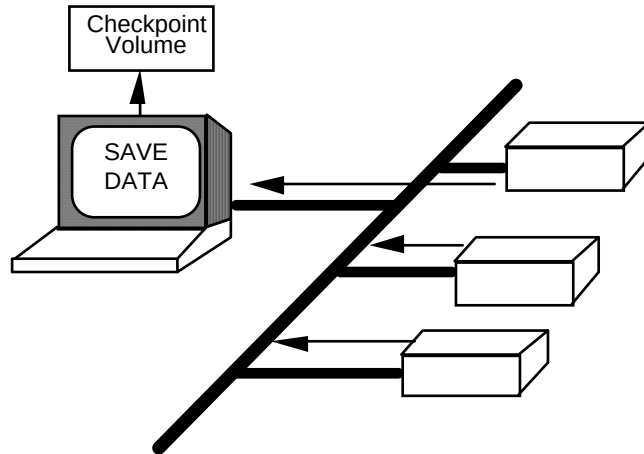
7. Use the Load Multiple command (7.2.4) to load the new points in the destination modules or gateways.

8. Repeat Steps 1, 2, 6 and 7 to build any other types of data points required.



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9. Checkpoint the database of each affected module and gateway immediately after loading new data points (see the *Engineer's Reference Manual*). This is essential in order to maintain a consistent database.



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SECTION 5 - ENGINEER'S KEYBOARD

The functions of the Engineer's Keyboard when the DEB is active are defined in this section.

The Engineer's Keyboard is required for interaction with the DEB. Most of the keys' function as you would expect; however, there are several function keys that have specific DEB functions. These functions are summarized on Figure 5-1.

Several keys that have dual functions have dual markings—a symbol or symbols on the top of the key and another symbol or symbols on the front edge of the key. The secondary function indicated on the front edge of each of these keys is obtained when one of the two CTL keys is held down while the function key is pressed.

The color and behavior keys (BKGND, BLINK, INTEN, WHT, BLK, CYAN, BLUE, MAGN, RED, YEL, GRN) do not operate when you are using the DEB.

There are two groups of function keys on the Engineer's Keyboard: F1 through F12 are generic function keys that have unique DEB functions. The second group consists of all of the other function keys.

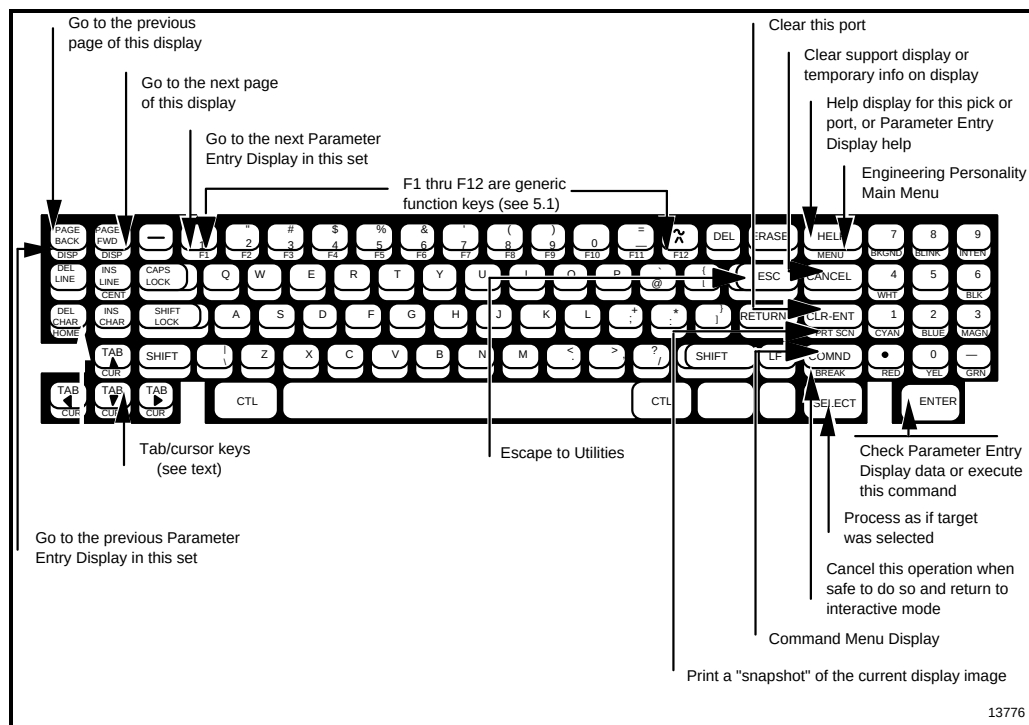


Figure 5-1 — Engineer's Keyboard, Summary of Key Functions

5.1 GENERIC FUNCTION KEYS

The prompter region of the DEB displays shows labels for the twelve generic function keys, except when that region is needed for prompter or error messages. When it is necessary to replace these labels with a prompter or an error message, the labels are restored as soon as possible. Figure 5-2 shows a simulation of these labels.

F1=PED	F3=ABORT	F5=OVERWRITE	F7=RECON	F9 =WLK BACK	F11=TAB
F2=RECALL DISP	F4=	F6=	F8=PED STATUS	F10=WRITE	F12=LOAD

Figure 5-2 — Generic Function Key

Note that F1 through F12 are on dual function keys and to get one of the generic functions, CTL must be held down while the key is pressed.

The function of each of these keys is as follows:

- **F1=PED** —Recalls the last Parameter Entry Display (PED). If no PED has been on the screen in this session, the following prompter appears: WARNING: NO PARAM. ENTRY DISPLAY TO RECALL.
- **F2=Recall Disp**—Recalls last support display (a PED help display, a display generated by a display command, or an error file display). If there is no such display to recall, the following prompter appears: NO DISPLAY TO RECALL.
- **F3=ABORT** —Deletes the selected area database workfile (AREAnn.WA) and exits the area database. Refer to Section 4.3.1 for a detailed description of the area database workfile abort function.
- **F4=** Reserved for future use.
- **F5=OVERWRITE**—Initiates an overwrite operation. When you attempt to write, load, reconstitute, or exception-build an entity, if that entity already exists in the destination, a prompter appears that asks if you want to overwrite it. See 7.1.6.
- **F6=** Reserved for future use.
- **F7=RECON**—Requests that the entity that is identified by the data ENTERed in the first few ports on the first page of the current PED (the ports that identify the entity) be reconstituted. This executes a Reconstitute command (see 7.2.5) without calling up the Reconstitute Command display. While this function works with data points, it is intended for use with reserved entities, where it lets you reconstitute without having to remember or to look up the entity name.
- **F8=PED STATUS**—Requests the status display for the current PED set. If no PED exists, the following prompter appears: NO PARAM. ENTRY DISPLAY STATUS TO RECALL.

- **F9=WALK BACK**—If a display other than a menu display is on the screen, this recalls the last menu display. If a menu is on the screen, this key recalls the preceding menu, if any. If a Parameter Entry Display is on the screen, and returning to a menu could cause data to be lost, the following prompter appears: WARNING! PED MODIFIED - REPEAT LAST KEYSTROKE TO IGNORE WARNING.
- **F10=WRITE**—Initiates a Write to IDF command (see 7.2.1). This executes a write to the last IDF indicated in the IDF pathname in the last Write to IDF command display, without the need to call up the command display. Thus, if you have updated information in a PED, you can write it to the IDF while the PED is still on the screen.
- **F11=TAB/CSR**—Refer to TAB/CUR in subsection 5.2.
- **F12=LOAD**—Requests that the entity in the current PED set be loaded in the appropriate module or gateway. If the data entered in the PED is not complete, the Load command is not executed and this prompter appears: PED NOT ENTERED. If the module or gateway the entity is to be loaded in is not operating, the prompter appears: ERR: NODE UNAVAILABLE, COULD NOT ESTABLISH ENTITY ON NET.

5.2 FUNCTION KEYS

The uses of the function keys other than the generic function keys are as follows (for functions on the front edge of a key, first press and hold one of the two CTL keys):

- **HOME**—Moves the cursor to the upper left corner of the screen (line 1, character position 1).
- **TAB/CUR (Arrow keys)**—These keys move the cursor. The following table explains how they work with the F11 function key. The current state of the F11 key (either F11=TAB or F11=CSR) is displayed at the lower right of the screen when the key is effective. Pressing F11 toggles between the two states.

F11 State	CTL Key Up	CTL Key Down
F11=TAB	Pressing an arrow key moves the cursor one position in the indicated direction. Two cursor keys can be simultaneously pressed to cause diagonal movement. The cursor continues to move as long as the key or keys are held down.	Pressing an arrow key moves the cursor to the next port or target in the direction indicated.
F11=CSR	Pressing an arrow key moves the cursor to the next port or target in the direction indicated.	Pressing an arrow key moves the cursor one position in the indicated direction. Two cursor keys can be simultaneously pressed to cause diagonal movement. The cursor continues to move as long as the key or keys are held down.

- **PAGE FWD, PAGE BACK**—These keys are used to page through displays that have multiple pages. If there is no next page or preceding page, one of the following prompters appears: NO FORWARD DISPLAY or NO BACKWARD DISPLAY.
- **DISP (FWD, BACK)**—If one of the CTL keys is held, the PAGE FWD and PAGE BACK keys become DISP FWD and DISP BACK keys, in which case they request the next or preceding display in the PED set. If there is no next or preceding display, one of the following prompters appears: NO FORWARD DISPLAY or NO BACKWARD DISPLAY.
- **CLR-ENT**—Clears all values in the port that the cursor is in. If the cursor is not in a port, the station "beeps."
- **CANCEL**—Clears help information, PED Set Status or a Command Display, from the screen. If no such displays are on the screen, the following prompter appears: NOTHING TO CANCEL.
- **ESC**—Escape to Utilities. Call up the Command Processor display, through which the file manager utilities and text editor can be used (7.1.13). To return to the DEB display, hold CTL and press HELP.
- **CENT**—Puts the line the cursor is on at the center of the screen. When the last full screen of a PED is displayed, and a few lines are left over on the next screen image, pressing CTL and CENT makes the partial screen join with the bottom of the last full screen, pushing an equal number of lines off the top. This is an easy way for you to see the last few lines in the context of the lines that precede them.
- **HELP**—Requests the display of help information for the pick or the port the cursor is in. If there is an error in the data in the port or an error has been made in selecting a pick, up to two additional lines of error information appear in the prompt region. To return to the PED, press CANCEL. If the cursor is not in a pick or a port, the following prompter appears: CURSOR NOT IN PORT OR PICK.
- **ENTER** (with a PED on the screen)—Requests that the data and selections in the PED set be processed and checked. If the cursor is in a port and no error is found, the cursor is tabbed to the next pick or port. If the cursor is not in a port, it is tabbed to the next pick or port. Should there be no selections or data, the following prompter appears: NOTHING TO ENTER.
- **ENTER** (with a Command Display on the screen)—Requests that the command be executed. If the command uses or will destroy parameter entry data not yet ENTERed, the following prompter appears: WARNING! PED MODIFIED - WRITE OR LOAD. REPEAT LAST KEYSTROKE TO IGNORE WARNING. If ENTER is pressed again, the command is unconditionally executed.
- **BREAK**—Requests that execution of the current command be terminated. If an entity or multiple entities are being processed, execution stops after the present entity is processed. This key can also be used to stop the printing of data requested by one of the print commands. Operates only during multiple entity operations and when "WAIT" or "PROCESSING ENTITY" is in the prompter region.

- **COMND**—Requests the Command Display.
- **PRT SCN**—Requests that the present screen image be copied and printed on a Matrix Printer. The printer must be connected to this Universal Station. This function does not print transitory images.
- **MENU**—Requests the Universal Personality's Engineering Main Menu. If data or selections have been entered but not yet filed or loaded, the following prompter appears: WARNING! DATA MODIFIED - WRITE OR LOAD. REPEAT LAST KEYSTROKE TO IGNORE WARNING.
- **SELECT**—Selects the pick the cursor is in. This duplicates touching a target (pick) with the touch screen option. An unselected pick has a green or yellow outline. When it is selected it fills in with light green with white or light blue letters showing through. Picks on menus do not fill-in when selected.

SECTION 6 - DATA ENTRY

This section explains how to enter data in the DEB's picks and ports.

The first activity in building an entity is the entry of at least some data in a Parameter Entry Display (PED). You see either the first display in the PED set or a menu that takes you to the first PED for the type of entity you are building when you touch or select one of these picks on the Universal Personality's Engineering Main Menu:

HIWAY GATEWAY	HM HISTORY GROUPS
APPLICATION MODULE	AREA DATA BASE
COMPUTING MODULE	BUILDER COMMANDS
NETWORK INTERFACE MODULE	

You do not have to enter data in the ports or make selections (picks) in any particular order. You can press ENTER to request that the data be processed and checked at any time, even if the data is not complete.

6.1 PARAMETER ENTRY DISPLAY FORMAT

Figure 6-1 shows the PED format. The parameter entry portion of the display is lines 3 through 26. The global description files (2.1) define what appears in this region, what data and selections are to be entered, and what to check.

Line 2 of all PEDs is a status line that gives you some very important status information. This example shows how it appears and what it means:

PED>>>>>> - POINT: FC100 UNIT 01 ERR UNENTERED PAGE 002 OF 008

Tag Name — FC100

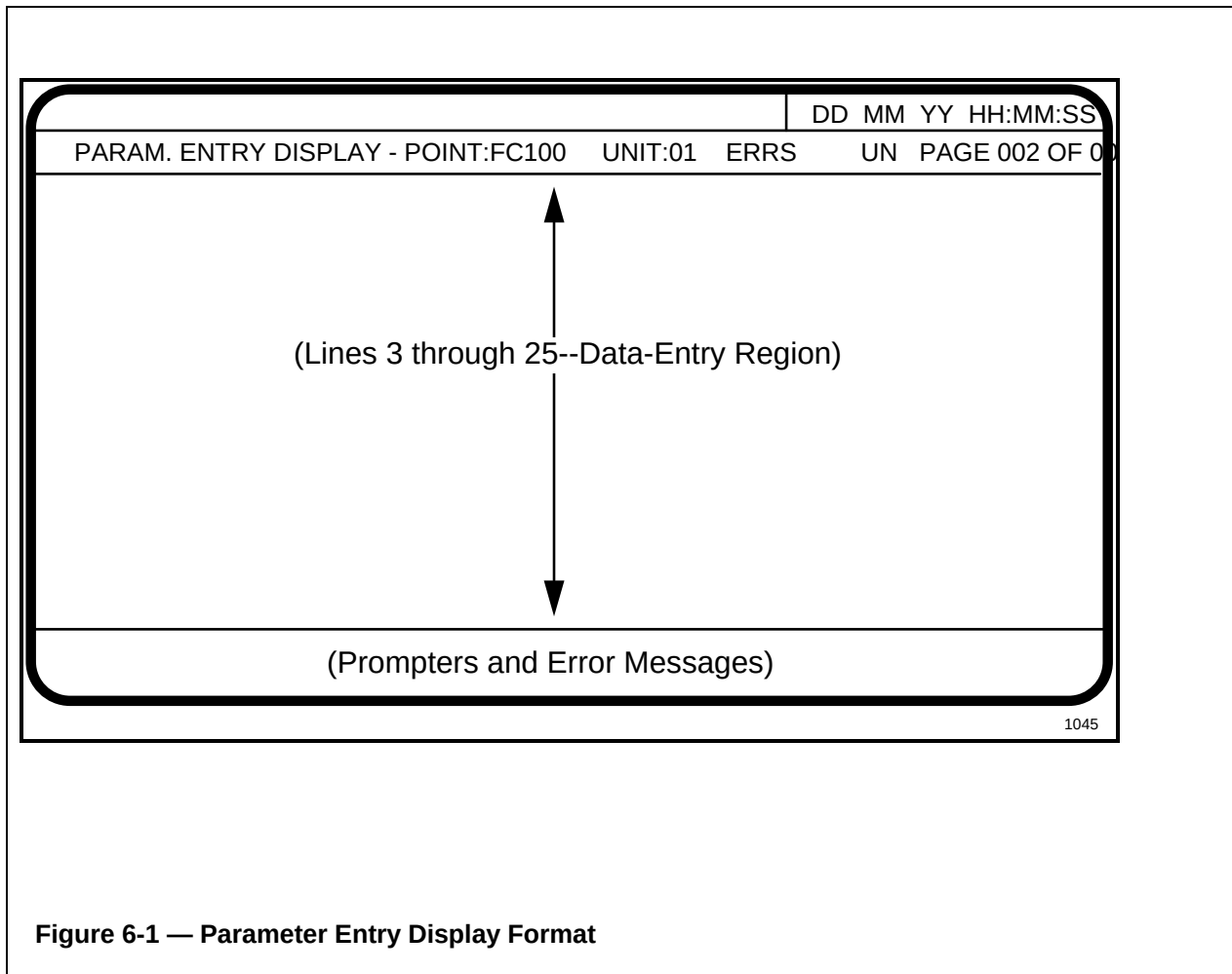
Unit ID — UNIT 01

Appears in red if the DEB detected one or more errors or missing data when ENTER was pressed. — ERR

Appears in white if selections have been made or data keyed in and not yet ENTERED. — UNENTERED

This is page 2 and there are 8 pages in this PED set. — PAGE 002 OF 008

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The two bottom lines of the PEDs normally show the generic function key labels (in green), but these are replaced by prompts and error messages when necessary. The color of each message in this region has meaning, as follows:

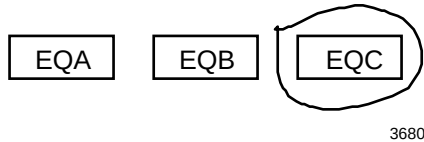
- Yellow—An informative message. Yellow prompts appear to confirm that a function is in progress. The most common of these is ****Busy****, which appears when something is happening that takes a noticeable amount of time.
- Red—All red messages are error messages.
- Green—The generic function key prompts appear here in green, most of the time. They remind you of the functions of keys F1 through F12 on the Engineer's Keyboard (5.2).

The red and yellow prompts are replaced by the generic function key labels (see Section 5, Figure 5-2) when the next operation begins. Pressing keys such as ENTER, CANCEL, COMND, F5, etc., causes new operations to begin.

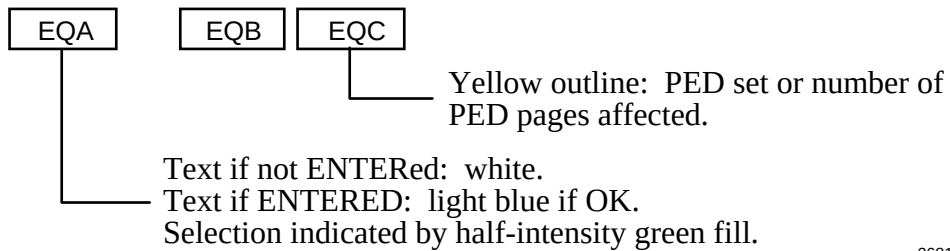
6.2 USE OF PICKS AND PORTS

You enter data in Parameter Entry Displays (PEDs) in picks and ports:

- **Picks**—You need to make choices when configuring an entity, such as selecting an equation, the type of initialization, or whether to enable an alarm or not. These choices are made through touch targets or picks. On a configuration form, indicate your choice by circling the item, like this:



On a PED display, an unselected pick appears with a green or yellow outline. Green means that selecting the pick does not affect the structure of the display set. Yellow means that it does affect the display set structure. You can also tell if the structure is affected by referring to the appropriate publication from the list under 1.3.2. To select a pick if you have the touch target option, you merely touch the appropriate pick. You can also tab to the pick and press SELECT. When a target (or pick) is selected, it fills in with light green. If a pick has not been ENTERed, its text is white; if it has been ENTERed, the text is light blue (cyan); if the selection is in error the outline is red.



The PED set is actually changed only when a yellow-outlined pick is selected and ENTER is pressed. Two types of changes are possible:

- 1) Picks or ports are added to the PED because of the selection of a pick.
- 2) Pages are added to the PED set because of the selection of a pick. If this happens, PAGE ### OF ### on the status line shows the change.

If you change such a selection after having made it, you might press ENTER and cause the current display to be removed from the set. If so, the present display is replaced by the PED Set Status Display.

- **Ports**—You type strings of displayable/printable characters in ports. The number of character positions in a port defines the maximum number of characters. If you do not know whether to enter numbers or letters or both, refer to the appropriate publication (1.3.1).

If you do not use the maximum number of characters, you can type the data anywhere in the port where it fits. For example,

LINEAR GAIN VALUE

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If the data keyed into a port affects the PED-set structure, the port is outlined in yellow; otherwise, it is outlined in green. Data in the port that has not been ENTERed is white. When it has been ENTERed it is light blue (cyan).

LINEAR GAIN VALUE

unENTERed data: white.

ENTERed data OK: light blue.

Yellow outline: display set affected.

Green outline: display set not affected.

Red outline: error in this port.

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Here is an example that shows how errors are indicated in the data keyed into a port:

For a port that requires a real number:

After ENTER is pressed:

ab appears in red because of a syntax error (only numbers are valid).

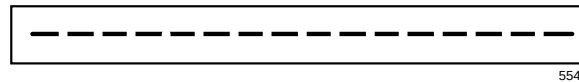
These characters are blue.

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If the value entered in a port is outside the acceptable range or is an illegal name, the port outline is red.

If an error is indicated, you can place the cursor in a port with a red outline or with red characters and press HELP to get additional information in the prompter region.

Most ports have default values in them (in white) that is typed over if a different value is needed. Sometimes the default value is



which indicates the value Not a Number (NaN). This means that value is not used. If you key in two or more dashes (--), you have entered NaN.

- The NAME port that appears on the first PED page for every data point has no default value and is initially filled with underlines. You must key a valid tag name into this port and replace all remaining underlines with spaces before pressing ENTER. The easiest way to do this is to press CLR-ENT, then key in the tag name.

6.3 DISPOSITION OF ENTERED DATA

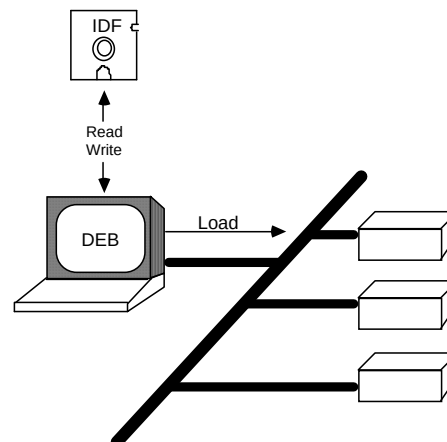
The data entered through the PED has no value until you dispose of it. It can be disposed of either by storing (writing) it in an Intermediate Data File (IDF) or by loading it in its destination module or gateway.

Refer to 9.6 for a list of the advantages and disadvantages of using IDFs as a storage medium for data entities. If you use IDFs and store them in a user volume on an HM, you should copy them to a removable medium because it is possible that some mistake or malfunction could destroy the data on the HM. You can use the Copy Volume command to do this. See *Command Processor Operation* in the Implementation/Startup & Reconfiguration - 1 binder.

Writing to an IDF can serve as a temporary disposal of entered data (complete or incomplete, and with or without errors), so that you can do data entry for another entity or perform one or more command functions.

Loading to the destination module or gateway causes the data owner in the module or gateway to validate the data and to return error information, if it finds errors. As it is validated, the data is loaded.

The procedures for writing to and reading from an IDF and for loading a module or gateway are in Section 7 of this publication.



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6.4 DISPLAY CONVENTIONS AND RELATIONSHIPS

One of the following character strings is displayed at the left of the status line of each DEB display to indicate the type of display:

Character Strings	Display Type
PARAM. ENTRY DISPLAY	PED
PARAM. ENTRY DISPLAY HELP	Support
PARAM. ENTRY DISPLAY SET STATUS	Demand
COMMAND DISPLAY	Demand
COMMAND DISPLAY HELP	Demand
COMMAND GENERATED DISPLAY	Support

As the **Display Type** column shows, there are three types of DEB displays:

- **Parameter Entry Displays**—There is a unique set of these displays for each type of entity you build. Further, the content of this set is affected by some of the selections (picks) you make. For example, if you choose an AM Regulatory Point, the set of displays is affected by the PV and control algorithms you select. In addition, some of the algorithm options affect the set of displays. PED displays are updated with entered data and error indications each time you press ENTER while a PED is on the screen.
- **Support Displays**—These displays provide auxiliary information, and once presented they are not updated. There are two types of these displays.

The PED Help displays are requested by pressing HELP while a PED display is on the screen with the cursor in a port or a pick. The Help display for that PED comes up on the screen and remains there until you press CANCEL or hold CTL and press F1, either of which brings back the PED.

A command generated display is a display generated by a Display File command or by a List command. These displays remain on the screen until you press CANCEL or use a generic function key to call up another display. If you press CANCEL, the last display that was on the screen before the command was selected, is recalled.

- **Demand Displays**—These displays are "demanded" from menus, by function keys, or by generic function keys. There are two types of these displays.

The PED Set Status display is called up by holding CTL and pressing F8. It displays the status of each of the PEDs and indicates whether the set has been ENTERed. You can use one of the PED picks on this display to select one of the PEDs in the set. See 6.6.

A command Help display is called up when you press the HELP key while a Command Display is on the screen with the cursor in a pick or port. The command Help display is replaced by the appropriate Command Display when you press CANCEL.

6.5 SELF-REFERENCES

The DEB uses an asterisk (*) to refer to parameters within the entity that is being built. When you need to refer to a parameter within the entity you are building you can use its own entity name or you can just use "*". When the DEB reconstitutes entities that have self-references (as in executing the Reconstitute command), it replaces any references to the same entity with "*".

As an example, for data point TC201, the following parameter designations are the same:

- TC201.PV
- *.PV

This self-referencing notation allows an entity to serve as an exception building template for other entities without rekeying the point tag-names in the body of the source file data.

6.6 USING THE PED SET STATUS DISPLAY AND PED HELP DISPLAYS

This PED Set Status display is very useful for determining which PED pages have errors or unENTERed data, and for going directly to those pages. In this example, the entity is an HG regulatory data point that has six PED pages.

Any time you need to know how many PED pages there are for this entity, and which have errors or missing data, you can look at this display to find out. To see it, hold CTL and press F8.

PARAMETER ENTRY DISPLAY SET STATUS		DD MM YY HH:MM 01
PARAMETER ENTRY DISPLAY - POINT FC100 UNIT 01		
Display Set ENTERed?	Yes	Error?
HG-POINT ASSIGNMENT		No
HG-PV CONFIGURATION		No
HG-ALARMING		No
HG-OPERATING CONFIGURATION		No
HG-MODE CONFIGURATION		No
F1=PED F3= F5=OVERWRITE F7=RECON F9=WLK BACK F11= F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD		

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If the PED Set Status display indicates that the display set has not been ENTERed, press ENTER to cause the DEB to check for errors and missing data. To go to a page that has errors, just touch its pick on this display or tab to the pick and press SELECT.

	DD MM YY HH:MM 01				
PARAMETER ENTRY DISPLAY HELP	PAGE 001 OF 001				
UNITS ARE NORMALLY ASSIGNED TO DISTINCT OPERATIONAL AREAS. ENTRY CONSISTS OF ONE OR TWO ALPHANUMERIC CHARACTERS.					
F1=PED F2 = RECALL DISP	F3= F4=	F5=OVERWRITE F6=	F7=RECON F8=PED STATUS	F9=WLK BACK F10=WRITE	F11= F12=LOAD

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SECTION 7 - COMMANDS

This section defines the functions of all of the DEB commands and provides step-by-step instructions for using them.

The DEB's Command Display (see Figure 7-1) has 22 picks, each of which selects a command and its picks and ports. At this time, the Command Display can be called up from any DEB display by pressing COMND on the Engineer's Keyboard. You can also call for the Command Display with the Builder Commands pick on the Engineering Main Menu.

While the 22 picks represent 22 different commands, many of these are variations of a command. There are only these 15 different types of commands:

- Write
- Load
- Reconstitute
- Repack IDF
- List
- Read to PED
- Print
- Install Area
- Select Area
- Display
- New
- Execute Command File
- Alter Parameters
- Delete
- Exception Build

COMMAND DISPLAY					DD MM YY HH:MM 01
					PAGE 01 OF 01
WRITE TO IDF	READ TO PED	DELETE SYS ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE	
ALTER PARAMETERS	EXCEPTION BUILD	BOX PARAM BACKUP RSTR	REPACK IDF	PRINT ENTITIES	NEW
F1=PED F2 = RECALL DISP	F3= F4=	F5=OVERWRITE F6=	F7=RECON F8=PED STATUS	F9=WLK BACK F10=WRITE	F11= F12=LOAD

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Figure 7-1 — Command Display

7.1 COMMON COMMAND FUNCTIONS

Several functions are common to more than one command. They are the following:

- Invoking Commands
- Termination of Commands
- The US Key Switch
- Command Pick and Port Behavior
- Pathnames
- Help Displays
- Command-Generated Files
- Use of the Text Editor
- Memory Space
- Generic Function Key Prompters
- Managing Removable Media
- Escape to Command Processor

These functions are defined in the following paragraphs.

7.1.1 How to Invoke Commands

All commands are invoked through the Command Display. This display is called to the screen from any DEB display by pressing the COMND key on the Engineer's Keyboard. This display can also be called up by selecting the Builder Commands pick on the Engineering Main Menu.

DD MM YY HH:MM 01
PAGE 01 OF 01

COMMAND DISPLAY					
WRITE TO IDF	READ TO PED	DELETE SYS ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE	
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW

F1=PED F3= F5=OVERWRITE F7=RECON F9=WLK BACK F11=
F2 = RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD

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The targets (picks) on the Command Display call up the corresponding command's picks and ports. Instructions for use of each of the commands are in 7.2. This shows what's displayed on the screen when you select the Write to IDF pick.

DD MM YY HH:MM 01
PAGE 01 OF 01

COMMAND DISPLAY					
WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE	
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW

WRITE BUILD ENTITY TO IDF

REFERENCE PATH NAME

with

pathname for IDF

F1=PED F3= F5=OVERWRITE F7=RECON F9=WLK BACK F11=
F2 = RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD

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7.1.2 Termination of Commands

When a command is executed without any errors, `OPERATION COMPLETE` appears in the prompter region, and the Command Display is usually replaced by the PED that was on the screen before the Command Display was called up. If the command was a Read to PED or Reconstitute command, the Command Display is replaced by the first display in the PED set for the entity read from the IDF or reconstituted from its module or gateway.

If one error (or more) is detected during an attempt to execute a command, an error message appears in red in the prompter region of the Command Display. The message remains on the screen until the command is executed or until the next DEB operation.

Some commands take several seconds to complete. During this time, one (or more) yellow prompter appears to indicate that the DEB is busy or to indicate the name of the entity that is being processed.

If `WAIT` is in the prompter region, command execution and printouts initiated by one of the Print Commands can be terminated by holding CTL and pressing BREAK. If one (or more) entity is being processed, command execution terminates after the present entity is processed. The green prompters reappear when the command is terminated. If `BUSY` is in the prompter region, BREAK causes the Universal Station to "beep" and the operation is completed.

7.1.3 Key Switch

All of the commands except the Display File, Print, and List Entities commands require the Engineer's key switch to be in the ENG position. If you press ENTER to execute any command that can change the process database, with the key switch in OPER or SUPVR, this message appears in the prompter region:

ENGINEER'S KEYLOCK MUST BE ENABLED TO USE THIS COMMAND.

7.1.4 Command Pick and Port Behavior

Picks (targets) and ports on the Command Displays behave almost as they do on PEDs (6.2). Ports have green outlines, unless they contain erroneous data. Picks have green or yellow outlines. A selected pick is filled with light green.

After pressing ENTER, if the information keyed in a port is acceptable, it appears in light blue (cyan). If the syntax of information in a port is not correct, invalid characters appear in red and the command is not executed. Errors other than syntax errors cause the port to be outlined in red, and the command is not executed. Once a command is invoked by pressing ENTER (with valid information in the ports), the Command Display remains on the screen until the command is executed.

If you go to another display before all data is keyed into a Command Display and all selections are made, you can recall that Command Display by pressing COMND.

7.1.5 Pathnames, Volumes, and File Names

You need to name and to keep track of the names of files that you use in entity building. To define the location of a file, the system uses a pathname that defines the path to the file. Complete pathnames are in this form:

\$F3>V123>FILE

File Name. (one to eight characters, \$ not allowed.)
This is the IDF name in most DEB commands. All entities in an IDF should be the same type and data points that refer to each other must be in the same IDF the first time they are loaded (see 7.2.3 and 7.2.4).

Volume Name. This is a user volume on a removable medium or on a HM. Removable medium volumes are established via the Utilities Function (see Task 4 in the *System Startup Guide*). HM volumes are established through Volume Configuration (see Task 10 in the *System Startup Guide*).

Logical Device Identifier. For a removable medium, \$F3 designates the third drive on the console, \$F4 would be the fourth drive, etc. For a History Module (HM), the Logical Device ID is NET.

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All of the Command Displays have a `REFERENCE PATHNAME` port. On the first Command Display that you use in a DEB session, you need to key in the device identifier and the volume name, for example,

NET>NORM

which says that user volume NORM is on an HM. After you execute this first command, the DEB remembers this reference pathname, and it appears on all subsequent Command Displays, unless you change it or return to the Engineering Main Menu. All commands that involve files also have ports that accept the names of the files. Files, other than IDFs and exception-build source files, usually require a suffix (see 7.1.8) but you do not have to enter a suffix for either IDFs or exception-build source files.

For example, suppose you key this Reference Pathname, plus other required information, in the first Command Display of your session:

\$F2>NORM>

You then execute the command, and the Command Display reappears with the same Reference Pathname:

\$F2>NORM>

You select READ TO PED. This display appears with the same Reference Path Name: \$F2>NORM>. You can now key in the Entity name and the same IDF pathname that you used last or you can key in another IDF pathname.

COMMAND DISPLAY

DD MM YY HH:MM 01
PAGE 01 OF 01

WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE	
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW

REFERENCE PATH NAME \$F2>NORM>

ENTITY name FC101

pathname for IDF HGPNTS

F1=PED F2 = RECALL DISP F3= F4= F5=OVERWRITE F6= F7=RECON F8=PED STATUS F9=WLK BACK F10=WRITE F11= F12=LOAD

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Once the command has been executed, you could decide to read another entity from another IDF on the same device. After you key in this new data, the display appears like this:

COMMAND DISPLAY

DD MM YY HH:MM 01
PAGE 01 OF 01

WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE	
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW

REFERENCE PATH NAME \$F2>NORM>

ENTITY name FC102

pathname for IDF AMPNTS

F1=PED F2 = RECALL DISP F3= F4= F5=OVERWRITE F6= F7=RECON F8=PED STATUS F9=WLK BACK F10=WRITE F11= F12=LOAD

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7.1.6 The "With Overwrite" Option

All commands that can change the process database or the content of an IDF require that you use the Overwrite option to make them overwrite existing data. These commands are:

- Write
- Load
- Load Multiple
- Reconstitute Multiple
- Exception Build

If you press ENTER to execute one of these commands which names an existing IDF or an existing entity in a module or gateway, the following message appears in the prompter region:

```
(entity or IDF) EXISTS - SELECT OVERWRITE OPTION IF DESIRED
```

When this message appears, consider carefully if you want to overwrite the existing entity or IDF. If you do, select the `With Overwrite` pick. The command is then executed.

You can also overwrite when using the generic function keys to execute a command from a PED. To do so, hold CTL and press F5.

Each execution of a Load or Load Multiple command that changes the existing database is recorded in the System Event Journal.

7.1.7 Help Displays for Commands

If, while you have a Command Display on the screen, you press the HELP key with the cursor in a command pick, the Command Display is replaced by a display that helps you to make selections on the Command Display and to key information in its ports. When a help display is on the screen, pressing CANCEL restores the Command Display.

7.1.8 Command-Generated Files and Other Files

Several of the commands automatically generate files. Table 7-1 lists each of these commands and the file types (suffixes) they generate.

Intermediate Data Files (IDFs) (.DB) have been described. Error files (.EF, .DY) are created when a data owner in a module or gateway detects errors in an entity sent to it by a Load command. Successful lists and unsuccessful lists list the names of entities that were handled successfully (.SL) and unsuccessfully (.UL) by commands that process multiple entities.

Another command-related file is the exception-build source file, which has a .EB suffix. These files can be built using the Print Entities command.

Table 7-1 — Command-Generated Files

COMMAND	FILE TYPE						NOTES
	.DB	.EF	.DY	.SL	.UL	IF	
Write to IDF	√						File Suffixes: .DB: IDF .EF, .DY: Error File .SL: Successful List .UL: Unsuccessful List If: List File: with a User-supplied suffix Beginning with .x. .y, or .z, or a .EL (edited list) file
Load		√					
Load Multiple		√		√	√		
Delete Multiple Entities				√	√		
Delete Multiple Entities from IDF				√	√		
Reconstitute Multiple	√	√		√	√		
List Entities in Module						√	
List Entities in IDF						√	
Print Entities in IDF				√	√		
Exception Build	√	√		√	√		

7.1.8.1 Entity Lists

Entity lists are lists of entity names created by or used by DEB commands that process more than one entity. These commands are

- Load Multiple
- Reconstitute Multiple
- Exception Build
- Delete (multiple entities)
- Print Entities
- Alter Parameters

Entity lists can be used for record keeping, for analyzing problems, and as selection lists. A selection list defines specific entities to be processed by one of the multiple-entity commands.

Entity lists are just lists of entity names that include both tag names for data points, and names beginning with "\$" that designate reserved entities. Names are separated by at least one space. A list of data points might look like this:

```
FC101 FC102 FC103 AB100 AB102 GG206A M205CD
AC4U11 MNR12A
```

A reserved entity list might look like this:

```
$HIWAY02 $HY02B05 $HY02B02 $HY02B03 $HY02B04
$HY02B06 $HY02B07 $HY03B04
```

Selection Lists

Selection lists tell the DEB which entities a multiple-entity command is to process. You can use the Text Editor to create selection lists, or you can use the Text Editor to edit any of the DEB-generated entity lists for use as a selection list.

If you only need a few entities in a selection list, you can key the entity names directly into the Command Display port labeled `pathname` for `SELECTION LIST` and thus avoid having to go to the Text Editor to create or to edit a list.

Selection-list file names are of this form:

```
name .EL
```

Selection-list files must have one of these suffixes:

.EL, .UL, .SL, or a user file suffix which begins with "X," "Y," or "Z," for example, .XX.
Lower case letters are also acceptable since they are treated as capitals.

Types of Entity Lists

- .EL Edited List—A list of entities created or edited for use as a selection list. See "How to Edit Entity Lists" below.
- .SL Successful List—A list of entities that were successfully processed during the execution of a command that processes multiple entities.
- .UL Unsuccessful List—A list of entities that were not successfully processed during the execution of a command that processes multiple entities. You could use a .UL file as a selection list; for example, you could use a .UL file with a Load Multiple command to load entities that were not processed by an earlier Load Multiple command, after you have corrected the problem that prevented them from loading.
- .lf List File—A list created by the List Entities command or created by a user through the Text Editor. These are user files that have a user file suffix, which begins with "x," "y," or "z," for example, .xx.

How to Create and Edit Entity Lists

1. With any DEB display on the screen press ESC. The Command Processor display appears.
2. Key in a command line to call up the Text Editor with the name of new, edited list, for example:

```
ED newname .EL
```

Note: If you have not yet used the SP command to set up a pathname to your user volume, do so (See the *Command Processor* manual in the Implementation/ Engineering Operations - 1 binder), or key in the full form of the pathname, for example

```
ED NET>USRV>newname .EL
```


3. See the *Text Editor Operation* manual for more detailed instructions. Read (F8, F2) the file to be edited into the editor's secondary buffer, Copy (F2, F1, F2) the entity names you want from the secondary buffer, Change (F8, F1) to the primary buffer and Paste (F2, F4) into that buffer.
4. Edit the file to add or delete entities as needed, and store your new file when you Quit (F1, F2) the Text Editor. The edited, selection-list file is now ready for use.
5. Hold CTL and press HELP (MENU). The DEB display that was present in step 1 reappears.

7.1.8.2 File Names Generated by the Data Entity Builder

When the DEB executes a command that generates a file, it names that file as follows:

- Commands that read from or write to an IDF use the IDF file name and the appropriate suffix; for example, `IDFA.EF`, `IDFA.SL`, or `IDFA.UL`
- Commands that do not use an IDF but have a selection-list pathname entered use the selection-list file name; for example, `sellist.EF`, `sellist.SL`, or `sellist.UL`.

CAUTION

Because the DEB commands create file names automatically, you may have unexpected conflicts with file names if you use one of the automatically generated names for another user file. For example, suppose you execute a Load Multiple command using an IDF whose name is `JOHN.DB`. If any errors are detected during execution of the Load Multiple command, the DEB creates error file `JOHN.EF`.

Now suppose you execute a command file whose name is `JOHN.EC`. If any errors are detected as `JOHN.EC` is executing, an error file named `JOHN.EF` is created, replacing the `JOHN.EF` file originally created by the Load Multiple command.

NOTE

A list of entities can be entered into the pathname for Selection-List port instead of the file name of a selection list. For example, `A110`, `B110`, `C110` or `$F1>test>LIST.EL` can be entered into the port.

7.1.9 Relation of the Text Editor to Commands

Any of the entity list files generated by DEB commands (see 7.1.8 for a list) can be copied and edited using the Text Editor to create selection lists to be used by other DEB commands. You give the edited files that result .EL (edited list) suffixes and you enter the suffixes with the file names in the pathname for SELECTION LIST ports on the Command Displays. .XX, .yy, and .Zz, are also legitimate suffixes.

Exception-build source files can be created through the Text Editor. These must be given .EB suffixes.

7.1.10 "Insufficient Media Storage"

The DEB makes the best use of memory and file space that it can. It's not likely, but in some circumstances it is possible that a command operation might not be completed because of a lack of storage space. In such a case, the following message appears in the prompter region:

```
INSUFFICIENT MEDIA STORAGE
```

The Universal Personality's Command Processor functions can be used to delete unneeded files (see *Command Processor Operation*), or you might use another removable medium to gain additional file space.

7.1.11 How to Get the Generic Function Key Prompters Back

If an error message or a prompter has replaced all or part of the generic function key prompters at the bottom of a Command Display, you can get them back by pressing the COMND key. This recalls the Command Display.

7.1.12 "Escape" to Command Processor Functions

You can go directly to the Command Processor from any DEB display by pressing the ESC key; then you can key in any of the Command Processor commands. You can also key in ED followed by a full pathname to enter the Text Editor to edit the file you named (see *Command Processor Operation* in the Implementation/Startup & Reconfiguration - 1 binder and *Text Editor Operation* in the Implementation/ Engineering Operation - 3 binder). To return to the DEB display, hold CTL and press HELP.

When a cartridge is needed that is not mounted in a drive, a message appears that asks you to mount it. One case where this occurs is during execution of a Reconstitute Multiple command for an area database entity, which at various times requires the &ARG GDF volume, the &Dnn volume, and a user volume with one or more IDFs.

7.2 COMMAND DESCRIPTIONS

7.2.1 Write to IDF

NOTE

We recommend that you use IDFs only for working copies of entities while you are building them. Exception-build (.EB) files are preferred as master copies of the entities you have built because they are readable text files and they facilitate installation of Honeywell-supplied software upgrades. Exception-build files can be used on other **TotalPlant** Solution (TPS) systems, IDFs cannot. See 7.2.7.2 for instructions on creating exception-build files.

Write to IDF stores the entity in the current Parameter Entry Display (PED) in an Intermediate Data File (IDF). The data is stored as is, complete or incomplete, with or without errors. The data must have been processed by the DEB (data is processed when you press ENTER after typing in ports or selecting picks).

There must be sufficient data to identify the entity. This means that for data points, you should enter at least a tag name. For reserved entities, the first several ports on the first PED page identify the entity and need to be ENTERed.

An IDF is established in an existing volume the first time you key its pathname into the Write to IDF Command Display and execute the command. You can add subsequent entities to that IDF by executing other Write to IDF commands with the same pathname. Volumes are established as defined under 3.3.

When data written in an IDF is recalled by a Read to PED command, errors and missing data remain, just as written.

NOTE

When the IDFs are upgraded from a previous revision to R620, or the R620 IDFs are updated, the file size of an IDF may increase if it contains procmod or array points. Because of the increased file size, the process may slow down. Refer 7.2.11.1 *Box/Module Parameters in EB* for details.

To execute a Write to IDF command:

1. On the Command Display, select the WRITE TO IDF pick. This Command Display appears.
2. If needed, key in a Reference Path Name (7.1.5). Key in the pathname for the IDF.

NOTE: If you are using removable media, the user volume needs to be in the drive indicated by the pathname while this command is executed.

DD MM YY HH:MM 01
PAGE 01 OF 01

COMMAND DISPLAY					
WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE	
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW

REFERENCE PATH NAME

with OVERWRITE

pathname for IDF

F1=PED F3= F5=OVERWRITE F7=RECON F9=WLK BACK F11= F12=LOAD
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE

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3. Press ENTER. When the data is stored with no errors, the Command Display is replaced by the PED that was on the screen before the Command Display was called up.*

If an error is detected, no data is stored in the IDF, the Write to IDF Command Display remains on the screen, and one of the following error messages appears in the prompter region:

(entity) EXISTS - SELECT OVERWRITE OPTION IF DESIRED

You tried to write over an existing entity tag name (entity) in the IDF. You can use the F5=OVERWRITE key to replace the data in the existing entity with the new data in the PED set. See 7.1.6.

INSUFFICIENT INFORMATION TO WRITE THIS DATA

You did not enter sufficient information in the ports in the PED to identify the entity.

NOTHING TO WRITE

You may have selected the Builder Commands pick to enter the DEB before creating a PED set.

* Once you have executed a WRITE TO IDF from the Command Display, you can use generic function key F10=WRITE, to write a PED to the same IDF.

PARAMETER ENTRY DATA NOT ENTERED

There is data in one or more PED picks and ports that have not been entered by pressing ENTER while a PED is on the screen.

IDF MAY HAVE BEEN CORRUPTED (pathname)

One or more of the entities in the IDF specified (pathname) has been corrupted. Use the Repack IDF command's Inspect IDF option (7.2.15) to determine which entities are corrupted, then use the Binary Repack option to gain back the space those entities occupied. Rebuild needed entities through a PED or via exception building (7.2.7).

IDF FULL (pathname) - REPACK OR START A NEW IDF

There is no room in the IDF specified (pathname) to accept another entity. Use another IDF or delete unneeded entities (7.2.8) and then use the Repack IDF command's Binary Repack option (7.2.15) to gain back the space those entities occupied.

VOLUME IS FULL - CANNOT WRITE IDF (pathname)

There is no room on the volume for the IDF specified (pathname) to accept another entity. Delete unneeded entities (7.2.8), and then use the Repack IDF command's Binary Repack option (7.2.15) to gain back the space those entities occupied. If your user volume is on an HM, you can name another volume (7.1.5), if one is available. If you are using removable media, you can insert a new disk, change the Reference Path Name to point to the new volume (7.1.5), and try again.

7.2.2 Read to PED

NOTE

We recommend that you use IDFs only for working copies of entities while you are building them. Exception-build (.EB) files are preferred as master copies of the entities you have built because they are readable text files and they facilitate installation of Honeywell-supplied software upgrades. Exception-build files can be used on other TDC 3000^X systems, IDFs cannot. See 7.2.7.2 for instructions on creating exception-build files.

Read to PED reads an entity from an IDF, to be used as the current PED set. The data is copied from the IDF, leaving the entity in the IDF. The data in the PED set is as it was copied from the IDF, with any errors, or possibly with missing data.

To execute a Read to PED command:

1. On the Command Display, select the READ TO PED pick. The READ ENTITY FROM IDF Command Display appears.

COMMAND DISPLAY						DD MM YY HHMM 01 PAGE 01 OF 01
WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA	
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA	
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE		
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW	

REFERENCE PATH NAME

ENTITY name

pathname for IDF

F1=PED	F3=	F5=OVERWRITE	F7=RECON	F9=WLK BACK	F11=
F2=RECALL DISP	F4=	F6=	F8=PED STATUS	F10=WRITE	F12=LOAD

2. If needed, key in the Reference Pathname (7.1.5).

NOTE: If you're using removable media, the user volume must be in the drive indicated by the pathname while this command is executed. The GDF, if not on the HM, must also be mounted. Refer to 3.5.

3. Key in the name of the entity you want to work on.
4. Key in the pathname for the IDF (7.1.5).
5. Press ENTER. When the command is executed with no errors, the first PED page for the entity you named appears on the screen.

If an error is detected, the entity is not read from the IDF, the Read Entity from IDF display remains on the screen, and one of the following messages appears in the prompter region:

ENTITY DOESN'T EXIST (entity)

You named an entity (entity) that is not in the IDF. Check your records to see if you can find whether you keyed in the wrong IDF name or the wrong entity name and try again. You can use the List Entities command (7.2.10) to see what entities are in the IDF.

FILE DOESN'T EXIST (pathname)

You named an IDF (pathname) that's not in the volume. Check your records and try again.

IDF DATA MAY BE CORRUPTED FOR (entity) FROM (pathname)

The specific entity (entity) you requested via the IDF pathname (pathname) has been corrupted during a write to the IDF. Use the Repack IDF command's Inspect IDF option (7.2.15) to determine which entities are corrupted, then use the Binary Repack option to gain back the space those entities occupied. Rebuild needed entities through a PED or through exception building (7.2.7).

7.2.3 Load

For all DEB-based activities not associated with the area database, the Load command loads the entity from the current PED into its destination module or gateway. The destination module or gateway is the one that contains the data owner software, which is the software that uses the entity to accomplish its functions. On the displays, modules and gateways are often referred to as "nodes."

For area database entities, the Load command loads the entity in the current PED into the area database work file, whose pathname is in this form:

```
NET>&Dnn>AREAnn
```

where nn is the area number, 01 through 10.

For data points, the DEB identifies the destination module or gateway through the Unit ID (and the hiway number and box number for HG points), which must be entered in the first PED. Network configuration must be completed earlier to define the module(s) or gateway(s) that contain data points for the unit. Reserved-entity destinations are determined by the entity name, and they also depend on network configuration being complete.

CAUTION

Entity Loading Precautions

The following are summaries of the precautions you should take before loading an entity. For more details on each of these precautions, see 9.2.1.

- All entities to which this entity is to be connected should already be loaded before you load this entity with a Load command. Use the Load Multiple command to load entities that have connections to each other.
- Before executing a Load or Load Multiple command to load new entities, all modules and gateways on the LCN (except USs) should be running with their on-process personalities. Otherwise, duplicate entities might be loaded and not be detected as they are loaded.
- Existing AM points cannot be reloaded unless the point in the AM is inactive. If an existing HG point is reloaded with its box active, data for the point may be deleted from operating displays. HG points should be loaded to the HG and box while the box in the Idle mode.
- The order in which entities are loaded is important. For entities in the same IDF that refer to each other, the Load Multiple command (a two pass operation) resolves the loading order. The Load command performs a single pass operation. To load an entity that references an entity in another IDF, the referenced entity should be loaded first to avoid errors. If you get such errors, reload the entities with the errors, and those errors should clear up.
- Alias points require special considerations before they are built and loaded. See 9.2.1.
- If you can, disable automatic checkpointing on the destination modules and gateways to which you intend to load points to avoid the possibility of generating any incorrect checkpoint data. This can happen if a point gets checkpointed in the midst of a number of parameter changes.
- If you are now, or will be using the Picture Editor or the Button Configurator, verify that you are not using one of the "semi-reserved" entity names for a point's Tag Name. The "semi-reserved" entities are listed in the *Picture Editor Reference Manual* in the Implementation/Engineering Operations - 2 Binder.

7.2.3.1 Execution of a Load Command for Non-area Database Entities

To execute a Load command for all types of entities **except** area database entities:

1. On the Command Display, select the **LOAD** pick. This Command Display appears.*
2. Press ENTER. If the data is stored in the destination module or gateway without errors, the Command Display is replaced by the PED that was on the screen before the Command Display was called up.*

DD MM YY HH:MM 01
PAGE 01 OF 01

WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE	
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW

LOAD BUILD ENTITY

REFERENCE PATH NAME

☐ with OVERWRITE

F1=PED F3= F5=OVERWRITE F7=RECON F9=WLK BACK F11=
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD

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If one error (or more) is detected, the data may or may not get stored in the module or gateway, and one of two things happens:

1. An Error File display automatically appears on the screen. See 10.4.1. This means that the data was transferred to the destination module or gateway but the data owner detected errors that are now recorded in the ERROR.EF file. You can select picks from this display to go to PED pages with errors and to correct the errors and retry. For more information on correcting errors, see Section 10.
2. The Load Command Display remains on the screen, and one of the following error messages appears in the prompter region (they may also appear after an unsuccessful F12=LOAD operation):

(entity) EXISTS - SELECT OVERWRITE OPTION IF DESIRED

You tried to load an entity (entity) that already has been loaded in the module or gateway. You can use the F5=OVERWRITE key to replace the data in the module or gateway with new data from the PED set.

NOTHING TO LOAD

There was no loadable data in the PED set.

PARAMETER ENTRY DATA NOT ENTERED

There is data in one or more PED picks and ports that have not been entered by pressing ENTER while a PED is on the screen.

* You can also hold CTL and press F12 (LOAD) to execute a Load command that loads the entity in the current PED without showing the Command Display.

ERR: NODE UNAVAILABLE

You attempted to load into a module or gateway that does not exist or is not operating. Check to see if the unit name you entered in the first PED display is correct. If so, go back to your network configuration records to determine which module or gateway is handling that unit, then determine why that node is not operating.

7.2.3.2 Execution of a Load Command for Area Database Entities

The Load command loads an area database entity into the work file for the area (.WA). After all newly built or changed entities are in the work file, the Install Area command (7.2.13) is used to install them in the area database data file (.DA).

To execute a Load command **for** an area database entity:

1. Execute a `SELECT AREA` command (7.2.12).*
2. Check that the Area Database pathname is correct. It must be in this form:

`NET>&Dnn>AREAnn`

where nn is the area number, 01 through 10.

NOTE: If you are using removable media, the `&Dnn` volume must be in the drive indicated by the pathname while this command is executed.

COMMAND DISPLAY DD MM YY HH:MM 01 PAGE 01 OF 01

WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE	
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW

REFERENCE PATH NAME

The pathname must take the form logical_device_id>&Dnn>AREAnn: where 'nn' denotes an area number 01 - 10.
Give path OR select area number and give logical device id for data base

current AREA DATA BASE pathname

AREA NUMBER

device ID for AREA DATA BASE FILE

F1=PED F3= F5=OVERWRITE F7=RECON F9=W/LK BACK F11=
F2= RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD

3. Select the `LOAD` pick, then press ENTER. If the data is stored in the work file without errors, the Command Display is replaced by the PED that was on the screen before the Command Display was called up.*

If one error (or more) is detected, the data is not stored in the area work file and one of the following two things happens:

1. An Error File display automatically appears on the screen (data is stored in the work file, except for items with errors). This means that the data may have been transferred to the area work file but errors were detected and recorded in the ERROR.EF file that is now on the screen. You can select picks from this display to go to the PED pages with errors and to correct the errors and retry. See Section 10 for additional information.

2. The Load Command Display remains on the screen (nothing is stored in the workfile), and one of the following error messages appear in the prompter region:

NOTHING TO LOAD

There was no loadable data in the PED.

PARAMETER ENTRY DATA NOT ENTERED

You have not pressed ENTER to request the DEB to process the data check for definition and syntax errors.

7.2.3.3 Loading an HG Point in a PIU or DHP

The flow chart on Figure 7-2 depicts a special situation for Load or Load Multiple commands executed through the DEB: loading one or more HG points that also load data in a PIU or a DHP. If the data points are configured to load both an HG and a box, and the box is a PIU or a DHP, the special checks shown on the flow chart are made by the HG, as the points are loaded.

In order for data for certain point types to be loaded in a PIU or DHP, the PIU or DHP must first be reset by the HG. The checks shown on the flow chart ensure that a PIU or DHP will not remain reset more than 1.5 minutes, because of the load operation. If for some reason, it is not possible to complete the load operation within 1.5 minutes, loading is discontinued, an Error File is generated, and the PIU or DHP is enabled, thus it is not left reset for more than 1.5 minutes. If the load completes normally in 1.5 minutes or less, the PIU or DHP is enabled.

With R682, an option is added to suppress the system alarm for the box state changes or change of address related parameters. This option is not applicable to the Hiway box state changes caused by manual RESET and box failure. This option is applicable only to the box state changes caused by the following:

- DHP/PIU point load
- Multiple loads from IDF or EB file or EB file generated by TPS builder or DEB EC
- Box Configuration Change

You can select this option in the NCF, using System Wide Values ->HG OPTIONS ->SYSTEM ALARM ON DHP/PIU POINT LOAD. The available options are INHIBIT and ENABLE. The default option is ENABLE. (See *Process Operations Manual* for additional information.)

* If a Select Area command (7.2.12) has been executed for this area, you can hold CTL and press F12 (LOAD) to execute a Load command to load the entity in the current PED in its area working file (.WA), without showing the Load Command Display.

7.2.3.4 HG Points Not Reloaded if the Point is in Service

The flow chart on Figure 7-3 depicts another special situation for HG points. If the point to be reloaded is a box, some additional checks, as shown on the flow chart, are made.

If an existing point is being reloaded, and the it's not in an idle MC or another type of box and it is in MANual mode (in service), this error message appears in an Error File display:

```
ERR: POINT IN SERVICE, COULD NOT RELOAD.
```

7.2.4 Load Multiple

NOTE

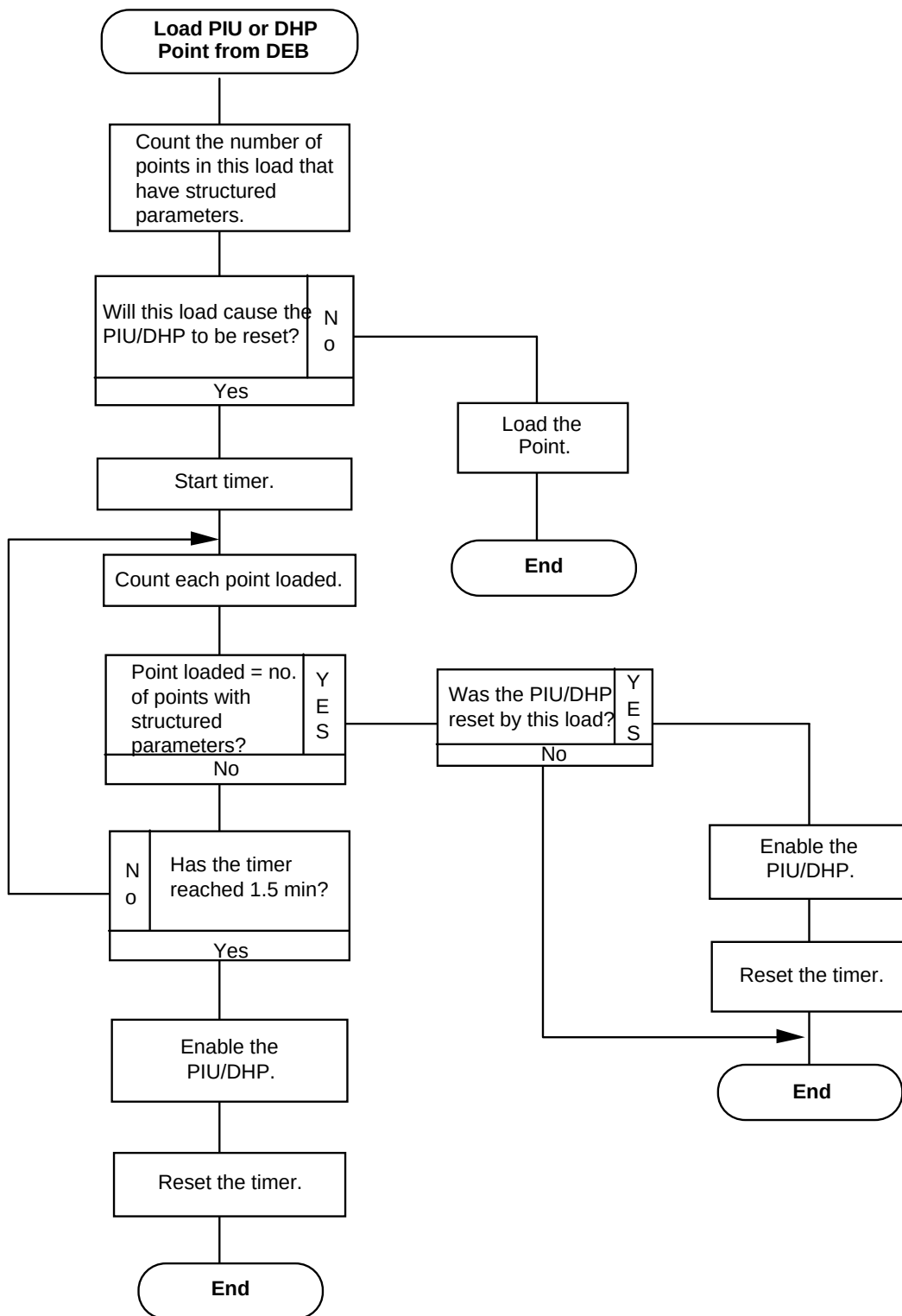
We recommend that you use IDFs only for working copies of entities while you are building them. Exception-build (.EB) files are preferred as master copies of the entities you have built because they are readable text files and they facilitate installation of Honeywell-supplied software upgrades. .EB files can be used on other TDC 3000^X systems, IDFs cannot. See 7.2.7.2 for information on creating .EB files.

For all but area database entities, Load Multiple loads several entities in an IDF into their destination module or gateway. A destination module or gateway is the one that contains the data owner software, which is the software that uses the data in the entity to accomplish its functions. On the displays, modules and gateways are often referred to as "nodes."

For area database entities, Load Multiple loads several entities in an IDF into the area database work file, whose pathname is in the form in this example:

```
NET>&D01>AREA01
```

The checks under 7.2.3.3 and 7.2.3.4 apply to points being loaded by either a Load command or a Load Multiple command.



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Figure 7-2 — Loading an HG Point in a PIU or DHP

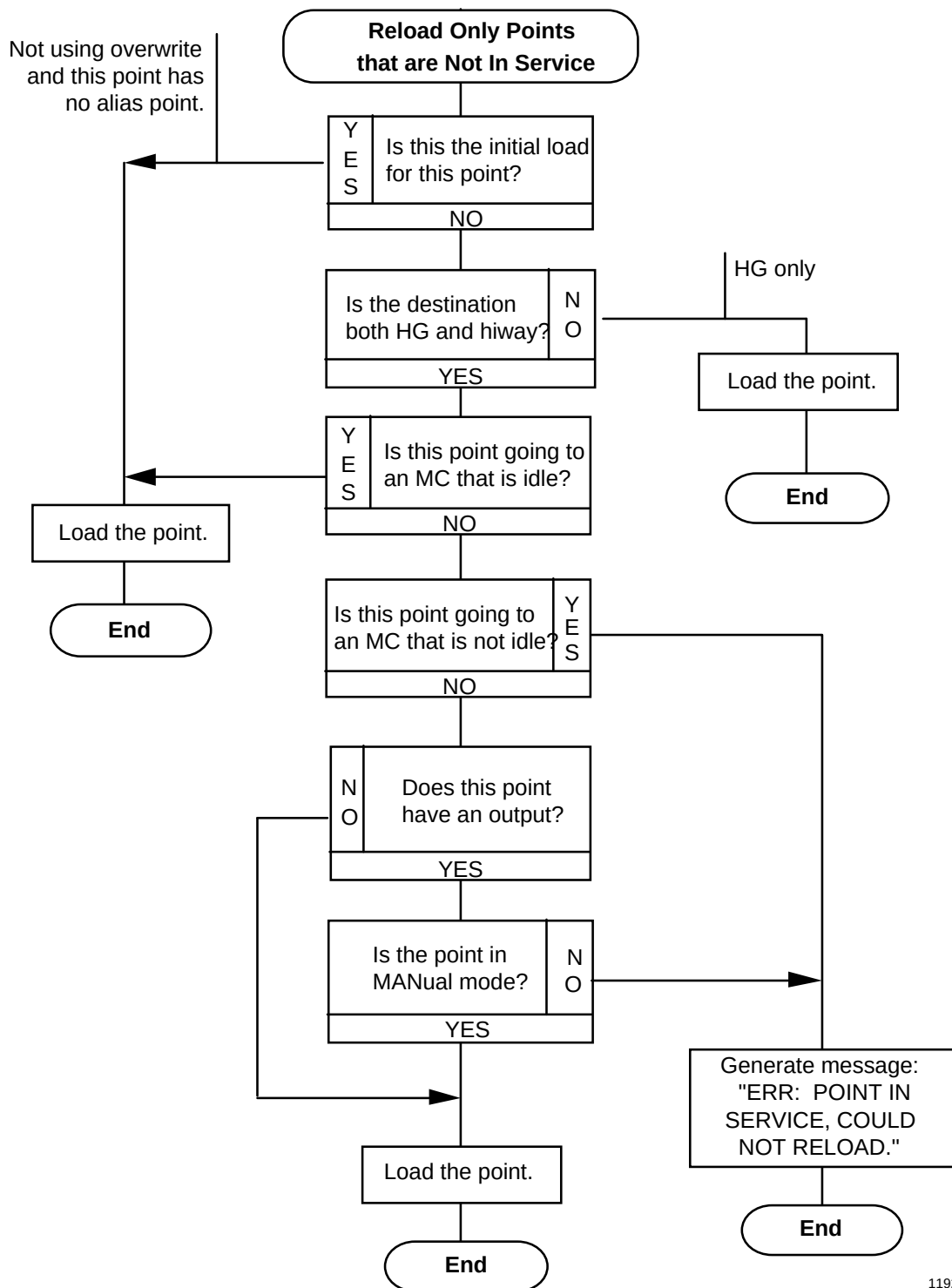


Figure 7-3 — HG Points not Reloaded if the Point is in Service

NOTE

Before executing a Load Multiple command, all modules and gateways on the LCN except USs should be running with their on-process personalities. If any is not operating (status -- other than OK), the uniqueness of the entities cannot be checked and difficult problems can arise when the inoperative module(s) is restarted.

The order in which entities are loaded is important. If data points that refer to each other are in the same IDF and that IDF is used in executing a Load Multiple command, the entities can usually be loaded regardless of their order in the IDF. Certain entities must be loaded in a proper order. You can control the order in which such entities are loaded by listing them in that order in a selection-list file. For information on entities that must be loaded in a prescribed order, see 9.5.

7.2.4.1 Execution of Load Multiple for Non-Area Database Entities

Load Multiple loads data points contained in an IDF into their destination modules or gateways. The destination modules and gateways are those that contain the data owner software, which is the software that uses the data point to accomplish its functions. On the displays, modules and gateways are often referred to as "nodes."

To execute a Load Multiple command for all entities except area database entities:

1. If you want to load all the entities in the IDF, go to Step 2. If you want to load only some of the entities in the IDF, create a selection-list file (7.1.8.1) and then proceed with Step 2.
2. Select the **LOAD MULTIPLE** pick on the Command Display. This Command Display appears.
3. If needed, key in the Reference Path Name (7.1.5). Key in the pathname for the IDF. Key in the pathname for the Selection-list **if** you're using one. Be sure to key in the suffix.

COMMAND DISPLAY						DD MM YY HH:MM 01
						PAGE 01 OF 01
WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA	
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA	
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE		
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW	
REFERENCE PATH NAME with OVERWRITE ☐ pathname for IDF pathname for SELECTION LIST						
F1=PED F3= F5=OVERWRITE F7=RECON F9=WLK BACK F11= F2= RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD						

NOTE: If you are using removable media, the user volume has to be in the drive indicated by the IDF pathname while this command is executed.

4. Press ENTER. If the data is stored in the destination modules or gateways without errors, the Command Display is replaced by the display that was on the screen before the Command Display was called up.

Entities that were successfully loaded without errors are listed in a (filename).SL file, where (filename) is the name of the IDF. Entities that were not successfully built or had errors are listed in a (filename) .UL file. You can use the Display File command to look at the .SL and .UL files

If one (or more) errors are detected, the data may or may not have been loaded in the module or gateway, and one of two things happens:

1. The Load Multiple Command Display remains on the screen and the following error message appears in the prompter region:

```
FILE NOT FOUND (pathname)
```

One of the files you named in the IDF or Selection-list (pathname) cannot be found on the named volume.

2. An Error File display automatically appears on the screen that has either entity error data, general error data, or both. Data with errors is not loaded. You can use picks in the entity error region to call up PED pages that have picks or ports with errors (see Figure 7-2 and Section 8). Under "General Error Data," one of these messages may appear:

```
(entity) EXISTS - SELECT OVERWRITE OPTION IF DESIRED
```

You tried to reload an entity (entity) that already is loaded in its module or gateway. You can use the F5=OVERWRITE key (7.1.6) to replace the data in the module or gateway with new data from the IDF.

```
INCOMPLETE INFORMATION - CANNOT LOAD DATA FOR (entity)
```

You did not enter enough information to identify the destination for the entity.(entity). You may have omitted the unit ID or, for an HG point, the hiway or box number.

```
NAMED ENTITY (entity) DOESN'T EXIST IN FILE (pathname)
```

An entity (entity) was named in the Selection-list (pathname) that was not found in the IDF.

```
ERR: NODE UNAVAILABLE
```

You attempted to load an entity into a module or gateway that doesn't exist or is not operating. Check to see if the unit ID is correct and, if the entity is an HG point, whether the correct hiway and box number were entered. You may need to go to your network configuration records to determine which unit IDs were assigned to which nodes, and which hiway number is assigned to each HG.

7.2.4.2 Execution of Load Multiple for Area Database Entities

Load Multiple loads area database entities into the work file for the area (.WA file). When all newly built or changed entities are in the work file, the Install Area command (7.2.13) is used to load them into the area database data file (.DA file).

To execute a Load Multiple command for area database entities:

1. If you want to load all the entities in the IDF, go to Step 2.
2. If you want to load only some of the entities in the IDF, create a selection-list file (7.1.8.1) and then proceed with Step 2.
2. Select the Load Multiple pick on the Command Display. This Command Display appears.
3. If needed, key in the Reference Path Name (7.1.5). Key in the pathname for the IDF. Key in the pathname for the Selection List **if** you're using one. Be sure to key in the suffix.

DD MM YY HH:MM 01
PAGE 01 OF 01

WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE	
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW

REFERENCE PATH NAME

pathname for IDF

pathname for SELECTION LIST

F1=PED F2=RECALL DISP F3=F4= F5=OVERWRITE F6= F7=RECON F8=PED STATUS F9=WLK BACK F10=WRITE F11=F12=LOAD

5587

NOTE: If you are using removable media, the user volume has to be in the drive indicated by the IDF pathname when this command is executed.

4. If it's not already filled in, key in the pathname for the Area Database File. It is in this form:

```
NET>&Dnn>AREAnn
```

where nn is the area number, 01 through 10.

NOTE: If you are using removable media, the area database volume &Dnn must be in the drive indicated by the Area Database File pathname when this command is executed.

5. Press ENTER. Each entity is displayed in yellow in the prompter region as it is processed, and each entity is processed twice. If all of the entities are stored in the work file without errors, the Command Display is replaced by the display that was on the screen before the Command Display was called up.

If one (or more) error is detected, the data may or may not have been stored in the area database work file, and one of two things happens:

1. The Load Multiple command display remains on the screen and one of the following error messages appears in the prompter region:

FILE NOT FOUND (pathname)

One of the IDF, Selection-list, or Area Database Files you named (pathname) could not be found on the named volume.

AREA DATA BASE FILE MUST BE SPECIFIED

The pathname in the port for the Area Database File is not valid. Key in the correct pathname or execute a Select Area command (7.2.12) and return to the Load Multiple command display.

2. An Error File Display (10.4.1) automatically appears on the screen that has either entity error data, general error data, or both. You can use picks in the entity error region to call up PED pages that have picks or ports with errors (see Figure 7-2). Under General Error Data, one of these messages may appear:

INCOMPLETE INFORMATION - CANNOT LOAD DATA FOR (entity)

You did not enter complete information in the Command Display before pressing ENTER. For the entity (entity), either the Reference Path Name is incomplete or wrong or the IDF pathname is missing or has invalid characters in it.

WARNING - USING UNINSTALLED WORKFILE

One of the entities in this load is going to the workfile (.WA) the first time. If the workfile is uninstalled, it may not match the area database file (.DA), so reconstituting from the .WA file after this load will not create IDFs or PEDs that represent the area database in the .DA file.

NAMED ENTITY (entity) DOESN'T EXIST IN FILE (pathname)

The entity named in (entity) is in your selection-list but isn't in the specified (pathname) IDF. See Step 4, above.

7.2.5 Reconstitute

For all entities except area database entities, Reconstitute reads the specified entity from the module or gateway it resides in (was loaded in), converts the data to PED form, and displays it as the current PED set. You can then make changes in the picks and ports in the PED set, and use the Load command with the Overwrite option to reload the entity. It is also possible to use the Write to IDF command to store the reconstituted entity in an IDF.

When data points are reconstituted, the resultant PED displays have the "live" parameter values. These are not necessarily the values stored in an IDF, because they may have been changed by normal processing of the entities. If you want the values from an IDF, use a Read to PED command (7.2.2).

Reconstitute commands can be executed by calling up the Reconstitute Command Display or through generic function key F7=RECON. The F7=RECON key is intended for use with reserved entities. It lets you execute a Reconstitute command without remembering the reserved-entity name. Procedures for the two methods are under 7.2.5.1 and 7.2.5.2.

For area database entities, the Reconstitute command retrieves the specified entity from the area database work file (.WA), converts it to PED form, and displays the resultant PED set. Appendix C contains the names of the area database entities. If an area database entity is for the first time reconstituted from a .WA file that has not yet been installed, the message below appears in the prompter region.

WARNING - USING UNINSTALLED WORKFILE

If the work file is uninstalled, it may not match the area database file (.DA) in the USs. Reconstituting from the .WA file may not create a PED that represents the area database that is in the .DA file.

7.2.5.1 Reconstitute an Entity through the Command Display

To execute a Reconstitute command through the Command Display:

1. On the Command Display, select the **RECONSTITUTE** pick. This Command Display appears.
2. If this is an area database entity and a Select Area command (7.2.12) has not been executed, key in the pathname for the area database file in this form:

`NET>&Dnn>AREAnn`

where nn is the area number, ranging from 01 through 10.

COMMAND DISPLAY DD MM YY HH:MM 01
PAGE 01 OF 01

WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE	
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW

REFERENCE PATH NAME

ENTITY name

F1=PED F3= F5=OVERWRITE F7=RECON F9=WLK BACK F11=
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD

3. Key in the **ENTITY** name.

If you are using removable media, the area database volume &Dnn must be in the drive specified by the pathname when the command is executed. The GDF must also be mounted (3.5).

4. Press ENTER. When the data is retrieved, the Command Display is replaced by the first page in the PED.

In the new PED set, any references to parameters in the same entity are converted to the self-reference form. For example, if this is data point FC001, and the port contained a reference to FC001.PV, the reconstituted reference appears as *.PV.

If errors are detected, one of the following two things happens:

1. An Error File display appears on the screen (10.4.1).
2. The Command Display remains on the screen and the following error message appears in the prompter region:

`ENTITY NAME DOESN'T EXIST (entity)`

The entity you named in Step 2 (entity) could not be found in any of the modules and gateways on the LCN. Check the name you entered and make sure that is actually loaded in its destination module or gateway.

7.2.5.2 Reconstitute Using F7=RECON

This method can be used for area database entities only if a Select Area command (7.2.12) has been executed to set up the area database pathname.

To execute a Reconstitute command with generic function key F7:

1. Call up the appropriate entity building activity from the Engineering Main Menu.

2. Call up the first PED page for the type of entity you need to reconstitute. For example, this display was called up by selecting AREA DATA BASE, which brought up the Area Configuration Menu, and from that menu we selected GROUP

3. Key enough data into the first few ports to identify the entity that you need to reconstitute. In this example, the group number is sufficient to identify the \$OGROUP1 entity. For data points, you need a tag name and the unit ID.

NOTE: If this is an area database entity and you are using removable media, the &Dnn volume must be in the drive indicated by the pathname when this command is executed.

4. Press ENTER. The display reappears with blue data in the ports.

5. Hold CTL and Press F7. When the Reconstitute command is executed without errors, the PED reappears with the data read from the area workfile (.WA).

In the new PED set, any references to parameters in the same entity are converted to the self-reference form. For example, if this is data point FC001, and the port contained a reference to FC001.PV, the reconstituted reference appears as *.PV.

If errors are detected, one of the following two things happens:

1. An Error File display appears on the screen (10.4.1).
2. One of the following error messages appears in the prompter region:

MISSING DATA

You did not key data into enough ports on the first PED page to identify an entity or you missed a port needed to identify the entity. Check the first PED page and try again. This message also appears if you key in this data but forget to press ENTER.

ENTITY NAME DOESN'T EXIST (entity)

The entity you identified in Step 3 (entity) could not be found in any of the modules and gateways on the LCN. You should recheck the first PED page, make changes and try again. You may also need to check your records to make sure that you are requesting an entity that has been built and loaded.

7.2.6 Reconstitute Multiple

For all entities except area database entities, Reconstitute Multiple reads one or more entities that you name in a selection-list, from the modules or gateways they reside in (were loaded in), converts them to PED form, and places them in an IDF. You can then use the Read to PED command to bring each entity into the current PED set to make changes in the picks and ports, and use the Load or Load Multiple command with the Overwrite option to reload the entities in their destination module or gateway.

It is also possible to use the Write to IDF command to store a reconstituted entity in an IDF.

For area database entities, Reconstitute Multiple reads entities from an area database data file (.DA), converts them to PED form, and stores them in an IDF.

To execute a Reconstitute Multiple command:

1. To define the names of the entities that you want to reconstitute, either create a selection-list file (7.1.8.1) to define the names or use an existing IDF. If you use an IDF, the names of **all** of the entities in the IDF will be reconstituted.

2. On the Command Display, select the RECONSTITUTE MULTIPLE pick. This Command Display appears.

COMMAND DISPLAY						DD MM YY HH:MM 01	PAGE 01 OF 01
WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA		
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA		
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE			
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW		

REFERENCE PATH NAME

☐ with OVERWRITE

pathname for IDF

pathname for SELECTION LIST

F1=PED F2=RECALL DISP F3= F4= F5=OVERWRITE F6= F7=RECON F8=PED STATUS F9=WLK BACK F10=WRITE F11= F12=LOAD

3. If needed, key in the Reference Path Name (7.1.5). If you are reconstituting area database entities, also key in the area database file name in this form;

NET>&Dnn>AREAnn

where nn is the area number, ranging from 01 through 10.

If you are using removable media, the &Dnn volume must be in the drive indicated by this pathname while this command is executed.

4. Key in the pathname for the IDF you want the data to be stored in.

NOTE: If you are using removable media, the user volume needs to be in the drive indicated by the IDF pathname while this command is executed. The GDF must also be mounted (3.4).

5. Type in the pathname for the Selection-list. Be sure to type in the suffix.
6. Press ENTER. The transfer of data begins and a WAIT message appears in the prompter region. When the transfer is completed without errors, the generic function key labels reappear in the prompter region, and the reconstituted entities are in the IDF, available for use with any of the commands that use IDFs or affect the content of IDFs.

If an error is detected, one of the following two things happens:

1. The Reconstitute Multiple Command Display remains on the screen and the following error message appears in the prompter region:

```
FILE NOT FOUND (pathname)
```

One of the IDF or selection-list files you named (pathname for IDF or pathname for Selection-list) cannot be found on the named volume.

2. An Error File display automatically appears on the screen that has either entity error data, general error data, or both (10.4.1). Data with errors is not loaded. You can use picks in the entity error region to call up PED pages that have picks or ports with errors.

Under General Error Data, one of these messages may appear:

```
NAMED ENTITY (entity) DOESN'T EXIST
```

One of the entities in the selection-list you defined (pathname for Selection-list) could not be found in any of the modules and gateways on the LCN. Check the name you entered and make sure that it is actually loaded in its destination module or gateway.

```
(entity) EXISTS - SELECT OVERWRITE OPTION IF DESIRED
```

One of the entities in the selection-list you defined (pathname for Selection-list) was found to be already present in the IDF you specified (pathname for IDF). You can use the F5=OVERWRITE key to replace the data in the existing entity with the data from the module or gateway.

7.2.7 Exception Build

The Exception Build command is used to create (and load) similar entities. We recommend that you use Exception Build whenever you need to build more than a few points of a type. Your source file can specify that the entities to be built be of a given type such as AM regulatory, CG ACDIP, Process Manager, analog input, etc; or it can name a template point in an IDF to be used to create the new point(s). Exception-build (.EB) files are preferred over IDFs for master copies of points because they facilitate software upgrades and can be easily transported to other systems.

7.2.7.1 .EB Source File Form and Rules

An example of a .EB source file is in Table 7-2. The source file does not have many syntax or format rules. It consists of a mix of five types of control items (&T, &X, &M, &N, and &E) plus parameter changes. It is easy to create a correct source file with the Print Entities command (see 7.2.7.2 and 7.2.11).

You can give the .EB file any name, but the name must include a ".EB" suffix. It's easiest if the user volume that contains the .EB file is the same on that has the IDF(s) you are using.

Extra spaces in the file are ignored and you can continue input that is too long to fit on a line, on the next line. You cannot break names or values at the end of a line, so if you see that this is going to happen, go on to the next line.

You can include comments in the file to explain its content. Comments are enclosed in brackets like this

```
{This is an exception-build source file comment.}
```

Each source file must begin with an &T, &X, or &M control item (control items are defined next). There can be more than one new entity specified in each source file and the portion for each entity must begin with "&N" followed by the new entity's tagname. Once an entity type (&T) or template point (&X or &M) is named, that item remains in effect until a new &T, &X, or &M item specifies something else. Each source file must end with an &E control item.

The five control items, their purpose, and the information associated with them are as follows (see Table 7-2 for examples of their use):

&T (point type) Specifies the type of point to be built. For example, &T REGAM, specifies that an AM Regulatory data point is to be built. The accepted names for all entity types are listed in Appendix B.

If you do not specify any parameter value changes in your source file, an entity will be built using the default values that appear in a new PED for the point type. However, it will not be a loadable entity because the default values do not provide enough information to allow the entity to be loaded. If you use &T, the effect of any previously used &X or &M items is cancelled.

The easiest way to create an &T source file is to select an existing entity of the type you want built and with values similar to those the new entity will have. Use the Print Entities command (see 7.2.7.2 and 7.2.11) to create the source file and then use the Text Editor to change any values that are different.

&X (template tagname) Specifies the tagname of a point in an IDF to be used as a template point for building new points. &X indicates to the DEB that the IDF that contains the template point is located where the OPTIONAL pathname for TEMPLATE port on the Exception Build Command Display points. If you use &X, you must enter a pathname in this port. Using &X cancels the effect of any previous &T or &M items.

&M (template tagname) Specifies the tagname of a point in an IDF to be used as a template point for building new points. &M indicates to the DEB that the IDF that contains the template point is the same IDF identified in the PATHNAME for IDF port. Using &M cancels the effect of any previous &T or &X items.

&N (new entity tagname) Specifies the tagname for the new point that is to be built. There must be an &N control item for every entity in a .EB source file. The default (unchanged) parameter values in the new entity are determined by the last &T, &X, or &M item.

&E This item indicates the end of the .EB source file. It is required at the end of every source file.

Parameter value changes are specified in this form:

```
SPEULO=003
```

Where the parameter name is on the left of = and the new value is on the right of the =. Each such specification is separated by at least one space or line return. The changes can be listed in any order. Parameters not listed are not changed—they remain the same as in the original (Print Entities or template) entity.

All user-defined character strings, such as for PTEDSC, KEYWORD, EUDESC, and all user-defined state descriptors, must be enclosed in quote marks, for example:

```
PTDESC="FUEL FLOW 01"
```

You can not embed quotes (") in the middle of a string and then exception build and load. Quotes can be entered using the DEB entry ports and the point will reconstitute and load.

In previous releases an "&C" control item was required just before the parameter changes. This is no longer required and if "&Cs" are still in your source files, they (but not the parameter changes) are ignored.

Table 7-2 — Source File Examples

Names of New Points	Source File
FL101	<pre>&T REGAM {Build an AM regulatory point} &N FL101 {The tag name is FL101} UNIT = 01 PTDESC = "TEMPERATURE" EUDESC = "KBTU/MCF" KEYWORD = "TEMPE" (etc.) &E {This is the end of the source file}</pre>
PR102	<pre>&X PR101 {Use PR101 as a template point} &N PR102 {The tag name is PR102} UNIT = 01 PTDESC = "PRESSURE" EUDESC = "PSI" KEYWORD = "PRESB" (etc.) &E {This is the end of the source file}</pre>
TNK1B	<pre>&M TNK1A {Use TNK1A as a template point} &N TNK1B {The tag name is TNK1B} UNIT = 01 PTDESC = "TANK 1B" EUDESC = "GAL" PVEULO = 0.0 PVLOTP = 10 PVALGID = DATACQ (etc.) &E {This is the end of the source file}</pre>

Explanation of Table 7-2

Three source file examples are in Table 7-2. While each of the source files specifies the creation of one new entity, all three entities, or more, could be specified in one source file, in which case, there would be only one &E control item and it would be the last item in the file.

- **FL101**

The &T item specifies that an AM regulatory point is to be built (see Appendix B for other entity-type names). The &N item names the new point "FL101." All parameter values for the point type specified are listed below the &N item. Any that are to differ from those in the point used by Print Entities must be changed through the Text Editor.

This file appears in the source-file form created by the Print Entities command, except for the comments, which would have been entered through the Text Editor.

- **PR102**

The &X item specifies point PR101 as the template point. In this case, the DEB will look for the IDF that contains PR101 through the pathname in the OPTIONAL pathname for TEMPLATE port on the Exception Build Command Display.

This file appears in the source-file form created by the Print Entities command, except for the comments, which would have been entered through the Text Editor, and except for the &X control item. The Print Entities command always uses an &T item at the beginning of a source file, to specify type of entity to be built.

- **TNK1B**

The &M item specifies point TNK1A as the template point. In this case, the DEB will look for the IDF that contains TNK1A in the same IDF that you specify to receive the newly built points. The pathname for IDF port points to this IDF.

This file appears in a free form, as might have been created by a control engineer through the Text Editor. Note that while the Print Entities command lists all parameters in the source files it creates, only those whose values are to be different than in the source point need to be listed.

While use of the &X and &M control items result in smaller .EB files, use of the &T control item is preferred because the .EB file entities that use &T facilitate software upgrades and are easily transportable to other systems.

NOTE

Prior to R620 release, an engineer could edit, view and save an .EB file with a maximum size of 32K records (each record corresponds to one line in the Text Editor). With R620 an engineer can edit, view and save an .EB file with a maximum size of up to 64K records. The engineer can also back-build the .EB file of the same size (increased size).

7.2.7.2 How to Create Exception-Build Source Files

The first step in exception building is to establish the source file that defines the new entities you need to build and the parameter value changes for those entities. You need one source file each time you execute the Exception Build command. This source file can define any number of new entities, up to the storage capacity of the IDF that will store them.

Using the Print Entities Command or the Text Editor to Create Source Files

In either case, you need an entity that is the same type of entity that you require and is already built. It must have the same structure and same set of parameters as the new entity. If you use the Print Entities command to create the source file, you will not use a template point. If you create the source file from scratch using the Text Editor, you should use a template point, rather than the &T control item, to insure that you have a set of parameters that defines a loadable point with a correct structure.

CAUTION

EB source file entries must be precise. If you make a mistake, such as typing in too many characters (exceeding the maximum defined for a parameter), a truncation character is inserted in the beginning of the field and the EXCEPTION BUILD command is not executed successfully. The IDF will be built containing the bad parameters and an error message will appear. The entity will not be loaded, even if the LOAD ENTITIES pick was selected.

To use Print Entities to create a source file:

1. Execute a Print Entities command (refer to 7.2.11) with the pathname for your source file in the PRINT device ID or DESTINATION pathname port (it must have a .EB suffix). You can use any of the print options to select your source; a screen entity (PED), an IDF, or a system entity (a point in a node). If successful, this creates a copy of all the parameter values for the point in the source file with the &T control item at the beginning of the file, followed by the name for the point type (see Appendix B).
2. Press ESC to go to the Command Processor display and then go to the Text Editor. To bring up the Text Editor with your new source file, key in a command line, as shown in the following example:

```
ED SOURCE.EB
```

Note: If you have not yet used the SP command to set up a pathname to your user volume, do so (see *Command Processor Operation* manual), or key in the full form of the pathname, for example

```
ED NET>USRV>SOURCE.EB
```

3. Press ENTER. The Text Editor display appears with the first several lines of your source file.

4. Change the tag name that follows the &N item (just below the &T) to your new tag name (see *Text Editor Operation* manual).
5. Change any parameter values that need to be changed.
6. If you need to build additional points of this same type and structure, Mark and Copy everything from the &T item to the last parameter value, then Paste at the end of the file to make another entity in the source file. Then edit this copy to rename the &N item and to change parameter values. To help in reading the file, insert one or more returns between the entities (copies).
7. Do a Quit, then an Exit to store the edited source file and return to the Command Processor display.
8. Press CTL and HELP |MENU simultaneously. The Command Display is restored.

You can now use your source file to execute an Exception Build command (7.2.7.3).

To use the Text Editor to create a source file:

1. On the Universal Personality's Engineering Main Menu select COMMAND PROCESSOR, or from any DEB display, press ESC to go to the Command Processor display and then to the Text Editor. To bring up the Text Editor with your new source file, key in a command line as in this example:

```
ED SOURCE.EB
```

Note: If you have not yet used the SP command to set up a pathname to your user volume, do so (see *Command Processor Operation* manual), or key in the full form of the pathname, for example

```
ED NET>USRV>SOURCE.EB
```

2. Press ENTER. The Text Editor display appears with the first several lines of your source file.

NOTE

With R620 an engineer can edit, view and save an .EB file with maximum size of up to 64K records. If the file size exceeds 64K records, the error "Record Number exceeded 65535 limit" is displayed.

- Key in the required control items for the first new entity as in the following example. If the destination IDF is not the IDF that contains the template point, use &X instead of &M. (For operating instructions, refer to *Text Editor Operation* manual).

```
&M FC101 {FC101 is the template point for this entity}
&N FC102 {This entity's tag name is FC102}
```

- Key in parameter value statements for all new entity parameters that are to be different from those of the template entity.
- If you need to build additional points of this same type and structure, Mark and Copy everything from the &M (or &X) item to the last parameter value, then Paste at the end of the file to make another entity in the source file. Then edit this copy to rename the &N item and to change parameter values. To help in reading the file, insert one or more returns between the entities (copies).
- Do a Quit, then an Exit to store the edited source file and return to the Command Processor display.
- Press CTL and HELP |MENU simultaneously. The Engineering Main Menu or the DEB display that was present at Step 1 is restored.

You can now use your source file to execute an Exception Build command (7.2.7.3).

7.2.7.3 Executing an Exception Build Command to Load Entities

To execute an Exception Build command to load entities:

- Create a .EB source file according to 7.2.7.1. This file defines the template entities, each new entity's name, and parameter value changes.
- If you want to build only a subset of the entities in your source file, use the Text Editor to create or to edit a selection-list (.EL) file (see 7.1.8.1).
- On the Command Display, select the **EXCEPTION BUILD** pick. This Command Display appears.
- If needed, key in the Reference Path Name (7.1.5).
- Key in the pathname of the source file created in Step 1.
- Key in the pathname of the IDF into which entities that did not load due to errors will be stored. This can be a new IDF or same as the template IDF.
- If you created a selection-list in Step 2, key in the Selection-list pathname.

COMMAND DISPLAY DD MM YY HH:MM 01 PAGE 01 OF 01

WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE	
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW

REFERENCE PATH NAME

If load pick is selected, non-loadable entities will be written to IDF.

pathname for SOURCE file

pathname for IDF

pathname for SELECTION LIST

OPTIONAL pathname for TEMPLATE

F1=PED F3= F5=OVERWRITE F7=RECON F9=WLK BACK F11= F12=LOAD
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE

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8. If you referenced any template entities in the .EB source file by using &X control items, key in the Optional pathname of the file that contains these entities.
9. Select the LOAD ENTITIES pick.
10. Press ENTER. As the command is being executed (new entities are being built and loaded), a yellow WAIT appears in the prompter region, then Processing Entity (name) appears as each entity is processed.
11. If the exception building is completed without errors, the generic key prompters reappear. New entities that were built successfully and loaded into the appropriate module or gateway are listed in a (filename).SL file, where (filename) is the name of the IDF.
12. If errors are detected, the Error File (filename).EF is displayed where (filename) is the name of the IDF. The errors are recorded in this file. Entities that were not built or loaded successfully are written to the IDF and listed in a (filename).UL file, where (filename) is the name of the IDF. Refer to Subsection 10.4 for help on how to correct the errors.

NOTE

When the Exception Build uses a file which was updated in R620, the time taken to back build the points may increase if it contains procmod, array points or AM points with CDS parameters of type Time. Refer 7.2.11.1 *Box/Module Parameters in EB* for details.

7.2.7.4 Executing an Exception Build Command to Build IDFs

Since you can use .EB files directly to load entities, make sure that you really need to build IDFs before you execute this command.

NOTE

With R620 an engineer may find an increased IDF file size if it contains the procmod and array points, in cases where the pre R620 IDFs are upgraded or R620 IDFs are updated.

To execute an Exception Build command to build IDFs:

1. Create a .EB source file according to 7.2.7.1. This file defines the template entities, each new entity's name, and parameter value changes.
2. If you want to build only a subset of the entities in your source file, use the Text Editor to create or to edit a selection-list (.EL) file (7.1.8.1).

3. On the Command Display, select the **EXCEPTION BUILD** pick. This Command Display appears.
4. If needed, type in the Reference Path Name (7.1.5).
5. Key in the pathname of the source file created in Step 1.
6. Key in the pathname of the IDF where you want the entities to be stored. This can be either a new IDF or the IDF where the templates are stored.

COMMAND DISPLAY						DD MM YY HH:MM 01 PAGE 01 OF 01
WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA	
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA	
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE		
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW	

REFERENCE PATH NAME:

If load pick is selected, non-loadable entities will be written to IDF.

with OVERWRITE
Load Entities

pathname for SOURCE file:

pathname for IDF:

pathname for SELECTION LIST:

OPTIONAL pathname for TEMPLATE:

F1=PED F3= F5=OVERWRITE F7=RECON F9=WALK BACK F11=
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD

7. If you created a selection-list in Step 2, key in the Selection-list pathname.
8. If you referenced any template entities in the .EB source file by using &X control items, key in the Optional pathname of the file that contains these entities.
9. Press ENTER. As the command is being executed (new entities are being built and stored in the IDF), a yellow **WAIT** appears in the prompter region, then **Processing Entity (name)** appears as each entity is processed.
10. If the exception building is completed without errors, the generic key prompters reappear. New entities that were built successfully and stored in the IDF are listed in a (filename).SL file, where (filename) is the name of the IDF.
11. If errors are detected, the Error File (filename).EF is displayed where (filename) is the name of the IDF. The errors are recorded in this file. Entities that were found to have syntax or data-type errors are not written to the IDF, while those with any other kinds of errors or with missing data will be written to the IDF. Entities for which **any** type of error has been detected are listed in a (filename).UL file where (filename) is the name of the IDF. Refer to Subsection 10.4 for help on how to correct the errors.

If while processing a source file any of the following occurs:

- the DEB is unable to access selection-list data,
- the DEB is unable to read a template entity,
- the DEB is unable to update a .SL or a .UL file,
- there is not enough US memory to complete the operation,

The message **ABORT ON ERROR** appears in the prompter region.

7.2.8 Delete

CAUTION

Do not attempt to alter a point by deleting it and then reloading it. This can cause very serious problems with references to or from other entities. Use a Reconstitute command to change existing points.

Reserved entities cannot be deleted from modules or gateways because all reserved entities exist all the time, in all systems. Loading a reserved entity does not cause it to exist, but it does change values in the entity.

If you delete a data point that is referred to by another data point and that point is online and on-control, the point that remains will encounter bad data the next time it is processed.

AM and CM60 points must be inactive before they can be deleted. HG points must be in manual mode before they can be deleted.

If the system uses PRIMMOD (Primary Module) points, run the Find Names Function or Documentation Tool to determine if the point you want to delete is being used as a PRIMMOD name. If so, then all system references to that point as a PRIMMOD should first be deleted or changed to a different point.

If the point being deleted is the primary of a contact cutout alarm scheme, run the Find Names Function or Documentation Tool to locate all secondary contact cutout points that refer to this primary. All secondary point references to the primary contact cutout point must first be deleted, or changed to a different point, before the primary contact cutout point is deleted. Otherwise, a “point in service” error will occur when there is an attempt to delete the point.

There are four types of Delete commands:

- **Delete System Entity**—Deletes a data point that has been loaded in a module or gateway. Normally, there will be only one point with each name, in all of the modules and gateways on the LCN, and only the `ENTITY` name port needs to be filled in. If, however, there are two data points with the same name in two different modules or gateways, the `MODULE` number and `UNIT` ports need to be filled in to identify which of the duplicated points to delete.*
- **Delete Mult Entities**—Deletes multiple data points that have been loaded in modules or gateways. The entity names are provided in a selection-list (.SL) file.
- **Delete Entity from IDF**—Deletes an entity from an IDF.
- **Delete Mult Entities from IDF**—Deletes multiple entities from an IDF. The entity names are provided in a selection-list (.SL) file.

The names of the entities that are deleted are recorded in the system-event journal.

To delete an entity:

1. If you are going to use one of the multiple-entity deletes, create a selection-list file according to 7.1.8.1.
2. Select the appropriate Delete command pick on the Command Display. The appropriate version of the Command Display appears. They are pictured below.
3. If needed, key in the Reference Path Name (7.1.5).
4. Key appropriate information in the remaining ports on the Command Display. Notes on their content are provided beside the Command Display images below.

* Normally, it is not possible to load two points with the same Tag Name, but two points with the same name might exist because one of them was loaded while the second node was not operating.

- Press ENTER. While the multiple-entity deletions are taking place, yellow prompts appear. After the deletion is completed without errors, the generic-function key labels reappear in the prompter region.

This Command Display appears when you select **DELETE SYSTEM ENTITY** on the Command Display. Key the name of the entity you want to delete into the Entity Name port.

If a duplicate tag name is found, a **DUPLICATE ENTITY** message identifying the duplication appears in the prompter region (see 10.3). Using this information, key in the Module number and the Unit to identify which one of the entities is to be deleted.

COMMAND DISPLAY						DD MM YY HH:MM 01 PAGE 01 OF 01
WRITE TO IDF	READ TO PED	DELETE SYS ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA	
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA	
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE		
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW	

DELETE SYSTEM ENTITY

REFERENCE PATH NAME

ENTITY NAME

The following information is required ONLY for deletion of duplicate entities.

MODULE number

UNIT

F1=PED	F3=	F5=OVERWRITE	F7=RECON	F9=WLK BACK	F11=
F2=RECALL DISP	F4=	F6=	F8=PED STATUS	F10=WRITE	F12=LOAD

5593

This Command Display appears when you select the **DELETE MULT ENTITIES** on the Command Display. A selection list is needed, so key in the Selection-list pathname.

COMMAND DISPLAY						DD MM YY HH:MM 01 PAGE 01 OF 01
WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA	
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA	
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE		
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW	

DELETE SYSTEM ENTITIES USING SELECTION LIST

REFERENCE PATH NAME

pathname for SELECTION LIST

F1=PED	F3=	F5=OVERWRITE	F7=RECON	F9=WLK BACK	F11=
F2=RECALL DISP	F4=	F6=	F8=PED STATUS	F10=WRITE	F12=LOAD

5594

This Command Display appears when you select **DELETE ENTITY FROM IDF** on the Command Display. Key the name of the entity you want to delete into the Entity name port. Key in the pathname for the IDF which contains this entity.

COMMAND DISPLAY						DD MM YY HH:MM 01 PAGE 01 OF 01
WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA	
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA	
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE		
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW	

DELETE ENTITY FROM IDF

REFERENCE PATH NAME

ENTITY name

pathname for IDF

F1=PED	F3=	F5=OVERWRITE	F7=RECON	F9=WLK BACK	F11=
F2 = RECALL DISP	F4=	F6=	F8=PED STATUS	F10=WRITE	F12=LOAD

5595

This Command Display appears when you select **DELETE MULT ENTITIES IDF** on the Command Display. Key in the pathname for the IDF which contains the entities you want to delete. If you created a selection list, key the file name into the Selection-list pathname port.

COMMAND DISPLAY						DD MM YY HH:MM 01 PAGE 01 OF 01
WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA	
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA	
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE		
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW	

DELETE ENTITIES FROM IDF USING SELECTION LIST

REFERENCE PATH NAME

pathname for IDF

pathname for SELECTION LIST

F1=PED	F3=	F5=OVERWRITE	F7=RECON	F9=WLK BACK	F11=
F2 = RECALL DISP	F4=	F6=	F8=PED STATUS	F10=WRITE	F12=LOAD

5596

If an error is detected, one of the following two things happens:

1. An Error File display automatically appears (10.4.1).
2. One of the following error messages may appear in the prompter region:

ENTITY DOESN'T EXIST (entity)

The entity named in (entity) was not found in any module or gateway in the system. Check your records to make sure you named the correct entity name and try again.

DOESN'T EXIST IN (pathname) .DB

The entity was not found in the specified (pathname) IDF. Check the pathname to make sure you entered (or listed in the .SL file) the correct entity name(s), device, volume, and filename, and try again.

7.2.9 Display File

Display File displays the contents of a selected file on the screen to provide useful entity-building information. You can display the following types of files:

- .EF—Error files
- .EB—Exception-build source files
- .SL—Lists of entities successfully handled by a multiple-entity command
- .UL—Lists of entities not successfully handled by a multiple-entity command
- .EL—An edited list of entities, created with the Text Editor, for use as a selection-list file for multiple-entity commands
- .xx—A user file that can contain entity lists or text. "xx" is an example. The suffix begins with "X," "y," or "z" (upper case or lower case).

To execute a Display File command:

1. On the Command Display, select the **DISPLAY FILE** pick. This Command Display appears.
2. If needed, key in the Reference Path Name (7.1.5).
3. Key in the pathname for the File you want to display. Often, the file name is the same as the IDF file name used by the DEB command that generated the file. However, when you build a .EL file with the Text Editor, you define the file name.

COMMAND DISPLAY						DD MM YY HH:MM 01
WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA	
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA	
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE		
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW	

DISPLAY FILE	
REFERENCE PATH NAME	
pathname for FILE	

Standard Text File Suffixes...

.EB - Exception Build Source	.EC - Command file
.EF - Error Files	.EL - Edited Entity List
.UL - Unsuccessful Entity List	.SL - Successful Entity List

F1=PED	F3=	F5=OVERWRITE	F7=RECON	F9=WLK BACK	F11=
F2 = RECALL DISP	F4=	F6=	F8=PED STATUS	F10=WRITE	F12=LOAD

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4. Press ENTER. The contents of the file you requested is displayed.

7.2.10 List Entities

There are two forms of the List Entities command:

- **List Entities in Module**—displays a list of specified entities, and, if requested, will also generate a selection-list file that contains a list of these entities. The entities can be in a module (node) or they can be HG entities related to a box on a Data Hiway.
- **List Entities in IDF**—displays a list of all or selected IDF entities in a specified IDF, and, if requested, will also generate a selection-list file that contains a list of these entities. Refer to 7.1.8.1 for details on selection-list files.

To execute a List Entities command:

1. On the Command Display, select the **LIST ENTITIES IN MODULE** pick or the **LIST ENTITIES IN IDF** pick. The appropriate Command Display appears.
2. If you are listing IDF entities, key the name of the IDF into the pathname for the IDF port.
3. If you want to generate a selection-list file, key in the pathname for the Selection List (7.1.8.1). This can be any name you choose. Key in an appropriate suffix.*

The screenshot shows the 'COMMAND DISPLAY' window. At the top right, it displays 'DD MM YY HH:MM 01' and 'PAGE 01 OF 01'. The main area contains a grid of command options. The 'LIST ENTITIES IN MODULE' option is highlighted. Below the grid, there are input fields for 'REFERENCE PATH NAME' and 'pathname for SELECTION LIST'. There are also checkboxes for 'MODULE or BOX number' and 'If BOX, specify HIWAY/UCN number'. Below these are buttons for 'BUILD TYPES of Entities to List': 'ALL' (selected), 'AM', 'HG', 'CIU', and 'NIM'. At the bottom, there is a row of function key definitions: F1=PED, F2=RECALL DISP, F3=, F4=, F5=OVERWRITE, F6=, F7=RECON, F8=PED STATUS, F9=WLK BACK, F10=WRITE, F11=, F12=LOAD.

COMMAND DISPLAY						DD MM YY HH:MM 01	PAGE 01 OF 01
WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA		
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA		
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE			
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW		

REFERENCE PATH NAME

pathname for SELECTION LIST

MODULE or BOX number ☐ If BOX, specify HIWAY/UCN number ☐

BUILD TYPES of Entities to List:

F1=PED F3= F5=OVERWRITE F7=RECON F9=WLK BACK F11=
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD

* If you do not need to edit the list, enter a filename with a .EL suffix and use the file directly. If you need to edit the list, enter a two character suffix beginning with .X, .Y, or .Z, and use the Text Editor to edit the resultant file. The file name must also be chosen carefully. If the file specified already exists, new entity names will not be appended. Instead, the existing file will be overwritten.

4. If the entities are in a module, key the node number into the Module or Box port. If they are in a box or an IOP, key that number in the Module or Box port **and** key the hiway or UCN number into the Hiway/UCN Number port.

5. Select the BUILD TYPE pick for the type of entities you want to list (All, AM, HG, CIU, NIM). A new set of picks listing the point type appears.

6. Select the pick for the point type you want to list.

7. Press ENTER. A WAIT message appears in the prompter region while the list is generated. When it is completed without errors, the list appears on the screen (and in the selection-list file if you specified one), and the generic function-key labels reappear.

The following error message may appear in the prompter region when executing a List Entities (in Module) command:

ERR: Node unavailable.

COMMAND DISPLAY

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WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE	
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW

REFERENCE PATH NAME

pathname for SELECTION LIST

MODULE or BOX number ☐ If BOX, specify HIWAY/UCN number ☐

BUILD TYPES of Entities to List: ☐ ALL ☒ AM ☐ HG ☐ CIU ☐ NIM

☐ ALL ☒ REGULATORY ☐ NUMERIC ☐ COUNTER ☐ CUSTOM

☐ FLAG ☐ TIMER ☐ SWITCH ☐ BCHPROT

F1=PED F3= F5=OVERWRITE F7=RECON F9=WLK BACK F11=
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD

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7.2.11 Print

There are two forms of the Print command:

- **Print Entities**—prints the contents of selected entities in the current PED, an IDF, or in the modules and gateways on the LCN, on a printer, or copies the information into a file. This command can also generate exception-build files.
- **Print File**—prints the contents of a file on a printer.

The Print Entities Command Display provides picks to let you select the screen (PED set) entity, entities in an IDF, or system (module or gateway) entities.

You can use the Print File command to print the content of any of the command-generated files defined under 7.1.8.

All of the Print commands print on the specified printer if you enter a logical device ID (form: &P1) into the Print device ID or Destination pathname port. If, instead, you enter the pathname for a new file in a user volume into that port, the Print Entities commands store the information in the file. The suffix for the pathname must be .EB or must begin with "x," "y," or "z" (for example, .XX, .yy, or .zz). If you use a .EB suffix, the entities are stored in exception-build source-file form, complete with the appropriate &T control item. The information in the .xx, .yy, and .zz files is in the same form as if it had been printed.

To execute a Print Entities command:

1. On the Command Display, select the **PRINT ENTITIES** pick. This Command Display appears.
2. To print the current PED set, or to copy it to a file, go to Step 3. To print or copy IDF entities, go to Step 4. To print or copy module or gateway entities, go to Step 5.

COMMAND DISPLAY				DD MM YY HH:MM 01 PAGE 01 OF 01	
WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE	
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW

PRINT ENTITIES FROM SCREEN, IDF, OR SYSTEM

REFERENCE PATH NAME

PRINT screen entity ☐ **PRINT IDF entities** ☒ PRINT system entities ☐

pathname for IDF

pathname for SELECTION LIST

PRINT device ID or DESTINATION pathname

F1=PED	F3=	F5=OVERWRITE	F7=RECON	F9=WLK BACK	F11=
F2=RECALL DISP	F4=	F6=	F8=PED STATUS	F10=WRITE	F12=LOAD

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3. Select the PRINT SCREEN ENTITY pick. Key in the Print device ID or Destination file pathname and then press ENTER. The selected entities are either printed or copied into the file.

COMMAND DISPLAY

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WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE	
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW

PRINT ENTITIES FROM SCREEN, IDF, OR SYSTEM

REFERENCE PATH NAME

PRINT screen entity

PRINT IDF entities

PRINT system entities

PRINT device ID or DESTINATION pathname \$P1

F1=PED F3= F5=OVERWRITE F7=RECON F9=WLK BACK F11=
F2 = RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD

5601

4. Select the PRINT IDF ENTITIES pick. To use all of the entities in the IDF, leave the pathname for Selection-list port blank. To use only some of the entities, enter either the pathname for a selection list or the actual entity names into the pathname for Selection-list port. Key in a Print device ID or a Destination pathname and then press ENTER. The selected entities are either printed or copied into the file.

COMMAND DISPLAY

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WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE	
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW

PRINT ENTITIES FROM SCREEN, IDF, OR SYSTEM

REFERENCE PATH NAME

PRINT screen entity

PRINT IDF entities

PRINT system entities

pathname for SELECTION LIST

current AREA DATA BASE pathname \$I1>&D01>AREA01

PRINT device ID or DESTINATION pathname \$P1

F1=PED F3= F5=OVERWRITE F7=RECON F9=WLK BACK
F2 = RECALL DISP F4= F6= F8=PED STATUS F10=WRITE

5602

5. Select the PRINT SYSTEM ENTITIES pick. Enter the pathname for a selection list, of the system entity names, into the pathname for Selection-list port. Key in a Print device ID or Destination pathname and then press ENTER. The selected system entities are either printed or copied into the file.

COMMAND DISPLAY

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PAGE 01 OF 01

WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE	
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW

PRINT ENTITIES FROM SCREEN, IDF, OR SYSTEM

REFERENCE PATH NAME

PRINT screen entity

PRINT IDF entities

PRINT system entities

pathname for SELECTION LIST

current AREA DATA BASE pathname \$I1>&D01>AREA01

PRINT device ID or DESTINATION pathname \$P1

F1=PED F3= F5=OVERWRITE F7=RECON F9=WLK BACK
F2 = RECALL DISP F4= F6= F8=PED STATUS F10=WRITE

5602

To execute a Print File command:

1. On the Command Display, select PRINT FILE. This Command Display appears.
2. Key in the pathname for the File to be printed, its suffix, and, if it's not already filled in, the ID for the printer (form: \$Pn, where "n" is the printer number, as listed on the Console Status Display).
3. Press ENTER. The contents of the file are printed.

COMMAND DISPLAY						DD MM YY HH:MM 01	PAGE 01 OF 01
WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA		
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA		
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE			
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW		

PRINT FILE

pathname for FILE

PRINT device ID or DESTINATION pathname

Standard Text File Suffixes...

.EB - Exception Build Source	.EC - Command file
.EF - Error Files	.EL - Edited Entity List
.UL - Unsuccessful Entity List	.SL - Successful Entity List

F1=PED	F3=	F5=OVERWRITE	F7=RECON	F9=WLK BACK
F2 = RECALL DISP	F4=	F6=	F8=PED STATUS	F10=WRITE

5603

If an error is detected when executing a Print Entities or Print Files command, one of the following messages may appear in the prompter region:

MISSING DATA

There is not sufficient information in the Command Display ports to execute the command.

FILE NOT FOUND (pathname)

The file named in one of the pathname ports (pathname) ca not be found on the named removable medium or on the network. If more than one of these messages appears, each one represents one of the three possible pathnames on the Command Display.

INVALID PATHNAME SUFFIX (pathname)

A pathname in one of the pathname ports (pathname) did not have a valid suffix (.EB, .EF, .UL, .EC, .EL, or .SL). If more than one of these messages appears, each one represents one of the three possible pathnames on the Command Display.

\$FMS Logical_Device_Not_Found(hh:mm) \$P1

The printer was not connected to this console or was not operating. Other similar messages may indicate other printer problems.

NOTE

You can stop a printout by holding CTL and pressing BREAK. For single-entity printouts or file printouts, the printout is canceled only if WAIT is in the prompter region when BREAK is pressed. Multiple-entity printouts stop when the current entity has been printed.

A Screen Entity printout is as in this example:

```
SCREEN ENTITY FL101
NAME                "FL101      "
UNIT                02
PTDESC              "FLOW CONTROLLER 101  "
KEYWORD              "FLOW FL11"
PRIMOD
HWYNUM              02
PNTBOXTY            CB
```

An IDF Entity printout is as in this example:

```
IDF NET>NORM>HGPNTS.DB, ENTITY FC0101
NAME                "FC1010      "
UNIT                01
PTDESC              "FUEL FLOW C0101"
KEYWORD              "FUEL C1  "
PRIMOD              COOK
HWYNUM              01
PNTBOXTY            MC
```

The List File (.EB, .EL, .SL, .UL) printout is a list of entity names as in this example:

```
FILE NET>NORM>HGPID.SL
FC501      ,      AAC10      ,      WGT5      ,
STEAMFL1    ,      HGTSTPID
```

An Error File (.EF) printout is described under 10.4.1.

7.2.11.1 Box/Module Parameters in EB

Before R620 software revision, while using the DEB Print Entities command to create an exception build file for process module points or array points, the numerics, flags, strings, and time parameters for those points were not included in the .EB file.

With R620 revision, the .EB file generated using the Print Entities command includes the numerics, flags, timers, and string parameters for procmod and array points. The size of the .EB file increases according to the parameters associated with the points.

NOTE

The size of .EB files for the procmod and array points increases significantly as it contains the numeric, flag, string and time parameters. Also the non-visible CDS parameters of type time are treated uniquely and written to .EB files. If the file size is too large, the .EB file should be split into two or more files. The end user scripts and entity lists used to generate and load these files may have to be modified accordingly.

For example, if an entity list used to print to an .EB file includes 25 array points, a print entities EC file that uses the list would print all the 25 points to an .EB file. With R620 revision, the .EB file may grow too large for the system to handle (e.g., the text editor may not be able to read it) and the user has to split the entity list into two or more lists and change the associated EC files accordingly.

To reduce the size of .EB files, only those elements that are exceptions to the default values are printed. For example, the default for flags is OFF and there are 1000 flags configured on an array point, and only 50 of those flags are ON. Only the flags that are not at the default value (OFF) are printed as multiple entries per line (wherever applicable) and thus the space for printing the default values for 950 flags in the .EB file is saved.

A sample line format of a .EB file for Procmod Point Types containing Flags, Numerics, Strings and Time are given below.

```
FL(043)=ON FL(077)=ON FL(104)=ON FL(106)=ON FL(107)=ON FL(127)=ON
NN(01)=2.6    NN(02)=3.2    NN(03)=4.8    NN(80)=-10.9
STR8(01)="WHITE  " STR8(02)="GREEN  " STR8(03)="PURPLE  "
STR8(16)="PURPLE  "
STR16(1)="A 16 char string" STR16(8)="A 16 char string"
STR32(4)="A 32 character string, with fill"
STR64(2)="A 64 character string, with fill and more fill and fill
and fill"
TIME(1)="09Sep98 11:59:38" TIME(4)="10Sep98 09:21:37"
```

A sample line format of a .EB file for Array Point Types containing Flags, Numerics, Strings and Time are given below.

```
FL(0043)=ON FL(0077)=ON FL(0104)=ON FL(1006)=ON FL(1007)=ON FL(1023)=ON
NN(001)=2.6    NN(002)=3.2    NN(006)=4.8    NN(240)=-10.9
STR8(001)="WHITE  " STR8(002)="GREEN  " STR8(240)="PURPLE  "
STR16(240)="A 16 char string"
STR32(240)="A 32 character string, with fill"
STR64(240)="A 64 character string, with fill and more fill and
fill and fill"
TIME(001)="09Sep98 11:59:38" TIME(240)="10Sep98 09:21:37"
```

7.2.12 Select Area

The Select Area command establishes the pathname for the area database files (working file .WA and data file .DA) that appears on all subsequent Load Area (7.2.3), Reconstitute Area (7.2.5), and Install Area (7.2.13) Command Displays, and puts the DEB in area-build mode. Select Area provides a way to enter the pathname only once during a DEB session.

To execute a Select Area command:

1. Select the **SELECT AREA** pick on the Command Menu. This Command Display appears.
2. Select the **AREA NUMBER** pick to get the system to fill in the current Area Data Base pathname port or key it in yourself. This pathname must be set up in one of the following two formats:

NET>&Dnn>AREAnn

OR

\$Fn>&Dnn>AREAnn

COMMAND DISPLAY						DD MM YY HH:MM 01	PAGE 01 OF 01
WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA		
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA		
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE			
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW		
REFERENCE PATH NAME NET>norm> The pathname must take the form logical_device_id>&Dnn>AREAnn: where 'nn' denotes an area number 01 - 10. Give path OR select area number and give logical device id for database AREA NUMBER 1 2 3 4 5 6 7 8 9 10 OTHER current AREA DATABASE pathname NET>&D01>AERA01 pathname for WORKFILE							
F1=PED F3= F5=OVERWRITE F7=RECON F9=WLK BACK F11=TAB F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD							

51945

where NET = History Module

\$Fn = Removable Media Drive No. n

nn = Area Number, 01 through 10

3. Press ENTER. If a valid area database pathname is established, an **OPERATION COMPLETE** message appears.

If an error is detected, an **INVALID PATHNAME** message may appear in the prompter region. Key in the correct pathname, and press ENTER again.

NOTE

If you try to select an area for which no Area Database exists, an error message appears telling you so. To create a new blank Area Database, refer to Task 29 in the *System Startup Guide* and use the steps that copy a blank Area Database.

7.2.13 Install Area

The Install Area command transfers the content of the area database working file (.WA) to the area database data file (.DA). The .WA file is used to collect area database entities as they are built. When all new or modified entities are built, this command is used to install them in the .DA file. The .DA file contains the actual database used by the Universal Stations assigned to this area. The .DA file is loaded when the Operator Personality is loaded in the US(s). See the *System Startup Guide* in the *Implementation/Startup & Reconfiguration - 1* binder.

To execute an Install Area command:

1. Select the **INSTALL AREA** pick on the Command Display. This Command Display appears.
2. Select the **AREA NUMBER** pick to get the system to fill in the current Area Data Base pathname port or key it in yourself. This pathname must be set up in one of the following two formats:

NET>&Dnn>AREAnn

OR

\$Fn>&Dnn>AREAnn

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WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE	
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW

REFERENCE PATH NAME

The pathname must take the form logical_device_id>&Dnn>AREAnn: where 'nn' denotes an area number 01 - 10.
Give path OR select area number and give logical device id for database

AREA NUMBER

current AREA DATABASE pathname

Pathname for WORKFILE

F1=PED F3= F5=OVERWRITE F7=RECON F9=WLK BACK F11=
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD

13778

where NET = History Module

\$Fn = Removable Media Drive No. n

nn = Area Number, 01 through 10

3. Press ENTER. A **BUSY** message appears in the prompter region and when .WA is successfully transferred to .DA, the Command Display is redisplayed.

If errors are detected, an **INVALID PATHNAME** or **VOLUME NOT FOUND** message appears in the prompter region. Check the pathnames and verify that the files are on an HM (NET) or on the removable medium indicated (\$Fn), make corrections if necessary, and retry.

7.2.14 New

The New command terminates use of the current Parameter Entry Display (PED) set and returns to the first display in the set with any entered data replaced by standard default values and selections. Any entered data is lost unless it is first written in an IDF or loaded in a module or gateway.

When a PED is redisplayed after execution of any other command, the entered values are displayed, rather than the default values. The New command provides a way to redisplay the PED with all default values.

To execute a New command:

1. Select the NEW pick on the Command Display. This Command Display appears.
2. Press ENTER. The first PED in the current set appears with default values in the picks and ports.

The screenshot shows a terminal window titled 'COMMAND DISPLAY' with a date and time stamp 'DD MM YY HH:MM 01' and 'PAGE 01 OF 01'. The screen is divided into a grid of command options. The 'NEW' option is highlighted in the bottom right corner. Below the grid, the text 'NEW - START ANEW, DISCARDING CURRENT ENTRIES' is displayed, followed by a prompt 'REFERENCE PATH NAME' and a text input field containing 'NET>nom>'. At the bottom of the screen, there is a row of function key shortcuts: F1=PED, F2=RECALL DISP, F3=, F4=, F5=OVERWRITE, F6=, F7=RECON, F8=PED STATUS, F9=WLK BACK, F10=WRITE, F11=, and F12=LOAD. The number '5605' is visible in the bottom right corner of the terminal window.

COMMAND DISPLAY				DD MM YY HH:MM 01 PAGE 01 OF 01	
WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE	
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW

NEW - START ANEW, DISCARDING CURRENT ENTRIES

REFERENCE PATH NAME

F1=PED F3= F5=OVERWRITE F7=RECON F9=WLK BACK F11=
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD

If any data was entered in the original PED set that had not been written to an IDF or loaded in its module or gateway, the following message appears in the prompter region: WARNING! DATA MODIFIED - WRITE OR LOAD. REPEAT LAST KEYSTROKE TO IGNORE WARNING. Press ENTER again if you want to erase the data and start over.

7.2.15 Repack IDF

This command lets you make the best use of IDF space, provides a method to update IDFs when revised GDFs are issued by Honeywell, and allows you to create a new IDF with only the entities that you select from an older IDF.

When you select the Repack IDF pick on the Command Display, these additional picks appear:

Three rectangular buttons are displayed side-by-side. The first button is labeled 'Inspect IDF', the second is labeled 'Binary Repack', and the third is labeled 'Upgrade IDF'.

Inspect IDF

Binary Repack

Upgrade IDF

2313

7.2.15.1 Inspect IDF

Use this function to determine if an IDF is a good candidate for binary repacking, because entities have been deleted or because entities have been corrupted. Even though an entity has been deleted by one of the Delete commands, the space occupied in the IDF is still not available. Use Binary Repack to gain it back.

To inspect an IDF

1. On the Command Display, select **REPACK IDF**. This display appears.
2. Select **INSPECT IDF**.
3. Key in the pathname for IDF you want to inspect and press **ENTER**.

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WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE	
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW

REFERENCE PATH NAME

pathname for IDF

F1=PED F3= F5=OVERWRITE F7=RECON F9=WLK BACK F11=
F2 = RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD

13779

4. This display appears. It tells how many entities have been deleted from this IDF, how many are valid entities, and how many have been corrupted because of a File Manager error that occurred when data was last written in the file.
5. To return to the Command Display, press **COMND**.

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COMMAND GENERATED DISPLAY

VALID ENTITIES IN IDF	2
NULL ENTITIES IN IDF	0
CORRUPT ENTITIES IN IDF	0
TOTAL ENTITIES IN IDF	2

F1=PED F3= F5=OVERWRITE F7=RECON F9=WLK BACK F11=
F2 = RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD

5607

7.2.15.2 Binary Repack

Use this function to copy an IDF with deleted or corrupted entities to a new IDF. You do not have to repack a whole IDF. You can specify a starting entity, or you can use a selection-list to select entities to be copied to the new IDF. As the IDF is repacked and copied, the storage space occupied by deleted or corrupted entities is eliminated.

To use the Binary Repack Function (after selecting REPACK IDF)

1. Select BINARY REPACK.
2. Key in the pathname for the IDF to be copied.
3. Key in the pathname for the destination (new) IDF.
4. If you are using a selection list, key in the Optional pathname for Selection List.
5. If you want to specify a starting entity, key in the Optional starting Entity name.

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WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE	
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW

REFERENCE PATH NAME

Inspect IDF Binary Repack Upgrade IDF

pathname for IDF

pathname for destination IDF

OPTIONAL pathname for SELECTION LIST

OPTIONAL starting ENTITY name

F1=PED F3= F5=OVERWRITE F7=RECON F9=WLK BACK F11=
F2 = RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD

13780

6. Press ENTER. When the repacked IDF has been copied to the destination IDF, an OPERATION COMPLETE message appears. If there is not enough space on the destination volume to take all of the entities, the next entity to be processed is displayed in the STARTING ENTITY port. You can finish the operation by using a new destination volume.

To gain back the space occupied by the old IDF, use the Utilities' Delete command to delete it.

7.2.15.3 Upgrade IDF

Use this function to update an IDF when Honeywell has issued a revised GDF. The original IDF is read, subjected to the changes in the revised GDF, and copied to a new IDF. You can specify the starting entity, if you wish, or you can use a selection-list to specify the entities to be upgraded. The revised GDF must be available on a removable medium or an HM.

To upgrade an IDF (after selecting REPACK IDF)

1. Select UPGRADE IDF.
2. Key in the pathname for the IDF to be copied.
3. Key in the pathname for the destination (new) IDF.
4. If you are using a selection list, key in the Optional pathname for Selection List.
5. If you want to specify a starting entity, key in the Optional starting Entity name.

The screenshot shows a terminal window titled 'COMMAND DISPLAY' with a date and time stamp 'DD MM YY HH:MM 01' and 'PAGE 01 OF 01'. The screen is divided into a grid of function keys. The 'REPACK IDF' key is highlighted. Below the grid, there are input fields for 'REFERENCE PATH NAME', 'pathname for IDF', 'pathname for destination IDF', 'OPTIONAL pathname for SELECTION LIST', and 'OPTIONAL starting ENTITY name'. At the bottom, there is a row of function key definitions: F1=PED, F2=RECALL DISP, F3=, F4=, F5=OVERWRITE, F6=, F7=RECON, F8=PED STATUS, F9=WLK BACK, F10=WRITE, F11=, F12=LOAD.

COMMAND DISPLAY						DD MM YY HH:MM 01	PAGE 01 OF 01
WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA		
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA		
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE			
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW		

REFERENCE PATH NAME

pathname for IDF

pathname for destination IDF

OPTIONAL pathname for SELECTION LIST

OPTIONAL starting ENTITY name

F1=PED F3= F5=OVERWRITE F7=RECON F9=WLK BACK F11=
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD

13781

6. Press ENTER. When the repacked IDF has been copied to the destination IDF, an OPERATION COMPLETE message appears. If there is not enough space on the destination volume to take all of the entities, the next entity to be processed is displayed in the Starting Entity port. You can finish the operation by using a new destination volume.

To gain back the space occupied by the old IDF, use the Command Processor's Delete command to delete it.

7.2.16 Execute Command File

This command enables you to execute a number of DEB commands by first listing them in a command file and then executing them all via a single Execute Command File request. Command files are defined in Section 8.

To execute an Execute Command File command:

1. On the Command Display, select the EXECUTE COMMAND FILE pick. This Command Display appears.
2. Key in the pathname for the Command File (8.2).
3. Press ENTER. The Command File is executed. As each command line is executed, each line in the file and its line number appears in the prompter region.

COMMAND DISPLAY						DD MM YY HH:MM 01 PAGE 01 OF 01
WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA	
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA	
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE		
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW	

REFERENCE PATH NAME

pathname for COMMAND FILE

F1=PED	F3=	F5=OVERWRITE	F7=RECON	F9=WLK BACK	F11=
F2 = RECALL DISP	F4=	F6=	F8=PED STATUS	F10=WRITE	F12=LOAD

13782

If any errors are detected as the Command File executes, an Error File is created and its content is displayed when execution of the Command File is complete. The filename of the Error File is the same as that of the Command File but it ends with a .EF suffix.

7.2.16.1 Initiation Of DEB EC File From Command Processor

EC files can be invoked from the Command Processor command line (see *Command Processor*).

7.2.17 Alter Parameters

The Alter Parameter command changes parameter values in the node(s) for one or more entities. You create a Parameter Assignment file that contains statements that define the value changes to be made in all of the entities. You name the entities to be changed in a selection list (7.1.8.1).

The Parameter Assignment file is a user file whose content is in the following form:

```
alprior = hi alenbsts = enable  
  
pvformat = d2
```

where the name of each parameter to be changed is to the left of the equal sign and the new value is to the right of the equal sign. The file can list as many as 1000 parameter changes.

As an example, to modify a text string: ptdesc = "new description" Value changes that alter the structure of a data point are not accepted. For example, a parameter like NOINPTS (number of inputs) cannot be changed.

Use the Text Editor to create this file (see *Text Editor Operation* manual). You can give it any name you choose and any suffix that begins with "X," "Y," or "Z." For example, CHANGE.XL.

When Alter Parameters is executed, the values are changed without reloading, without using the Overwrite option, and regardless of the position of the OPER/SUPVR/ENG keyswitch. Because the keyswitch must be in the ENGR position to enter the Text Editor, Parameter Assignment files can be created or changed only when the switch is in that position.

To execute an Alter Parameters command:

1. On the Command Display, select the ALTER PARAMETERS pick. This Command Display appears.
2. If needed, key in the Reference Path Name (7.1.5).
3. Key in the pathname for the Selection List.
4. Key in the pathname for the Param-Value List.
5. Press ENTER. The parameters in all entities named in the Selection List are changed to the values in the Param-Value List (Parameter Assignment file).

COMMAND DISPLAY						DD MM YY HHMM 01 PAGE 01 OF 01
WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA	
LOAD	LOAD MULTIPLE	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA	
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE		
ALTER PARAMETERS	EXCEPTION BUILD		REPACK IDF	PRINT ENTITIES	NEW	

REFERENCE PATH NAME

pathname for SELECTION LIST

pathname for Param-Value list

F1=PED	F3=	F5=OVERWRITE	F7=RECON	F9=WLK BACK	F11=
F2=RECALL DISP	F4=	F6=	F8=PED STATUS	F10=WRITE	F12=LOAD

If an error is detected, both the Alter Parameters Command Display stays on the screen and an error message appears in the prompter region, or an Error File display appears.

7.2.18 Backup and restore box parameters

With TPN R682, you can take a backup of box parameters to a file and later use the file for restoring the values. The backup and restore option is available in xPM, SM and LM. This is also applicable for APM and HPM simulator box parameters. You can take a backup or restore box parameters through DEB and DEBEC.

The backup and restore option is available for the following box parameter type:

- STRINGS
- NUMERICS
- FLAGS
- TIMES



ATTENTION

SM does not allow box parameter stores when the box is in IDLE state. Attempting to backup or restore when the box is in IDLE will result in the creation of an unsuccessful list.

xPM and LM and SM must be in run state (IDLE, OK) to perform a backup/restore operation. If no index value is specified, the backup operation is performed for the maximum number configured for each parameter. If no index value is specified, the restore operation is performed for all the applicable parameters present in the file, with the maximum configured parameter limit.

The maximum limit for box parameters that can be configured is as follows:

Parameter type	PM	APM	HPM	SM	LM
Flags	1023	4095	4095	2000	1024
Numerics	2047	4095	4095	1000	1024
Times	NA	4095	4095	NA	NA
Strings	NA	4095	4095	NA	NA

The starting and ending subscripts of the parameters for backup or restore operations can be specified. The default value for the index is blank. If index is not entered, the subscript value will be the actual number of box parameters configured. After the operation is complete the message 'OPERATION COMPLETE' appears. For aborting the operation use CTRL + BREAK keys, and then ABORT ON BREAK message appears.

Multiple users can save backups in different locations. You can take a backup of the same parameter from different workstations at the same time. Similar to other DEB functionalities, file manager errors are reported, if the same file is selected by more than one user at the same time.

Note: During simultaneous Back up operation, do not use the same file name with different extension, because the error file is created with the same name.

The errors that occur during the backup/restore operation are recorded in an error file with .EF extension. The error file name is the same as the .EC user file and is created in the same location as the backup/restore file.

7.2.18.1 Initiation Of backup and restore functionality

To execute a backup and restore operation through DEB perform the following steps:

1. On the Engineering Menu, select the Builder Commands. The Command Display appears.
2. Click the BOX PARAM BACKUP RSTR pick.
3. Click BACKUP pick to perform a backup operation.
Click RESTORE pick to perform a restore operation.
4. Enter the Box number and the UCN number.
5. Click the Types of entities for the box parameter type. You can select multiple entities.
6. Enter the Starting Index and Ending Index of the parameter for the backup/restore operation.
7. Enter the Pathname for Backup/Restore.
8. For Restore, select the Default values not in file as “Yes” to default the values not present in the file but within the selected range.
9.
 - If the backup is to be taken to an emulated disk/physical disk, the drive/disk must be formatted and a volume needs to be created.
 - If the restore is to be performed from an emulated disk/physical disk, the drive/disk must be mounted.
10. Press ENTER. At the end of a successful operation, the OPERATION COMPLETE message appears.

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WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA
LOAD	LOAD	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE	
ALTER PARAMETERS	EXCEPTION BUILD	BOX PARAM BACKUP RSTR	REPACK IDF	PRINT ENTITIES	NEW

REFERENCE PATH NAME

BACKUP RESTORE

BOX number UCN number

TYPES of Entities : ALL NUMERICS FLAGS TIMES STRINGS

Starting Index Ending Index

PATHNAME FOR BACKUP:

F1=PED F3= F5=OVERWRITE F7=RECON F9=ULK BACK F11=TAB
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD

BOX PARAM BACKUP RSTR pick

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WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA
LOAD	LOAD	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE	
ALTER PARAMETERS	EXCEPTION BUILD	BOX PARAM BACKUP RSTR	REPACK IDF	PRINT ENTITIES	NEW

REFERENCE PATH NAME net>rrj

BACKUP RESTORE

BOX number 21 UCN number 03

TYPES of Entities : ALL NUMERICS FLAGS TIMES STRINGS

Starting Index Ending Index

PATHNAME FOR BACKUP: abc.xx

F1=PED F3= F5=OVERWRITE F7=RECON F9=ULK BACK F11=TAB
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD

Backup operation for single parameter

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WRITE TO IDF	READ TO PED	DELETE SYSTEM ENTITY	DELETE ENTITY FROM IDF	DISPLAY FILE	SELECT AREA
LOAD	LOAD	DELETE MULT ENTITIES	DELETE MULT ENTITIES IDF	EXECUTE COMMAND FILE	INSTALL AREA
RECONSTITUTE	RECONSTITUTE MULTIPLE	LIST ENTITIES IN MODULE	LIST ENTITIES IN IDF	PRINT FILE	
ALTER PARAMETERS	EXCEPTION BUILD	BOX PARAM BACKUP RSTR	REPACK IDF	PRINT ENTITIES	NEW

REFERENCE PATH NAME net>rrj

BACKUP RESTORE

BOX number 21 UCN number 03

TYPES of Entities : ALL NUMERICS FLAGS TIMES STRINGS

Starting Index 1 Ending Index 4095

Default Values not in file YES NO

PATHNAME FOR RESTORE: abc.xx

F1=PED F3= F5=OVERWRITE F7=RECON F9=ULK BACK F11=TAB
F2=RECALL DISP F4= F6= F8=PED STATUS F10=WRITE F12=LOAD

Restore operation for single parameter

7.2.18.2 File source/destination

For backup operation, you can specify the location where the backup file is to be stored. For restore, you can specify the location from where the file is to be restored. The file path can be:

- Any emulated disk
- Any volume in HM
- Any physical media

If the backup file is to be saved in...	Then the file path format is
removable medium and emulated disk	\$Fxx>DIR>FILE Where \$Fxx - Logical device identifier. "xx" indicates the drive number of the removable medium/emulated disk in the console. DIR - Directory name as configured on the removable medium/emulated disk.
HM	NET>DIR>FILE Where DIR - Directory name as configured on the HM.

You can specify a reference pathname so that the file name alone is specified in the backup/restore pathname port.

7.2.18.3 File format

The file name for backup and restore operations must have the following extensions.

“@#”

where:

@ can be ‘X’, ‘Y’, or ‘Z’, and

can be any alphabet from a-z or a number from 0-9.

You can edit and modify the values in the backup file. If a parameter appears more than once in a file, the latest value is restored on to the parameter. If there is a parameter mismatch in the file, the parameter cannot be restored. Then, an error file is created.

7.2.18.4 Change from UCN Node specific configuration to Command display

If the point mix is changed, then the backup and restore option defaults the value of the UCN number and the box number from the UCN node specific configuration.

SECTION 8 - DEB COMMAND FILES

This section defines DEB command files and their use. It also provides examples of these files.

8.1 PURPOSE OF DEB COMMAND FILES

A DEB command file is used in conjunction with the Execute Command File command described in subsection 7.2.16. It enables you to execute a series of DEB commands without using the Command Display repeatedly to execute each individual command. This saves time and effort when you have to execute the same series of DEB commands (such as exception-build a point, print the IDF, and then load it) for several points.

8.2 DEB COMMAND FILE CONTENT AND USE

The *Text Editor Operation* manual explains how you can create DEB command files. A command file name consists of one to eight alphanumeric characters followed by a .EC suffix; e.g., COMMAND.EC.

NOTE

The name of the command file must be different than any of the file names used within the command file. An error will occur if two identical names are used.

A command file is composed of the following items:

- comments
- device and volume/directory declaration and printer declaration (one per line).
- commands

{example of an Exception-build command file} Comment Line
|
Comments

V1=\$F1>XBPT
V2=\$F2>NIDF
V3=NET>TEMP
P1=\$P1

Device Declarations Line
Device Declarations Line
Volume/Directory Declarations Line
Printer Declarations Line

XB V1>SOURCEA V2>IDFA &B V3>TEMPIDFA

Command Line

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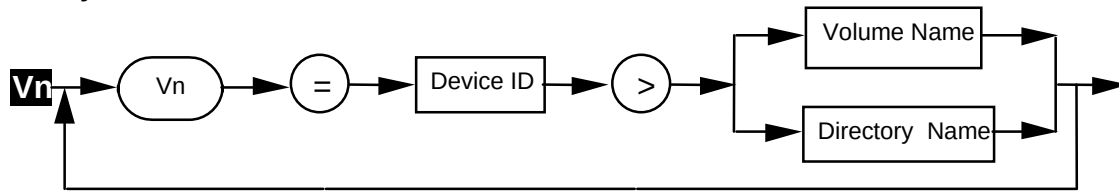
8.2.1 Comment Lines

Comment lines can be included anywhere in a DEB command file to describe the actions to be performed. Comments must be preceded by a "{" and ended with a "}". They cannot exceed the length of one line. If comments require more than one line, enter them in separate lines following the first, beginning each with a "{" and ending each with a "}".

8.2.2 Device and Volume/Directory Declarations

Device and volume/directory declarations specify the location of the files to be used during execution of the DEB commands which follow. They reduce the amount of typing needed to create a command file. The declarations line must precede command lines in a command file. Device and volume/directory declarations are designated by "Vn" where "n" is a number from 1 to 50.

8.2.2.1 Vn Syntax



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8.2.2.2 Vn Example

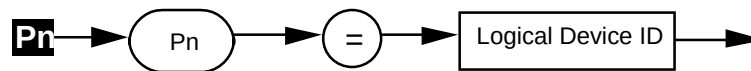
```
V1=$F1>DIM1
V2=NET>IDF1
```

In the above example, V1 points to removable medium 1, directory DIM1 and V2 points to the History Module, volume IDF1. Additional declarations would be V3, V4,...,V50.

8.2.3 Printer Declaration

A printer declaration defines the printer to be used during execution of the DEB commands listed in the command lines which follow. It is placed in the declaration line, following the device and volume/directory declarations. A printer declaration is designated by "Pn" where "n" is a number from 1 to 10.

8.2.3.1 Pn Syntax



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8.2.3.2 Pn Example

```
P1=$P2
```

In the above example, P1 points to printer 2.

8.2.4 Command Lines

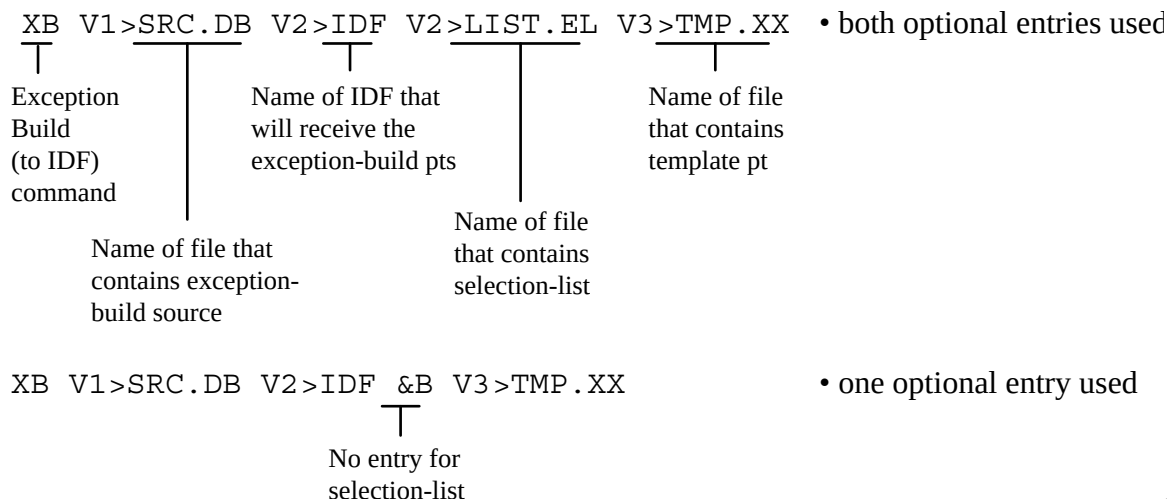
Command lines follow the device/volume/directory declarations in the declarations line. Each command line consists of an abbreviated command followed by one or more arguments containing the information required to execute the command. These arguments correspond to the information that has to be entered using the Command Display except for the Reference Pathname. No data is entered in a command line for the Reference Pathname. The arguments must appear in the command line in the order shown in the syntax diagrams in subsection 8.3.

A special “&B” control item can be used as a special directive to indicate “not making” an optional Command Display entry. If this is the last argument on a command line, you can just leave it blank, and omit the “&B.”

For example, there are four entries defined for an Exception Build (to IDF) command, excluding the Reference Path Name which is not used in command files.

- a source file pathname
- an IDF pathname
- an optional selection-list pathname*
- an optional template pathname

The following two command line examples describe the arguments and show how the “&B” control item is used. The device and volume/directory portions of the pathnames are defined by the Vn declarations in the declarations line, which always precedes the command lines. Note also that the same Vn declaration can be used to identify more than one file; e.g., both IDF and the selection-list file are on the same disk, V2.



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* Selection-list file names must have an appropriate suffix such as .EL, .UL, .SL or a user-file suffix such as .XX, .yy or .Za.

A list of the abbreviated command file commands and the corresponding Command Display commands follow:

IA	Install Area	AS	Alter Parameters
LI	List Entities in IDF	RPB	Repack IDF (Binary)
LS	List Entities in Module	RPU	Repack IDF (Upgrade)
PI	Print (IDF) Entities	XB	Exception Build (to IDF)
PS	Print (System) Entities	XL	Exception Build (and Load)
PF	Print File	DI	Delete Entity from IDF
RM	Reconstitute Multiple	DIM	Delete Mult Entities IDF
LM	Load Multiple	DS	Delete System Entity
		DSM	Delete Mult (System) Entities

The LI and LS commands list all entity types and do not accept entity-type information.

The following list of point types match those seen in the DEB interactive mode when executing a List Entities in IDF or List Entities in Module command, except that any imbedded spaces have been removed:

POINT REALM	ASSOCIATED POINT TYPES			
AM	REGLATRY	NUMERIC	COUNTER	CUSTOM
	FLAG	TIMER	SWITCH	
HG	COUNTER	DIGIN	DIGOUT	DIGCMP
	ANALOGIN	ANALOGOUT	ANALOGCOMP	FLAG
	TIMER	NUMERIC	LOGBLK	REGLATRY
	CTRLCNTR	BATCHASSN	PROCESSMOD	
CIU	ACIDP	CRDP		
NIM	ANALOGIN	ANALOGOUT	REGPV	REGCTL
	DIGIN	DIGOUT	DIGCOMP	LOGIC
	PROCESSMOD	FLAG	TIMER	NUMERIC
	ARRAY	DEVICECONTROL	CONTROLMODULE	

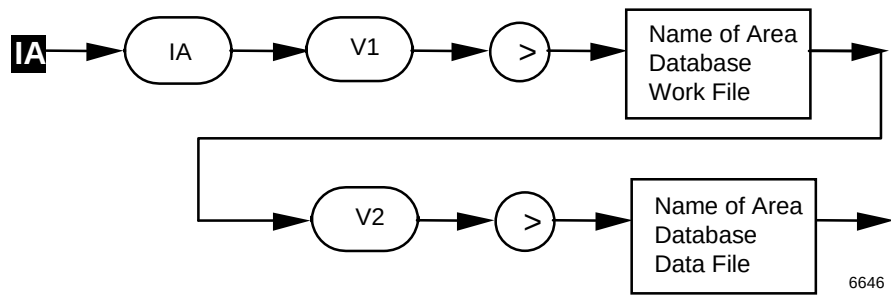
8.3 COMMAND FILE COMMANDS

The following section defines the syntax and lists an example of a typical command line for each of the eighteen command file commands.

8.3.1 IA Definition

This is the Install Area command which is used to transfer the content of the area database work file (.WA) to the area database data file (.DA).

8.3.1.1 IA Syntax



8.3.1.2 IA Example

IA V1>AREAnn V2>AREAnn

Where:

V1 = NET>&D01 (Work File (.WA)).

V2 = NET>&D01 (Data File (.DA)).

AREAnn is the name of the area database file 01 to 10.

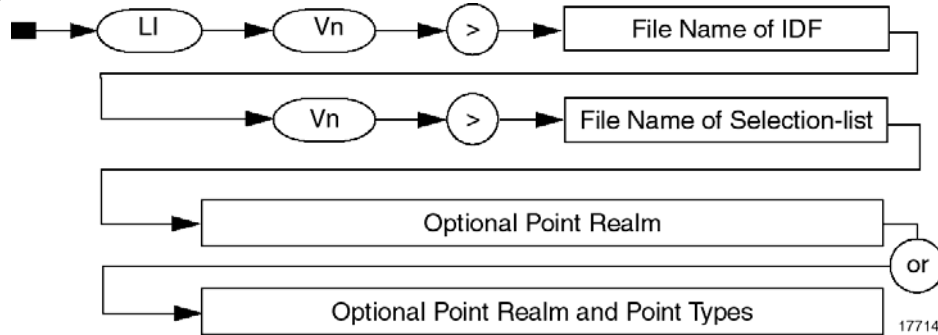
NET = either NET or \$Fn

n = drive number

8.3.2 LI Definition

This is the List Entities in IDF command used to display a list of the entities in an IDF. Release 610 extends the function of the existing DEB EC file LI command to enable restricting the entities listed to those meeting “point realm” or “point realm and point type” criteria.

8.3.2.1 LI Syntax



8.3.2.2 LI Example

```
LI V1>IDF V2>LIST.EL
```

Where:

V1 and V2 represent volume/directory designations.

IDF is the name of the file that contains the IDF entities to be listed.

LIST.EL (optional) is the name of the selection-list file where the entities will be listed.

(see Vn example on section 8.2.2)

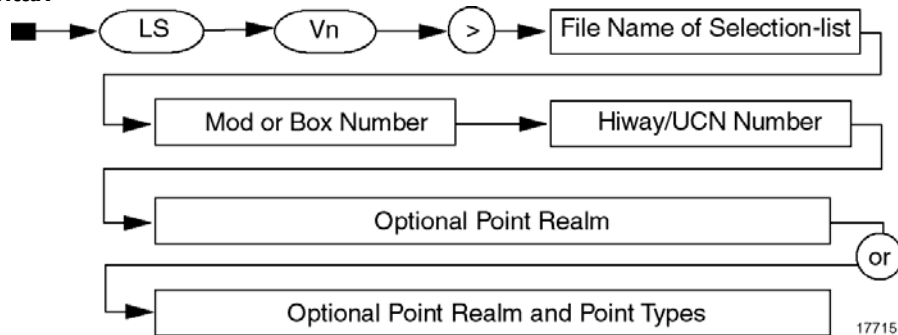
NOTE

Vn definitions must be declared at the front of the file before any commands are encountered.

8.3.3 LS Definition

This is the List Entities in Module command that is used to display a list of the entities in a module, or in a box, or an IOP on a Hiway or UCN. Release 610 extends the function of the existing DEB EC file LS command to enable restricting the entities listed to those meeting “point realm” or “point realm and point type” criteria.

8.3.3.1 LS Syntax



8.3.3.2 LS Example

```
LS V1>LIST.EL 5 1
```

Where:

V1 represents a volume/directory designation.

LIST.EL is the name of the selection-list file where the entities will be listed.

5 is the box number where the entities are located.

1 is the Hiway number where the box is located.

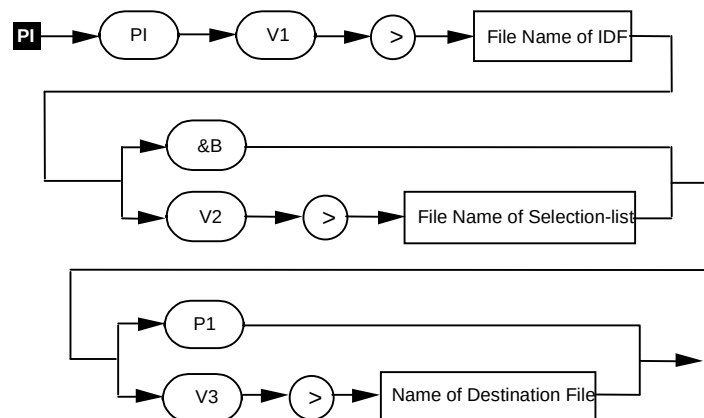
NOTE

- Use “&B” for Hiway/UCN Number when not applicable and it is not the last entry on the command line.
- Vn definitions must be declared at the front of the file before any commands are encountered.

8.3.4 PI Definition

This is the Print (IDF) Entities command that is used to print out the contents of either all or only a selected number of entities in an IDF on a designated printer or to copy the entities to a designated User or Exception-build source file.

8.3.4.1 PI Syntax



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8.3.4.2 PI Examples

Example #1

```
PI V1>IDF V2>AMPTS.EL V3>AMPTS.EB
```

Where:

V1, V2 and V3 represent prior device and volume/directory designations.

IDF is the name of the file that contains the entities to be printed or copied.

AMPTS.EL is the name of the selection-list file that lists the names of the entities to be printed or copied.

AMPTS.EB is the name of the file where the entities will be copied. Because the .EB suffix is used, the entities will be stored in exception-build source-file form.

Example #2

```
PI V1>IDF &B V2>AMPTS.EB
```

Where:

V1 and V2 represent prior device and volume/directory designations.

IDF is the name of the file that contains the entities to be printed or copied.

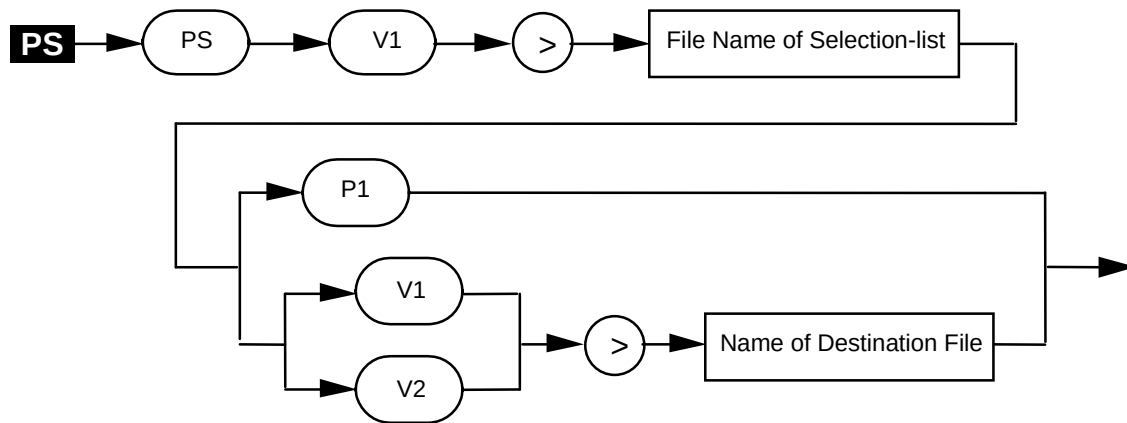
&B is the special control item used to indicate that no selection list is to be used.

AMPTS.EB is the name of the file where the entities will be copied.

8.3.5 PS Definition

This is the Print (System) Entities command that is used to print out the contents of specified system entities on a designated printer or to copy the contents of specified system entities to a designated User or Exception-build source file.

8.3.5.1 PS Syntax



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8.3.5.2 PS Example

```
PS V1>LIST.LS V1>LIST.XL
```

Where:

LIST.LS is the name of the User file which contains system entity names (for example: FIC100, TI201).

V1 represents a prior device and volume/directory designation.

(LIST.XL is the name of the User file where the entities will be copied)

To print Reserved System Entities, you must create a Selection List (SL) file. In the >EC, specify the SL file (which contains the Reserved System Entities) and use the command:

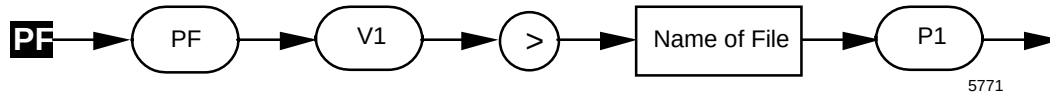
```
PS V1>HWY3BOX.SL V1>HWY3BOX.EB
```

If you use the actual Reserved System Entity name in the command, the error “ILLEGAL DEVICE ID” is returned.

8.3.6 PF Definition

This is the Print File command that is used to print out the contents of any command-generated file on a designated printer.

8.3.6.1 PF Syntax



8.3.6.2 PF Example

```
PF V1>FILE.EL P1
```

Where:

V1 represents a prior device and volume/directory designation.

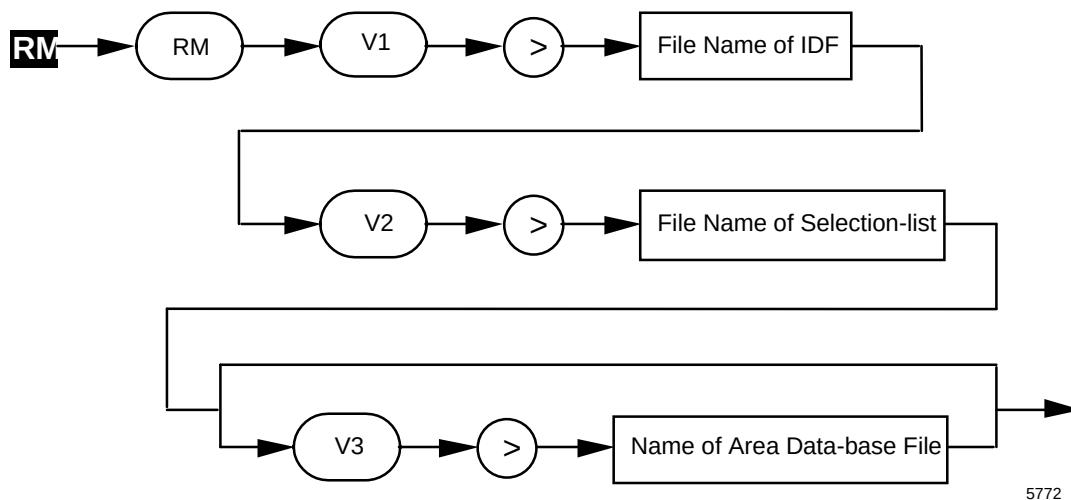
FILE.EL is the name of the selection-list file to be printed.

P1 represents a prior printer designation.

8.3.7 RM Definition

This is the Reconstitute Multiple command. For all entities except area database entities, it is used to read selected entities from the modules or gateways where they reside, convert them to PED form and then store them in a designated IDF. For area database entities, it is used to read the entities from the area database data (.DA) file, convert them to PED form and then store them in the designated IDF.

8.3.7.1 RM Syntax



8.3.7.2 RM Examples

Example #1 Process Data

```
RM V1>IDF V2>LIST.EL
```

Where:

V1 and V2 represent prior device and volume/directory designations.

IDF is the name of the IDF where the entities will be stored.
 LIST.EL is the name of the selection-list file that lists the names of the entities to be reconstituted.

Example #2 Area Data

```
RM V2>IDF V3>LIST.EL V1>AREA01
```

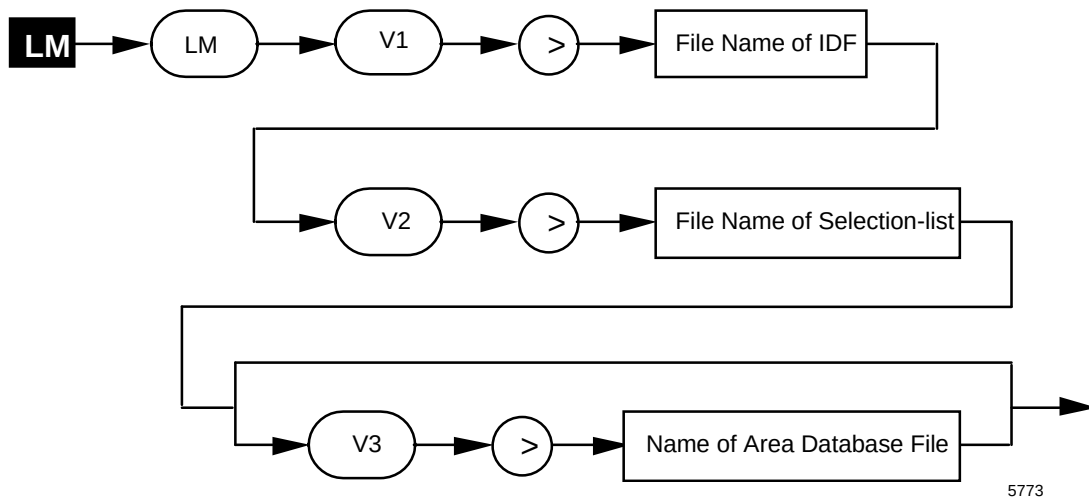
Where:

V1, V2, and V3 represent prior device and volume/directory designations where V1=NET>&D01.
 IDF is the name of the IDF where the area database entities will be stored.
 LIST.EL is the name of the selection-list file that lists the names of the area database entities to be reconstituted.
 AREA01 is the name of the area database file where the entities to be reconstituted reside.

8.3.8 LM Definition

This is the Load Multiple command. For all entities except area database entities, it is used to load selected entities from an IDF into their destination modules or gateways. For area database entities, it is used to read the entities from the IDF into the area database work (.WA) file, which can then be loaded into the area database data (.DA) file by using the Install Area command.

8.3.8.1 LM Syntax



8.3.8.2 LM Examples

Example #1 Process Data

```
LM V1>IDF V2>LIST.EL
```

Where:

V1 and V2 represent prior device and volume/directory designations.
 IDF is the name of the IDF that contains the entities to be loaded.
 LIST.EL is the name of the selection-list file that lists the names of the entities to be loaded.

Example #2 Area Data

```
LM V2>IDF V3>LIST.EL V1>AREA01
```

Where:

V1, V2, and V3 represent prior device and volume/directory designations where V1=NET>&D01.

IDF is the name of the IDF that contains the area database entities to be loaded.

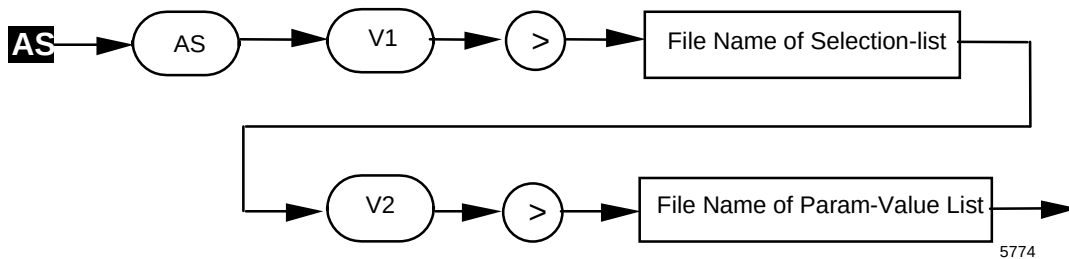
LIST.EL is the name of the selection-list file that lists the names of the area database entities to be loaded.

AREA01 is the name of the area database file where the entities will be loaded.

8.3.9 AS Definition

This is the Alter (System) Parameters command that is used to change the parameter values in a node for a selected number of entities. You use a Parameter-value list to specify the parameters and their new values.

8.3.9.1 AS Syntax



8.3.9.2 AS Example

```
AS V1>LIST.EL V2>PARAMS.XL
```

Where:

V1 and V2 represent prior device and volume/directory designations.

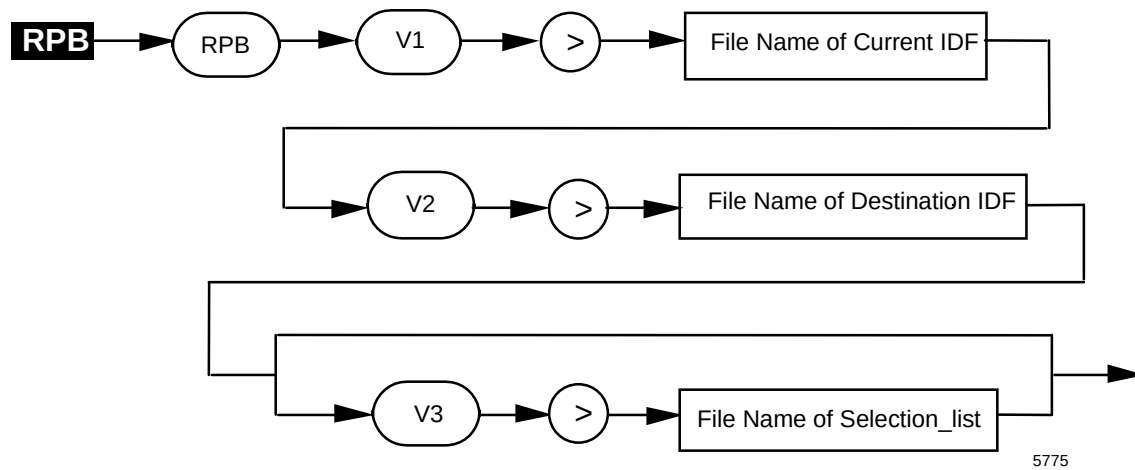
LIST.EL is the name of the selection-list file that lists the names of as many as 1000 entities to be altered.

PARAMS.XL is the name of the Parameter Assignment file that lists the names of the parameters and their new values.

8.3.10 RPB Definition

This is the Repack IDF (Binary) command that is used to repack an IDF with deleted or corrupted entities into a new IDF. You can repack either all or only a selected number of the entities in the IDF.

8.3.10.1 RPB Syntax



8.3.10.2 RPB Example

RPB V1>IDF V2>N_IDF V3>LIST.EL

Where:

V1, V2, and V3 represent prior device and volume/directory designations.

IDF is the name of the IDF that contains the IDF entities to be repacked.

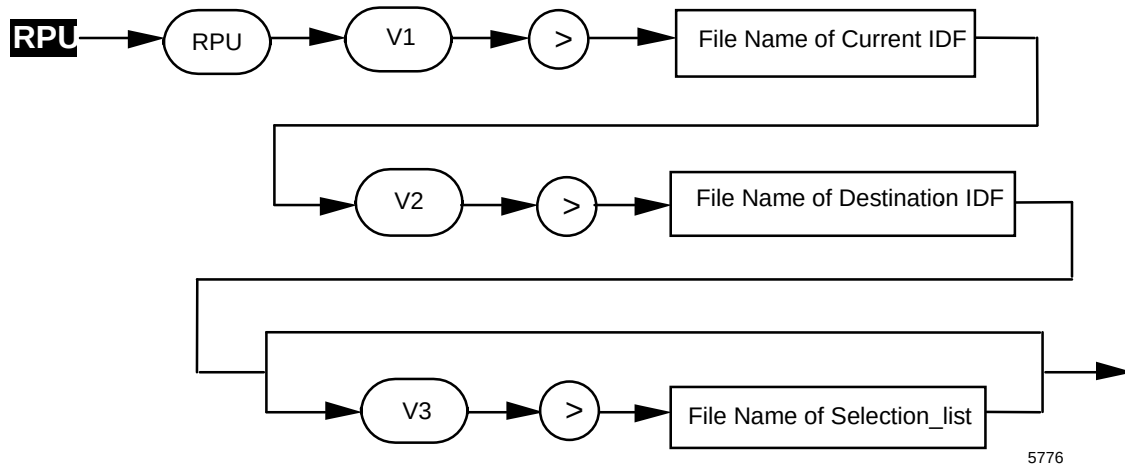
N_IDF is the name of the new IDF where the valid entities will be copied.

LIST.EL (optional) is the name of the selection-list file that lists the names of the entities to be repacked.

8.3.11 RPU Definition

This is the Repack IDF (Upgrade) command that is used to create a new, updated IDF when Honeywell issues a revised GDF. You can upgrade either all or only a selected number of entities in the IDF.

8.3.11.1 RPU Syntax



8.3.11.2 RPU Example

```
RPU V1>IDF V2>N_IDF
```

Where:

V1 and V2 represent prior device and volume/directory designations.

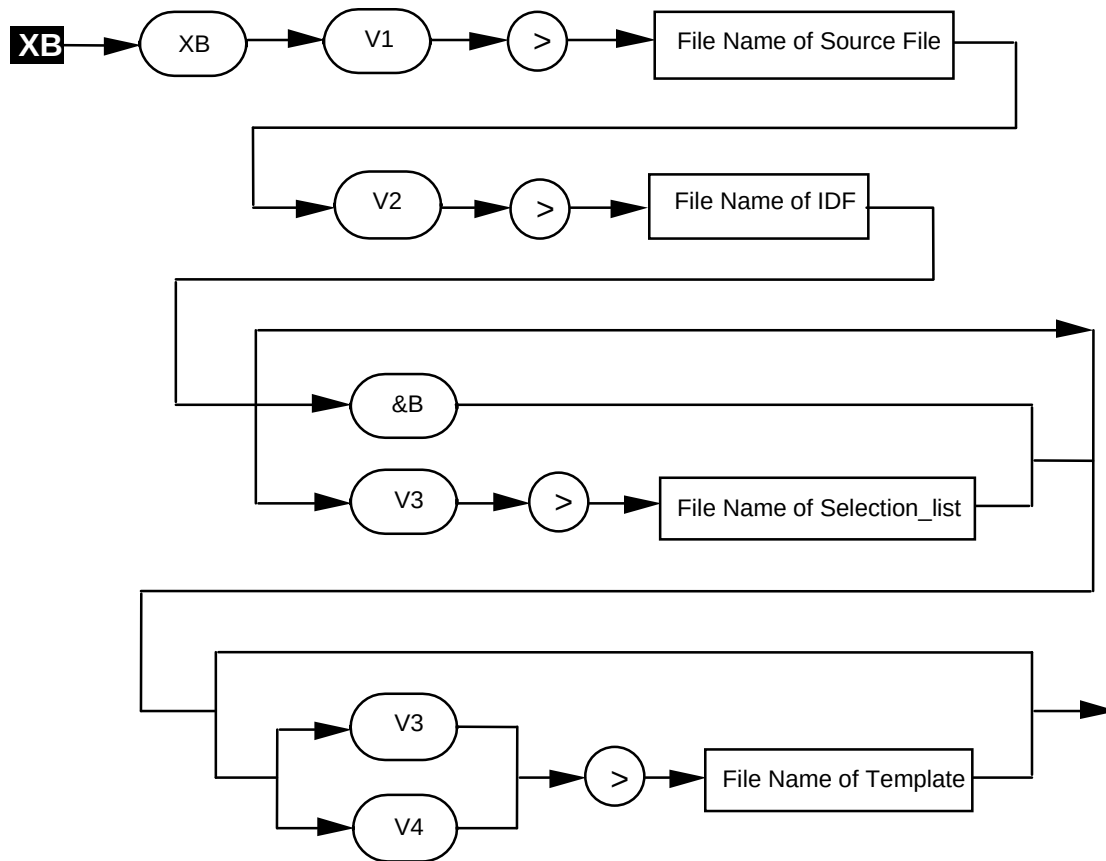
IDF is the name of the IDF that contains the IDF entities to be upgraded.

N_IDF is the name of the new IDF where the upgraded entities will be copied.

8.3.12 XB Definition

This is the Exception Build (to IDF) command. It is used to build an IDF by using either all or only a selected number of entities in an exception-build (.EB) source file. If required, an IDF containing additional template entities can also be used.

8.3.12.1 XB Syntax



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8.3.12.2 XB Example

XB V1>SOURCE.EB V2>IDF V3>LIST.EL V4>TEMP

Where:

V1, V2, V3 and V4 represent prior device and volume/directory designations.

SOURCE . EB is the name of the file that contains the template entities, the name of each entity and the parameter-value changes.

IDF is the name of the IDF where the newly built entities will be stored.

LIST . EL (optional) is the name of the selection-list file that lists the names of the entities to be built.

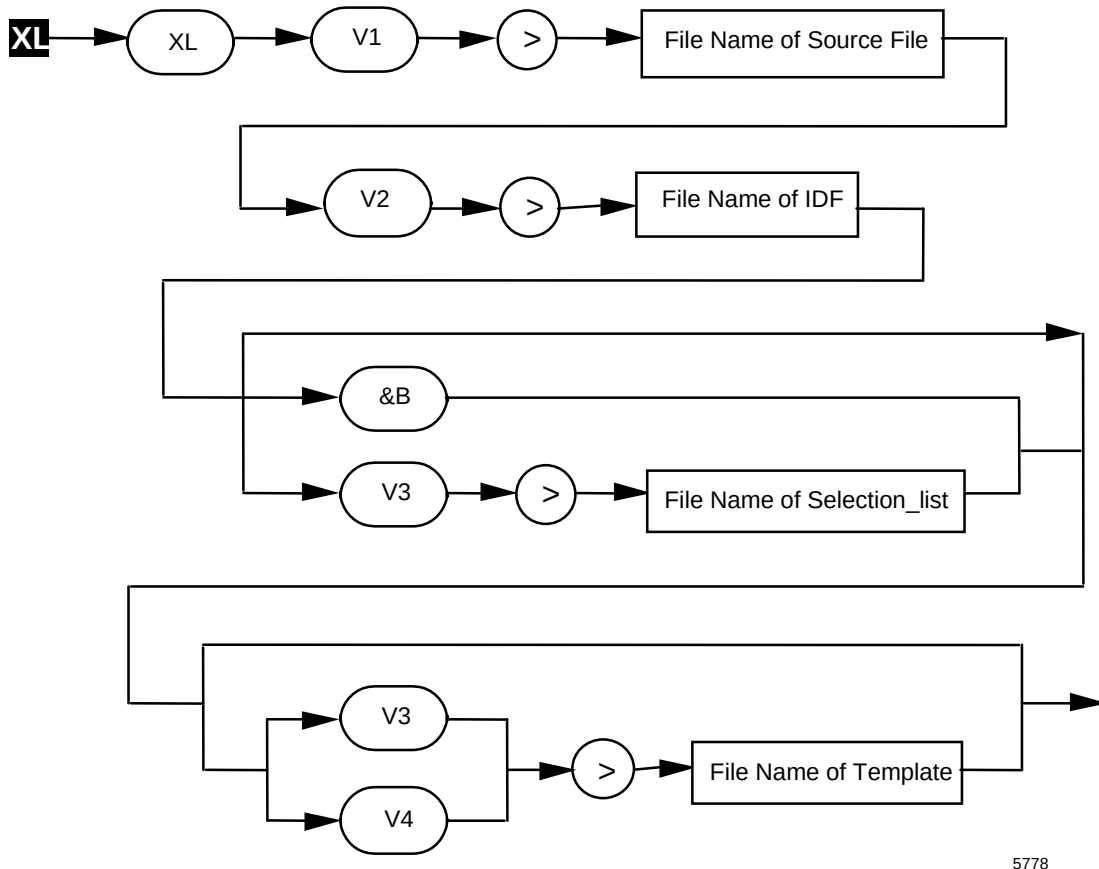
TEMP (optional) is the name of the IDF that contains any additional template entities that were referenced in the SOURCE . EB file by “&X” control items.

NOTE: With R620 an engineer can edit, view and save an .EB file with a maximum size of up to 64K records.

8.3.13 XL Definition

This is the Exception Build (and Load) command. It is used to build and load either all or only a selected number of entities in an exception-build (.EB) source file into their destination modules or gateways. If required, an IDF containing additional template entities can also be used.

8.3.13.1 XL Syntax



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8.3.13.2 XL Example

```
XL V1>SOURCE.EB V2>IDF V3>LIST.EL V4>TEMP
```

Where:

V1, V2, V3 and V4 represent prior device and volume/directory designations.

SOURCE . EB is the name of the file that contains the template entities, the name of each entity and the parameter-value changes.

IDF is the name of the IDF where entities that were not successfully loaded will be stored.

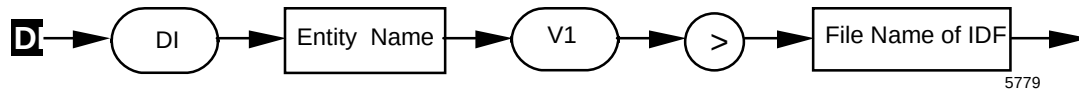
LIST . EL (optional) is the name of the selection-list file that lists the names of the entities to be loaded.

TEMP (optional) is the name of the IDF that contains any additional template entities that were referenced in the SOURCE . EB file by “&X” control items.

8.3.14 DI Definition

This is the Delete Entity from IDF command that is used to delete a specified entity from an IDF.

8.3.14.1 DI Syntax



8.3.14.2 DI Example

```
DI HG0501 V1>IDF
```

Where:

HG0501 is the name of the entity to be deleted.

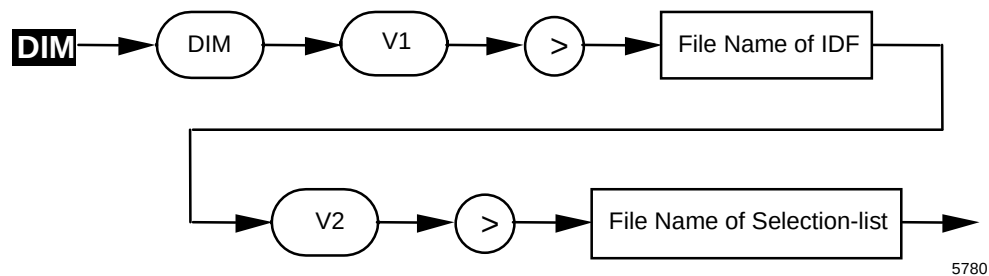
V1 represents prior device and volume/directory designations.

IDF is the name of the IDF that contains the entity to be deleted.

8.3.15 DIM Definition

This is the Delete Mult Entities IDF command that is used to delete selected entities from an IDF.

8.3.15.1 DIM Syntax



8.3.15.2 DIM Example

```
DIM V1>IDF V2>LIST.EL
```

Where:

V1 and V2 represent prior device and volume/directory designations.

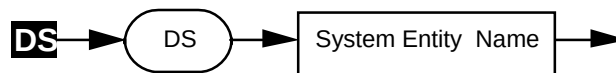
IDF is the name of the IDF that contains the entities to be deleted.

LIST.EL is the name of the selection-list file that lists the names of the entities to be deleted.

8.3.16 DS Definition

This is the Delete System Entity command that is used to delete a specified system entity from the module or gateway where it resides.

8.3.16.1 DS Syntax



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8.3.16.2 DS Example

DS HG0501

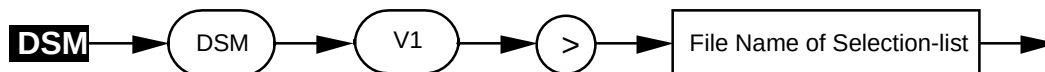
Where:

HG0501 is the name of the system entity to be deleted.

8.3.17 DSM Definition

This is the Delete Mult (System) Entities command that is used to delete selected system entities from the modules or gateways where they reside.

8.3.17.1 DSM Syntax



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8.3.17.2 DSM Example

DSM V1>LIST.EL

Where:

V1 represents a prior device and volume/directory designation.

LIST.EL is the name of the selection-list file that lists the names of the system entities to be deleted.

8.3.18 BPB Definition

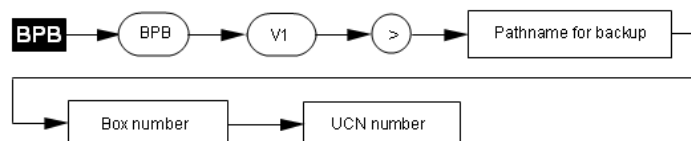
The BPB command is the backup command for box parameters using DEBEC. This is applicable for xPM, SM and LM.

The following parameters must be specified for the BPB command:

- UCN number
- Box number
- Pathname for backup

An error file with .EF extension with the same file name as the .EC user file is created to record the errors produced after an unsuccessful backup operation.

8.3.18.1 BPB Syntax



8.3.18.2 BPB Example

```
V1= NET>RRJ
BPB V1>ABC.XX 21 03
```

Where:

ABC.XX is the pathname for backup
V1 represents a volume/directory designation
21 is the box number,
03 is the UCN number

8.3.19 BPR Definition

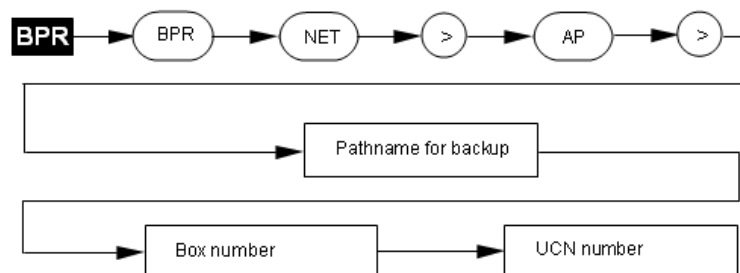
The BPR command is the restore command for box parameters using DEBEC. This is applicable for xPM, SM and LM.

The following parameters must be specified for the BPR command:

- UCN number
- Box number
- Pathname from where to be restored

An error file with .EF extension (with the same file name as the .EC user file) is created to record the errors produced after an unsuccessful restore operation.

8.3.19.1 BPR Syntax



8.3.19.2 BPR Example

```
BPR NET>AP>ABC.XX 21 03
```

Where:

NET>AP>ABC.XX is the pathname for restore
21 is the box number, 03 is the UCN number

8.4 DEB COMMAND FILE EXAMPLES

Example No. 1:

```
V1=$F1>XBPT
V2=$F2>NIDF
V3=NET>TEMP
P1=$P1
```

```

XB V1>SOURCEA V2>IDFA &B V3>TEMPIDFA
XB V1>SOURCEB V2>IDFB &B V3>TEMPIDFB
XB V1>SOURCEC V2>IDFC &B V3>TEMPIDFC

PI V2>IDFA.SL &B P1
PI V2>IDFB.SL &B P1
PI V2>IDFC.SL &B P1

```

The declarations line says V1 points to the removable medium 1, which contains volume XBPT; V2 points to removable medium 2, which contains volume NIDF; V3 points to user volume (or directory) TEMP, which is in a History Module; and P1 points to printer 1.

The first three command lines build data points, as directed by the source files (SOURCE_n) that are contained in volume XBPT. The resulting new points are stored in the IDFs (IDFA, IDFB, IDFC) in volume NIDF, and the template points used are in TEMPIDF_x files in volume (or directory) TEMP.

The last three command lines print on printer 1 the content of the three IDFs where the new points were stored.

Example No. 2:

```

V1=$F1>XBPT
V2=$F2>NIDF
V3=NET>TEMP
P1=$P1

LM V2>IDFA
LM V2>IDFB
LM V2>IDFC

```

In this example, the data points in IDFs A, B, and C are loaded in the appropriate nodes. The three IDFs are in volume NIDF, which is on removable medium 2. The V1, V3, and P1 declarations on the directive line are ignored because they are not used in the command lines. The directive line was copied from the preceding command file, for convenience.

Example No. 3:

```

V1=NET>AIDF
V2=NET>NIDF
P1=$P1

RM V1>AMIDF1
PI V1>AMIDF1 P1 V1>AMSEL1.SL

RM V2>NIMIDF1
PI V1>NIMIDF1 P1 V1>NIMSEL2.SL

LM V1>AMIDF1
LM V1>NIMIDF1

```

In this example, several entities are reconstituted, the content of the resulting IDFs is printed, and then the entities are reloaded in the appropriate nodes. Note that the selection-list file names

include suffixes. A user might want to use a command file like the one in this example to pick up Honeywell-provided GDF revisions, as the points are reconstituted and reloaded. Of course the revised GDF volumes (or directories) would have to be available on a removable medium, or on an HM, at the time the DEB command file is executed.

SECTION 9 - GUIDELINES AND PRECAUTIONS

This section provides guidelines and precautions for the building of data points and reserved entities. This includes discussions of the necessity for multiple-load operations, reconstituting existing data points and reserved entities, precautions in the use of alias points and guidelines for the use of Intermediate Data Files (IDFs) and Global Description Files (GDFs).

The Data Entity Builder (DEB) is the internal software that builds points that have tagnames, as well as other entities that have reserved-entity names. Using functions on the Universal Personality's Engineering main-menu, process entities can be built for: **Application Module (AM)**, **Network Interface Module (NIM)**, **Hiway Gateway (HG)**, **Computing Module (CG)**, **Process Manager Module (PM)**, **Advanced Process Manager Module (APM)**, **High-Performance Process Manager Module (HPM)**, **Safety Manager Module (SM)**, and **Logic Manager Module (LM)**. In addition, configuration data is built for the **Data Hiway**, **Area Database**, **HG Library**, and **HM History Groups**.

WARNING

Once data has been loaded, the external name for entities is converted into an internal ID, and all references are kept in terms of that internal ID, not the external name. **Do not delete and reload an entity to make a simple change, because the delete and reload breaks all references to the entity.** To change an entity, you should reconstitute it, make the needed changes in the PED, and then reload it.

9.1 SUGGESTIONS FOR IDF CONTENT AND SIZING

The size of an IDF is important when the files will be stored on removable media and depends on the media capacity. IDF size can be estimated from the number of displays shown in the Parameter Entry Display (PED) status of a typical entity at build time in the DEB. The PED is the forms that are displayed on the US and used to build entities.

An IDF is not in the form of text source, nor in the final form for loading. The structure of the file is known only to the DEB, and the DEB provides commands to display, print, and modify the IDF. The space used by IDFs is considerable.

You might also consider using exception-build (.EB) files instead of IDFs. See 9.6 for a list of the advantages and disadvantages of each file type.

An easy way to determine the number of PED displays is to select typical options in a PED for the point type, press ENTER, and look at the PED Status Display (hold CTL down and press F8) to see how many displays there are.

The size of an IDF in kilobytes (except for NIM* logic points) is approximately—

$$\text{kilobytes} = (0.75 * \text{No. Displays In Typical Entity}) * \text{No. Entities}$$

For NIM* logic points, the IDF size in kilobytes is approximately—

$$\text{kilobytes} = (1.75 * 7) * \text{No. Entities}$$

These are not hard and fast rules, but will help you to estimate IDF size. Particularly with the large and complex entity types.

You should also understand that the directory for the IDF contains a maximum of 255 entities (current file blocking is 3072 words and the directory is limited by file blocking size).

The following recommendations can be followed to help prevent problems with IDFs:

1. Do not mix module-build types in the same IDF (that is, use separate IDF files for each entity type in AM*, CG, NIM, and HG). This increases efficiency by reducing the number of accesses to the Global Description Files (GDFs) entity-by-entity during exception-building or multiple-entity operations. An exception is points that refer to each other—these must be in the same IDF regardless of type.
2. Use the Command Processor's `List (LS)` command to help you watch the file characteristics of sector utilization and file sizes. Remember, that if working on the HM, the maximum size should still be tracked because you may want to back up the file (or transfer it to a cartridge).
3. The following table contains statistics determined by building the point types indicated. You can use it and the PED Status Display for the type(s) of points you are dealing with to guide you in using IDFs. Six kb has been added for each entity in the "kb, Ent. Plus Suprt. Files" column to allow space for the support files (.SL, .UL, .EF, etc.) that are generated by the DEB as entities are built, and the values in that column are used to calculate the number of points per cartridge.

* For this discussion, NIM points include PM/APM/HPM/LM and SM points. AM points include AM and A^XM points.

Table 9-1 — IDF Sizing

Point Type	No. Ents. Built	Total Sec.	PEDs Per Ent.	Sec. Per Entity	kb per Entity	Sec. Per Display	kb, Ent. Plus Suprt. Files	Pnts. Per Cartridge
NIM Reg PV	11	145	5	13.2	3.4	2.6	9.4	2160
NIM Reg Ctl	11	193	7	17.5	4.5	2.5	10.5	1940
NIM Logic	11	529	7	48.1	12.3	6.9	18.3	1100
AM Flag	256	612	3	2.4	0.6	0.8	6.6	3080
AM Reg.	11	552	14	59.2	12.9	3.6	18.9	1080
HG Reg.	11	193	6	17.5	4.5	2.9	10.5	1940

Estimating IDF sizes can be complex. A good guide is the number of displays in the PED set. As the number of displays increases, so does the size and complexity of the entities.

WARNING

IDFs for AM data points with Custom Data Segments cannot be used on another system without the risk of unpredictable errors and node failures. This is because the IDFs contain internal name forms for units, segment names, and custom parameters. The only way to prevent such errors is to copy the custom-name library (the .SE and .SP files) on directory &ASY for use on the other system. This is impractical because this causes any CDSs operating on the other system to stop functioning. In addition to receiving the custom-name library from the source system, the destination system must have the same unit configuration as the source system. Exception Build files, however, are transportable.

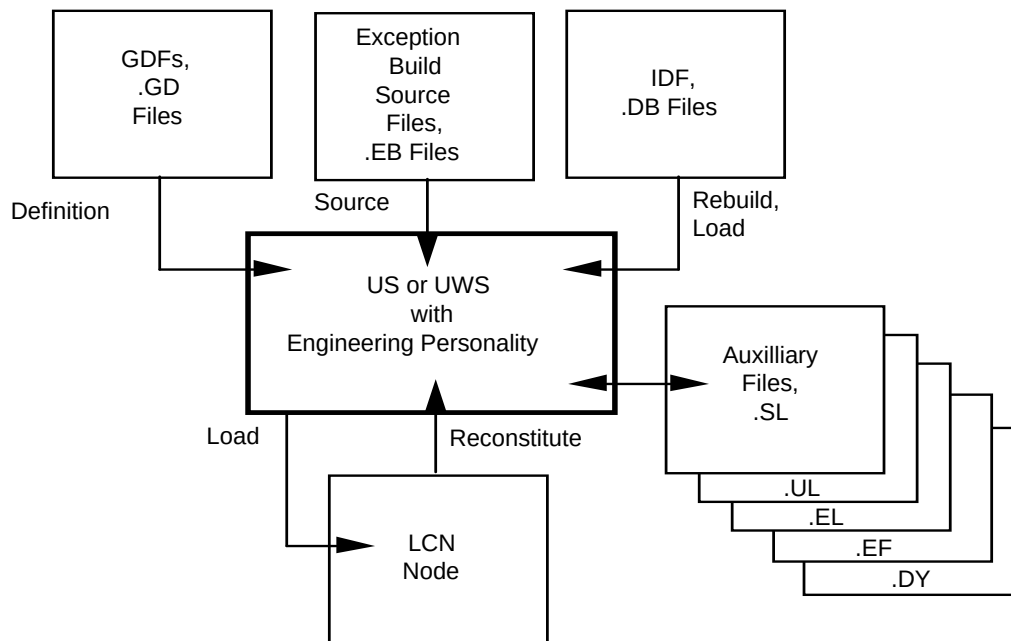
9.2 ENTITY BUILD/LOAD/RECONSTITUTE OPERATIONS

The file relationships during operation of the DEB are diagrammed in Figure 9-1. This figure shows the direction of data flow between the DEB, the files it uses, and the destination (owner) of an entity. Recommendations for most efficient processing are as follows:

- If you use IDFs, collect entities of one build type (for example, HG regulatory or regulatory control) in one IDF.
- When exception-building, use the DEB to build and write the template entities into the IDF to which the exception-built entities are stored.
- To allow getting prints of the data on the screen, use a US with the locally-connected printer. The DEB's Print Entities or Print File command is more efficient and allows using the printer anywhere on the console, but the printout is in summary form and is not as seen on the screen.
- Store all interactively built entities into an IDF, before loading the entity using the F12-load feature, or into a .EB file, for data backup and convenience.
- Watch the PED-status indicators shown on the right of the header line (second line on the screen).
- Always check the PED Status Display (F8) to be sure all data has been entered and no data is in error or is missing before attempting to perform any command.
- If you use IDFs, put entities that refer to each other in the same IDF and use the Multiple Load command to satisfy as many references as possible.
- If you use .EB files, put entities that refer to each other in the same .EB file so that they can

be loaded using the Exception Build command.

- Load entities that are used as references, either in other entities or in groups, logs, trends, schematics, etc., before trying to use them. If an entity refers to a parameter in another entity, the load can be done without error if both are in the same IDF. If in different IDFs, you can load the entities and then just reload any with errors to complete the cross-references.
- Remember that system-defined reserved entities start with "\$." You cannot build process entities whose names start with "\$", nor can you delete reserved-entity names. The reserved-entity names of interest to the engineer are listed in Appendix C. To reconstitute the reserved entities when using the PED, it is not necessary to know the actual entity name. Call for the type of data configuration desired, fill in the data on the first form (usually the first one or two items determine the reserved entity name), and press ENTER. The entity name appears in the header line of the display. Simply use the F7 function button to automatically reconstitute the entity as named in the header line. It is not necessary to use the Reconstitute command.
- To reconstitute a reserved entity that defines a specific log or report, it is necessary to know its number. We recommend that you make the number of the log or report a part of the log or report name and description.
- If using removable media, for faster operation, use a US that has the cartridge drives directly connected to it.
- Use separate HM-user volumes or directories for the IDFs.
- Use separate HM-user volumes or directories for the .EB source files.
- Use the Upgrade IDF option of the Repack IDF command (7.2.15) to update your IDFs when revised GDFs are issued by Honeywell.
- Use the Print commands (7.2 11) to create .EB source files directly from an IDF or from entities loaded in nodes. If you specify a pathname with a .EB suffix instead of the device ID (\$Fn) for a printer, the entity is copied in .EB source file form.



Note: The Load and Reconstitute operations shown here apply to data points and other entities that reside in an owning node. Area database entities don't reside in a node, but in Area Data Base files that reside in an HM or on floppies, and the Load and Reconstitute operations operate on that database, not on nodes.

The Auxilliary files contain lists of entities for which an operation was successful (.SL) or unsuccessful (.UL), edited entity lists (.EL), and error files (.EF and .DY).

Figure 9-1 — Data Entity Builder File Relationships

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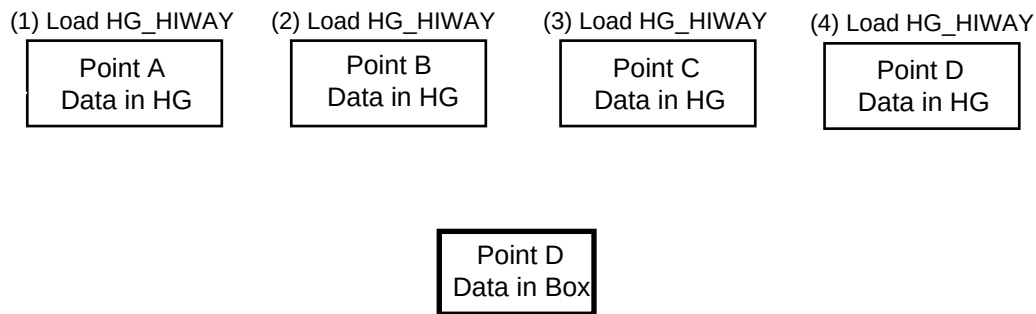
9.2.1 Precautions for Entity Building

Confusion and unexpected results can occur during the entity-building process, unless you understand what you are doing and its effect on other people and on the system. The following notes are based on experience during extensive entity building operations.

- **Alias Points**—These are virtually identical HG points that use the same HG, box, and slot, or subslot, but may have different unit assignments. This allows alarms for such points to be generated for more than one unit. You must take care to avoid unintended differences in these points that can cause very unexpected results, including **alarms that do not work properly** on any set of alias points.

To understand how unintended differences can occur, consider how the point data is handled when alias points are loaded:

Some of the point data goes to the HG and the remainder goes to the box. If you build alias points A, B, C, and D for the same slot and the same box, and use the DEB to load them in A, B, C, D, order, you end up with the data distributed like this.



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If you now reconstitute point A, you might expect to get what was loaded, but instead, the data received is the point A data from the HG and the point D data from the box.

If you load point A to the HG and the box, then load points B, C, and D to only the HG and then reconstitute point D, the reconstituted data will be that of the HG data for point D and the box data for point A.

These are the key factors:

- The last load of box data overwrites any previous data loaded in the slot.
- Loads to only the HG do not overwrite the box data.
- The HG data for each set of alias points is independent, except for the **Red Tag** status. This means that parameters, such as the mode attribute (MODATTER), may be different in alias points. The operators may think they have control of an output because of looking at one point's data, but a user program may think it has control of the output because of looking at the data for an alias of the first point.

NOTE

You must take considerable care in building and modifying alias points. You may have some very unexpected results if you do not plan carefully and consider the consequences of changes you may want to make. For example, the actual input connections used are those specified for the last alias point loaded in the HG, rather than those of any previously loaded alias point that uses the same box and slot. If an alias point is deleted, it cannot be reloaded without deleting all of its aliases and reloading all of them, in the proper order. Another example, if you build two or more alias digital output points, each parameter must have the same value in Output Indication, OUTIND, as each of the others.

- **Will This Affect Anyone Else?**—When you execute a Load command or a Load Multiple command, you should consider whether anyone on another US is using the entities involved for process operations or some engineering function. For example, when you load a data point that someone is observing on an on-process display, the point becomes inactive, and much of its data can change or disappear. You should not load a data point when data from the node is likely to be checkpointed, because the checkpoint data may include a partially built point or the checkpoint operation may abort the load. It's best to inhibit checkpointing for the node(s) to receive the data for a Load or Load Multiple.
- **Unconfigurable Parameters**—Many entities have several parameters that do not appear on the configuration forms nor on the PEDs, because they are not configurable. These are parameters that receive values that are calculated or derived by the system and are not changed by the user. When you reconstitute entities, these parameters are not read back from the node; however, they are listed in the Parameter Reference Dictionaries, and for data points, you can see them on the Detail Displays and any custom displays that present them.
- **"Semireserved" Entity Names**—Several entity names are reserved for use by the Picture Editor and the Button Configurator. In many cases, if you happen to use one of these names for a data point name (tag name), there will be no conflict. If you do use one of these names as a tag name, that data point cannot be accessed by the Picture Editor nor the Button Configurator. Therefore, if you are or will be using either the Picture Editor or the Button Configurator, you must refer to Appendix C of the *Picture Editor Reference* manual. There you will see the list of "semireserved" names to make sure that you do not use any of them as a tag name.

9.2.2 IOP Point Build

In order to prevent inadvertent changes to PM/APM/HPM points that are in service, active points must be made inactive or the IOP must be placed in the IDLE state in order to load the point. This function was installed for security.

NOTE

All DI, DO, and AO slots are defaulted to component **active** points. These points **must** be made inactive or the IOP IDLED before a successful point build can occur. Such points can be accessed by their hardware address.

See subsection 9.2.2.1 below for this procedure.

Attempts to load points to slots without placing points in the inactive state will result in the following error at the DEB: "PNT MUST BE INACT."

NOTE: Deleting a point will leave the slot **active** with the database initialized (component).

9.2.2.1 Method for Making Untagged Points Inactive

- Call up the PM/APM/HPM Box detail display
- Select the IOP containing the point to be built
- Pick the SLOT SUMMARY display target
- Select the slot of the point to be built (the Point Name for this slot will be blank for an untagged point).
- Select the PTEXECST target (should be ACTIVE)
- Pick the INACTIVE target
- Select the ENTER target
- The point can now be loaded from the PED

9.3 EXCEPTION-BUILDING TIPS

The following tips will help in setting up Exception-build (.EB) files:

- You can create source files for subsequent point-building sessions with the DEB's Print commands. If you specify a pathname with a .EB suffix instead of the device ID (\$Fn) for a printer, the point is copied in source-file form into a .EB file. You can then use the Text Editor to make any needed changes to these files.
- Based on the priority of the DEB's data requests, the HG (or any other data owner) may not honor all of the requests made by requests for exception lists, group or unit summaries, etc. It is suggested these tasks be done in relatively small increments and no problems will be encountered. For a small request, normally all data requests are available. However, if trying to do an entire hiway or an AM, some data may not be returned in time to correctly build the >EB files.

9.4 FIXING DUPLICATE ENTITIES AND DELETING ENTITIES

Because each owning node keeps the list of entities that have been built in it, if the node is not operating at the time a new entity is loaded, the DEB cannot verify uniqueness of the name. The DEB displays a caution message that other nodes are not operating when a new entity is loaded. Because of this, it is possible for you to create duplicate entity names in the system that will be discovered the first time an external-to-internal conversion of the entity name is attempted after both nodes are operating on the LCN. Typically, the conversion is attempted when a detail display of the entity is requested from the Operator's Personality or a reference to the entity is made by the Universal Personality functions.

Detection of the duplicate entity results in an error message, which in the DEB gives the unit and the node number for each duplicate entity. Determination of which entity is in error is based on your system criteria and the steps to fix the problem are as follows:

1. If not discovered while still using the DEB, use the DEB to attempt to reconstitute the entity to get the unit and node number for a delete operation.
2. If the entity is extraneous and not needed, simply use the DEB's Delete (entity) command (7.2.8).
3. If one of the entities is to be saved and renamed, you must either completely rebuild one of them or temporarily shut down the node that owns the entity with the correct name so that the entity to be renamed is unique on the LCN.
 - To completely rebuild an entity, delete the one to be renamed as in Step 1, then use the DEB to rebuild it with the correct name.
 - If you have reset the owning node of the entity that is to keep the duplicated name, use the DEB and reconstitute the entity to be renamed. No error occurs because the other node is shut down. Then rename the entity in the PED display, load it, and delete the entity that was reconstituted. This removes the duplicated name.

Restart the node that was shut down and the entity names are correct. Remember that all references to the deleted entity have been broken and you must rebuild all references for the new entity name.

In general, the system requires you to delete an entity if you wish to change its owning node, unit, or build type. In any delete operation, remember that references to the entity are not automatically deleted. If an entity is deleted and loaded again with the same node, the references that were built before the delete do not use the "new" entity data. This is because of an internal ID that is unique for each loaded entity, even if it has a name that was previously deleted. All prebuilt references are held in internal ID form and thus, the entity, when deleted and loaded again, does not have the same internal ID.

NOTE

The internal ID is guaranteed to be unique only if the owning node is not shut down and reloaded or if the database is consistently checkpointed and reloaded before entities are loaded or reloaded when the owning node is reloaded.

NOTE

To delete an entity from a NIM, the NIM parameter LOADSCOPE must be changed from NIM ANDPM to NIM ONLY. Then reload the NIM and delete the entities. When the deletes are completed restore the LOADSCOPE to NIM AND PM and reload.

9.5 DEB LOADING AND SYSTEM CHECKPOINTING RULES

To keep the system database consistent, follow these rules:

1. The order in which entities are loaded to a node is significant. If entities that refer to each other are in the same IDF and that IDF is used in executing a Load Multiple command, the entities can usually be loaded regardless of their order in the IDF (which you cannot control). There are, however, some situations that require certain entities to be loaded before others. You must consider what you are doing when you load entities. When you need to specify the order in which entities are loaded from an IDF, you can specify that order by listing the entities in the proper order in your Selection List. These are examples of situations that require a certain order:
 - A. HG data points depend on the Hiway entity (\$HIWAYn) and its associated Box/Slot entity (\$HYnBn) being loaded. Further, the Box/Slot entity depends on the Hiway entity being loaded. All three could be loaded at the same time if the Selection List specified the Hiway, Box/Slot, HG point order. They could also be individually loaded; however, the HG point might refer to another point not yet loaded and, therefore, need to be loaded with a Load Multiple command.
 - B. AM Points that are scheduled as members of Before/After sets must be loaded in a certain order. If point A is configured to be processed **before** point B, point B must be loaded before point A. If point B is to run **after** point A, point A must be loaded before point B. See 2.2.1.4 in *Application Module Control Functions*.

Always load nodes from the most current checkpoint, or if using the DEB always use the same IDF, in the same order, to ensure that the sequence of point loading is consistent and the internal entity IDs remain the same, for all references to properly work.

2. Always make changes to the existing database. Do not delete an entity and reload it to make a simple change. Instead, reconstitute and reload. Be sure to reload the last, correct, checkpoint data to reloaded nodes before doing any operation that loads or links to entities in the nodes.
3. The list of external entity names is distributed among the owning nodes; there is no central list. Thus, all nodes must be running in the normal on-process personality to ensure that points are not duplicated.
4. Remember that deleting a point and reloading it changes its internal ID and all references have been broken by the delete.
5. Do not load entities into a primary NIM or HG while starting up (loading) its secondary.
6. Because PM/APM/HPM point parameter values revert to the default values if the NIM is restarted (reloaded) with missing entities in the database in the checkpoint files, you should

always request a demand checkpoint (SAVE DATA) after loading new or rebuilt points from the DEB. To override the values in any existing point when a rebuilt point is loaded from the DEB, the node must be in the Idle state and the point's execution state in PTEXECST must be Inactive.

7. To delete entities from the PM/APM/HPM database, change the value in the NIM's LOADSCOP parameter (in the \$NMhhNnn entity) from NimAndPM to NimOnly. You can then reload the NIM and delete the desired entities.
8. If the MC box configuration is not present on the system, MC Analog Composite and Digital input/output/composite points cannot be loaded without error and a “PNT STATUS ERR” message appears. The MC box configuration must be present first.

WARNING

Do not put NIM/PM/APM/HPM/LM or SM configuration data in the same IDF as the data points. The DEB system tries to load the data points first, which causes errors.

Honeywell recommends that the NIM configuration be in a single IDF, the PM/APM/HPM/ LM/SM configurations in another IDF, and the data points in separate IDFs.

9.6 ENTITY DATA BACKUP TECHNIQUES

You can back up your TDC 3000^X entity data in three different file formats — Intermediate Data files (IDFs), exception-build (.EB) files and checkpoints. You can use the DEB and the Text Editor to translate IDFs to .EB files and vice versa without too much difficulty. A discussion of the primary uses, advantages and disadvantages of each format is presented in the following table.

Table 9-2 — Entity Data Backup Formats

Entity Data Storage Form	Primary Uses	Advantages	Disadvantages
Intermediate Data Files (IDFs)	These files may be used for the initial database load, for updates to the database, or for reloading the database when the checkpoint is no longer viable, as in some cases when a software upgrade is performed.	1) IDFs can be updated directly from a live database using the Reconstitute command. 2) IDFs are faster to load than .EB files. 3) The Write to IDF command creates an IDF and the Read to PED command uses an IDF. 4) Some database validity checking is done when IDFs are loaded. They are more likely to load the first time than .EB files.	1) IDFs are quite susceptible to becoming obsolete when moving to a new software release. In that event, however, a translator would be provided to perform the upgrade. 2) IDFs can only be viewed or altered through the DEB. 3) We do not recommend that a preexisting database be completely reloaded from IDFs because referencing structures, such as displays, logs, etc, will also have to be rebuilt. The Load with Overwrite command is appropriate and has no such side effects.

(continued)

Table 9-2 — Entity Data Formats (continued)

Entity Data Storage Form	Primary Uses	Advantages	Disadvantages
Exception-build (.EB) Files	These are text files, built using the Print Entities command. They may be used for the initial database load, for updating the database and for reloading the database when the checkpoint is no longer viable, as in some cases when a software upgrade is performed.	1) You can use .EB files to pre-build the system database before the system is installed. 2) These files are Flexible. Since they are source files, you can perform search and replace edits, view more than one entity at a time, etc. 3) EB files are easily transported to other systems, such as Personal Computers for storage or data manipulation or even to another LCN. 4) EB files are smaller than IDFs. 5) EB files are the least susceptible to becoming obsolete when moving to a new software release. 6) If you build .EB files via Workbook, database files are validity checked. 7) With R620 an engineer can view, edit and save an .EB file having a maximum size of 64K records.	1) No database validity checking is done unless you use Workbook. 2) You cannot do a Reconstitute directly from a live database. You must start with IDFs. The .EB format is the most difficult to keep synchronized with a live system database. 3) Loading from .EB files is slower than loading from IDFs. 4) The time taken for opening and editing an .EB file is proportional to the file size. When the file size is large, it takes more time and the process is slowed down. 5) The time taken for exception building of an .EB file is proportional to the file size. When the file size is large, it takes more time and the process is slowed down.

Entity Data Storage Form	Primary Uses	Advantages	Disadvantages
NOTE: While performing the backup operation, if the PRINT NIM PRIORITIES TO .EB FILE WHEN TP IS NAN option is set to YES, the .EB file contains the NIM priority parameters even though the trip point parameters are set to NaN. While performing the restore operation, if the option is set to NO, the NIM priority parameters are not restored irrespective of the values in the .EB file. While performing the backup operation, if the PRINT NIM PRIORITIES TO .EB FILE WHEN TP IS NAN option is set to NO, the .EB file does not contain the NIM priority parameters even though the trip point parameters are set to NaN. While performing the restore operation, if the option is set to YES and the .EB file does not contain priority parameters, then the default value is assigned to the NIM priority parameters of that point.			
Checkpoints	Taken periodically so that the database can be quickly restored to a recent version in case of a node failure. Checkpoints can be taken automatically at an interval of up to 20 minutes or can be demanded any time.	1) Most likely to be consistent and up to date. 2) It's automatic. No Operator interaction is required. 3) Easy to demand. Requires no setup after the system is configured.	1) Checkpoints should be backed up to removable media periodically in case the database gets corrupted. 2) Checkpoints are most susceptible to becoming obsolete when moving to a new software release, but Honeywell will provide a translator to perform the upgrade.

9.7 MOVING A POINT DATABASE TO ANOTHER SYSTEM

The preferred method for moving points from one system to another is through .EB files because:

1. Using .EB files is more secure and requires less disk space than IDFs.
2. Points that have custom segments cannot be moved in an IDF from one system to another, as the internal IDs of custom parameters will be different.

To transport such entities from one system to another, use the DEB's Print System Entities or Print IDF Entities command to write to an exception build source file. Enter the pathname for a file with a .EB suffix in the PRINT DEVICE ID or DESTINATION PATHNAME port. This creates an exception build source file with the points in it. This can be used on the other system to exception build and load the points.

SECTION 10 - ERROR INDICATION AND ERROR RECOVERY

This section describes how errors are indicated through the DEB and provides guidance for error recovery.

10.1 HOW ERRORS ARE INDICATED

You are notified of errors in several ways:

- Red pick and port outlines, red characters in ports. See 6.2 for data on usage.
- Red prompters that appear when a generic function key or a function key is pressed. See Section 5 for a description of the Engineer's Keyboard.
- Red prompters that appear as or after a command is executed. See the command descriptions under 7.2.
- Error messages in command-generated Error Files (.EF). See the command descriptions under 7.2 and Error File format under 10.4.1.

10.2 CORRECTING ERRORS IN PARAMETER ENTRY DISPLAYS (PEDS)

First, place the cursor in the pick or port that indicates an error and press HELP. The Help display may indicate what is wrong with the data or your selection. Then you should consult the applicable Configuration Form Instructions or the Parameter Reference Dictionary listed in section 1.3.1.

10.3 CORRECTING ERRORS INDICATED BY RED PROMPTERS

The following is an alphabetic list of the most common error messages that appear in the prompter region of DEB displays. Under each message is a paragraph that explains what causes the message, followed by a paragraph that suggests a remedy or remedies.

For a more comprehensive list of error messages and suggested remedies, refer to the *Messages Directory* manual.

ABORT ON BREAK

CTL was held and BREAK pressed, so command execution (or printing) was terminated.

If you want to execute the aborted command, press COMND to recall the command, check the data on the Command Display, and press ENTER. This starts command execution at the beginning, not where the break occurred.

ABORT ON ERROR

An error occurred during command execution that prevented completion of the operation.

Check for other error indications, including an Error File (.EF) (see Figure 10-1, Subsection 10.4.1) and any other error messages. As applicable, check your selection-list file, exception-build (.EB) source file, and template points in an IDF.

ABORT ON ERROR AT SOURCE LINE n

While executing an Exception Build command, line n in the .EB source file was found to have an error that prevented completion of the operation.

Carefully recheck your source file, particularly the line(s) indicated. Also check for any other error indications, including other error messages.

AREA DATABASE FILE MUST BE SPECIFIED

An attempt was made to execute a Load command (7.2.3), a Select Area command (7.2.12), or an Install Area Command (7.2.13) and something other than an area database name was found in the port where the area database file name was expected.

Check the pathnames on the Command Display and correct, as necessary. Refer to the command descriptions in 7.2 for a definition of the area database pathname form.

CANNOT DELETE RESERVED ENTITY ID

An attempt was made to delete a reserved entity. All reserved entities exist in all systems and they cannot be deleted.

Recheck the tag name for the point you were trying to delete. Only reserved entity names can begin with "\$."

CANNOT DO LOAD WITH OVERWRITE AND MAKE THIS CHANGE

While attempting to load an entity by overwriting an existing entity, a value change was encountered that prevented loading the entity.

Recheck the changes made to this entity. You cannot change the point type, the unit, or the hiway and then reload. You might be able to delete the existing entity and then reload, but if it is a data point, you must also delete and reload all of this point's secondaries and all points it refers to. See 9.4 for guidance in deleting and reloading entities.

CANNOT TRANSLATE INTERNAL ENTITY ID TO EXTERNAL

While trying to reconstitute an AM point with CDS data point parameters, at least one tag was deleted, so there was no external name. The parameter value will show up as blanks in the PED.

COMMAND LINE EXCEEDED 80 CHARACTERS (R610)

```
SrcFileLineNumb: 0004
LS V1>DEMOX.EL      88      07      AM SWITCH FLAG
COMMAND LINE EXCEEDED 80 CHARACTERS
```

A command line has exceeded 80 characters – rare, but a possibility.

CURSOR NOT IN PORT OR PICK

HELP was pressed when the cursor was not in a pick or a port.

The DEB must find the cursor in a pick or a port to determine what type of help information is desired.

CUSTOM SEGMENT NAME NOT FOUND

During execution of a Read to PED command, a custom segment that is referenced by one of the points in the IDF could not be found on this LCN. This is caused by using an IDF with custom names that was created on one system on another system. The custom files on the two systems will always be different, and IDFs with custom names cannot be transported to another system. For more information, see *Control Language/Application Module Data Entry*. Another reason is that the segment's .SP file differs from the AM's database. This can happen if you did not copy the SE and SP files after an upgrade. The error is caused because the custom segment name is stored in the point record of the AM's database.

CUSTOM SEGMENT TOO LARGE

1. A CDS was attached prior to R620 when changes were made as to how time variables were handled in the PED. Time variables now count against the total number of "loaded" parameters that the DEB can handle, which is approximately 1440. To fix this problem, you will have to load a station with a release prior to R620 and make the CDS size smaller.
2. A change was made to a CDS, compiled with the –OCD, and now the GD is too large. In both cases, you will need to divide the CDS into two CDSs.

DATA PNT TOO BIG

Trying to load an AM point larger than 16089 words.

DATA Point <pointname> NOT FOUND – ALL NODES NOT OPERATIONAL

The node was too busy to load the point.

DEVICE UNAVAILABLE (ldid)

The device named in a pathname on a Command Display cannot be found on the network.

Recheck device ID numbers, correct the pathname, and retry. It is possible that the device (a printer, a removable media drive, an HM) is not operating or is not connected to the network.

DUPLICATE ENTITY (entity) IN UNIT (uuu) ON NODE (nnn)

While executing a command such as Reconstitute or Print Entities, two entities were found with the same entity name.

You need to use the Delete (system entity) command (7.2.8) to delete one of these entities. If this is a data point to be renamed, you must delete all of this point's secondary points, all points it refers to, and then rebuild them. See 9.4.

DUPLICATE NAME – COULD NOT ESTABLISH ENTITY OFF NODE

The AM was too busy. Retry and enter again.

DUPLICATED PARAMETER ID

During execution of the Exception Build command, the source file was found to specify a change for the same parameter in a point more than once.

ENGINEER'S KEYLOCK MUST BE ENABLED TO USE THIS COMMAND

An attempt was made to execute a command that changes the process database with the key switch on the US in the OPER or SUPVR position.

Move the key switch to ENG and try again.

ENTITY DATA CORRUPTED—DELETE AND REWRITE (entity)

During an attempt to read an entity from an IDF, the entity name was found but the data could not be read.

If you have a backup copy of the IDF, you can use the Command Processor functions to copy it on your working IDF, or you can read it from the backup into a PED set and then write the PED to the working IDF. You might also be able to reconstitute the entity and then write its PED to the IDF.

ENTITY DATA IN IDF CORRUPTED (entity)

During an access to an IDF, the DEB detected corrupted entity data.

See the Repack IDF command under 7.2.15.

ENTITY DOESN'T EXIST (entity)

The named entity was not found during an attempt to read from an IDF or to reconstitute from its module or gateway.

Check that the module that "owns" the entity has a status of OK. Check the entity name. If an IDF is involved, check that you used the right pathname for the IDF that has this entity.

ENTITY ITEM EXPECTED

The Exception Build command expected an entity name item (&N (name)) following the template name (&X or &M), but the entity name was not found, so this point could not be processed.

Recheck your source file to see that you do have an &N (name) item for every new point you are building. Also verify that the name is legal (see Parameter Name in the Parameter Reference dictionaries).

ENTITY TYPE DIFFERS FROM ORIGINAL TYPE FOR ENTITY (entity)

During a load with the overwrite option, the entity being reloaded was found to be of a different type than the original entity, so this entity could not be loaded.

If you want the entity to be a different type, delete the original entity and then load the new one. If this is a data point, you must also delete and reload all of this point's secondaries. See 9.4 for guidance in deleting and reloading entities.

EQUAL SIGN EXPECTED

During exception building, in processing a parameter, an equal sign was expected to specify a parameter change but it was not found, so this point cannot be processed.

Check your source file to be sure that all parameter changes include the equal sign, and that parameter names and values are separated from other names and values by at least one space.

EXCEEDED MAXIMUM NUMBER OF ENTITIES ALLOWED IN IDF

While executing a command that writes entities in an IDF, the storage capacity of the IDF was exceeded.

Use the Inspect IDF function of the Repack IDF command (7.2.15) to determine if there are corrupted entities. If so, use the Binary Repack function to delete them and regain the space they occupy. Otherwise, write the data in another IDF.

(entity) EXISTS—SELECT OVERWRITE OPTION IF DESIRED

During an operation that loads an entity in a module or gateway or writes an entity in an IDF, the named entity was found to already exist.

If you want to change the data in this entity, select the With Overwrite pick on the Command Display and press ENTER.

FILE DOESN'T EXIST (pathname)

The file named in one of the Command Display ports was not found.

Check your pathname(s), check that you have the right removable medium mounted in the correct drive, and try again.

FUNCTION NOT IMPLEMENTED

A function was requested that is not presently available.

A DEB function that may be available in later releases, is not presently available.

HISTORY WILL BE LOST

For Release 510 and later systems, when loading a history group, if any point currently historized by the group would be deleted from the group as a result of the load, a warning is issued. The complete message identifies the unit number (xx) and group number (n):

```
$CHxx(n) HISTORY WILL BE LOST - Press <F5> Overwrite To  
Continue If Desired.
```

You can cancel the operation or press function key F5 to overwrite and continue.

IDF MAY HAVE BEEN CORRUPTED (pathname)

During access to an IDF, the DEB encountered corrupted data.

Use the Inspect IDF function of the Repack IDF command (7.2.15) to determine if there are corrupted entities. If so, use the Binary Repack function to delete them and regain the space they occupy. If necessary, restore the deleted entities in the IDF through Reconstitute commands (7.2.5 or 7.2.6), or by rebuilding each of them through a PED set and writing to the IDF.

INSUFFICIENT INFORMATION—CANNOT LOAD DATA FOR (entity)

Data was missing from the PED set when an attempt was made to load the data.

Go back to the PED Status Display (CTL/F8). Find the PED display that has missing data, key in the missing data, and press ENTER. When the PED status line indicates no errors and no missing data, retry the Load command.

INSUFFICIENT INFORMATION TO WRITE THIS DATA

There was not enough information on the first PED page to identify this entity, so it could not be written in the IDF.

If this is a data point, at least the entity name must be entered. If this is a reserved entity, enough information must be entered to identify which entity it is. The first few parameters are enough.

INVALID CONTROL ITEM

The Exception Build command found "&" in the source file, followed by an illegal character.

Check your source file. The valid control items are &T, &X, &M, &V, &C, and &E (see 7.1.2.1).

INVALID DEVICE VARIABLE

During execution of a DEB Command File an invalid device declaration was encountered. Device declarations must be one of V1 through V50 or P1 through P10 .

INVALID POINT REALM (R610)

```
SrcFileLineNumb: 0006
LS V1>DEMOX.EL 88 07 SWITCH
                        ^
```

INVALID POINT REALM

The specified point realm is not valid. Valid point realms are AM, HG, NIM and CIU. Point types may not be specified until a point realm has been identified.

INVALID POINT TYPE (R610)

```
SrcFileLineNumb: 0010
LS V1>DEMOX.EL 88 07 AM HG
                        ^
```

INVALID POINT TYPE

The specified point type is not one of the point type choices for the specified point realm.

NOTE

Caret (^) position indicates where error has occurred.

ITEM NOT PROCES

Cannot reduce the CLSLOT value for AM Regulatory, Switch, and custom points.

MEDIA IS NOT MOUNTED OR INITIALIZED

The media is not inserted or mounted, or during an operation that writes to the removable media, some malfunction caused the disk to be dismounted and the files closed. This could happen if the disk was removed from the drive before a write operation, such as a Multiple Load to an area database file, was completed. This corrupts the data on the disk. The disk should never be removed until the Operation Complete prompter appears.

Check that the disk is mounted in the drive. You may need to recreate the file that was being written. If this situation should corrupt an IDF, you can remove the corrupted IDF and repack the disk with the Repack IDF commands (7.2.15).

MISSING STRING DELIMITER

The Exception Build command found the number of characters that fill a field, but no space or quote sign (") that indicates where one field ends and the next begins.

Recheck your source file. Each parameter change must be separated by at least one space and all user-defined character strings must be enclosed in quotes (see 7.2.1.7).

MOD/BOX [xx] HWY/UCN [yy] COMBINATION INVALID (R610)

```
SrcFileLineNumb: 0002
LS V1>DEMOX.EL 88 07
MOD/BOX [88] HWY/UCN [07] COMBINATION INVALID
```

The combination of MOD/BOX and HWY/UCN is invalid, typically because one or both have not been configured into the system. In this example, 88 is not a value box address.

NAMED ITEM DOESN'T EXIST

A parameter named in a .EB source file does not exist in the new point. This could be because it is not in the template point or because another parameter controls how many of this type of parameter values exist, and that parameter's value has not been changed.

NO OFFNODE REF

Loading an AM point with a CDS data point parameter containing a tagname unavailable on the system. The missing tag may have been deleted or the data owner is isolated.

NOT A VALID PARAMETER ID

Using Alter Parameters with an incorrect parameter name.

PARAMETER (ppp) DOESN'T EXIST ON POINT (name)

During exception building, a parameter was found in the source file that was not found in the template point.

Carefully check your source file. Either an &C item was not provided for the parameters in a PED, or the parameter changes were not specified in the same order as the parameters appear in on the configuration forms and on the PEDs.

PARAMETER UNAVAILABLE FOR CHANGE

You selected a pick on an error-file display that is not a configurable parameter, based on current parameter values.

RECORD NUMBER EXCEEDED 65535 LIMIT

You have attempted to either open/save an .EB file from the text editor, or to load an EB file using the Exception Build command when the file has more than 64K records.

REV NO MISMATCH

You have attempted to load another point in that node with the same CDS descriptor, but different time stamp. There can not be two different CDS descriptions with the same name in the same node at one time.

Use Find Names to identify which points have CDS descriptors, and then use the List File Attributes (LS) to find out the time stamp of the files. The applicable point(s) must be deleted and rebuilt, which may require taking the process off-line. Refer to the Data Entity Builder manual for precautions in deleting points.

TEMPLATE ITEM EXPECTED

The Exception Build command expected an &X or &M item at the beginning of a source file, but did not find one.

Check your source file. When you use &X or &M, you must define a template entity as the first control item in a source file. See 7.2.7.1 for assistance.

UNASSIGNED DEVICE VARIABLE

During execution of a DEB Command File, a device was named that was not declared at the beginning of the Command File. For example, V3 may have been used when only V1 and V2 are declared above.

UNKNOWN ENTITY TYPE

Invalid entity type name in the &T command. Refer to Appendix B in this manual for a list of all valid entity types.

WARNING—USING UNINSTALLED WORKFILE

A load or reconstitute of an area database entity is to or from an area database work file (.WA) that has not been installed yet. This warning occurs the first time the .WA file is accessed in this session. The session ends when the .WA file is installed.

Recheck your records or use the Command Processor functions to look at the time stamps on your .WA and .DA area files. You may be loading a new area entity to a work file that does not match the area database now in use. If so, use the Command Processor to delete the old .WA file and load your new area entities into a new work file.

&E CONTROL ITEM MISSING

The Exception Build command found a new &T, &X or &M item, which indicates the start of a new source file, but it did not find a &E item to indicate the end of the old source file.

Check your source file, insert the &E item at the proper place, and retry.

10.4 CORRECTING ERRORS INDICATED BY ERROR FILE MESSAGES

Because there are so many different error file messages, it is not practical to analyze them here. You can obtain information that may help you to correct errors indicated by these messages from these sources:

- **Help Displays**—If the message pertains to a command, call up the appropriate Command Display, place the cursor in the appropriate pick or port, and press HELP. If the message pertains to an entity name or a parameter in an entity, select the entity name or the parameter name (they are picks). If the PED set is available in the US or in an IDF, the PED page with selected name or the selected parameter appears. You can then place the cursor in the pick or port affected, then press HELP to call up the Help display.
- **Parameter Reference Dictionaries**— You will find descriptions of the value types, value ranges, default values, access keys, and parameter functions in these publications. See subsection 1.3.1 for a list of these manuals.
- **Engineer's Reference Manual**—This manual provides guidelines and recommendations for engineers who are configuring or modifying the TPS system. The sections in the *Engineer's Reference Manual* with information pertaining to DEB activities are listed in subsection 1.3.2 of this manual.
- **Control Functions Manuals**—These manuals describe the data acquisition and control subsystem functions. They provide understanding of the relationships of the control-subsystem elements, including parameters. They are as follows:

System Control Functions

Process Manager Control Functions and Algorithms

Advanced Process Manager Control Functions and Algorithms

High-Performance Process Manager Control Functions and Algorithms

Logic Manager Control Functions

Safety Manager Control Functions

Hiway Gateway Control Functions

Application Module Control Functions

10.4.1 Error File (.EF) Format

The form and content of an Error File is shown in Figure 10-1 which follows. Three types of error information are shown. The Entity Error Directory and the Entity Error Data are placed in the Error File when data owners detect errors, usually as a result of executing a Load or Load Multiple command. Exception Build Source File Errors information is placed in the Error File when the Exception Build command detects an error in a .EB source file. If the Error File is too large for one screen, you can page through it by pressing PAGE FWD on the Operator's Keyboard. To return to the beginning of the Error File, press CTL and F2=SUP FULL keys simultaneously.

You can select picks in the Entity Error Directory to call up the Entity Error Data for each entity named in the directory. You can also select picks in the Entity Error Data to call up the PED page that contains the named parameter. The cursor goes to the pick or port on the PED that has the error. To return to the Error File display, press CTL-F2. If the Error File is not on the screen, it can be retrieved by executing the Display File command, as described in subsection 7.2.9.

Entity Error Directory

09 Sep 11:49 DATA ERROR FILE (pathname) RESPONSE

==== ENTITY ERROR DIRECTORY - Select Pick to See Entity's Errors ====

FC110

(name)

(name)

(name)

(name)

==== GENERAL ERROR DATA ==== 51979

Entity Error Data

ENTITY FC110 () ERROR DATA

BOXNUM

PARAM UNAVAILABLE

(name)

(error message)

(name)

(error message)

(name)

(error message)

(name)

(error message)

51980

Exception Building Source File Errors

09 June 12:30 DATA ERROR FILE (pathname) COMMAND

==== ENTITY ERROR DIRECTORY - Select Pick to See Entity's Errors ====

FL101

A100

==== GENERAL ERROR DATA ====

==== ERRORS ON ENTITY FL101 ====

SOURCE LINE 00007

KEYWORD=Flow FL11

MISSING STRING DELIMITER

SOURCE LINE (no.)

(copy of the source line)

^ (error pointer)

(error message)

SOURCE LINE (no.)

(copy of the source line)

^ (error pointer)

(error message)

Note:In this example, the

source line should read

KEYWORD="Flow F11"

The quote marks were the

missing delimiters.

51981

Figure 10-1 — Error File Forms and Content

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10.4.2 An Example of Error Correction

The following is an example of how to resolve errors detected when an Exception Build command encounters bad data in a .EB source file. The Command Display is replaced by a display of the Error File listing both entity errors and general errors.

To correct general errors, proceed as follows:

1. Note the description (line number, parameter name, location, etc.) of each error in the General Error Data segment of the Error File. If there are a number of errors and a hard copy of the data would be beneficial, press PRINT DISP on the Operator's Keyboard to get a print of the screen.
2. If the error is not immediately obvious, you can consult the appropriate *Parameter Reference Dictionary* to get parameter information such as value types, value ranges, default values, access keys, etc. See Subsection 1.3.1 for a list of these manuals.
3. Press COMND to restore the Command Display.
4. Press ESC to call up the Command Processor.
5. Use the Text Editor to correct the errors in the .EB source file. Refer to the *Text Editor Operation* manual for operating instructions.
6. Press CTL-HELP|MENU to restore the Command Display.
7. Execute the Exception Build command again, with the Overwrite option selected.
8. If required, repeat steps 1 through 7 until all general errors are corrected.

To correct entity errors, proceed as follows:

1. From the Entity Error Directory segment of the Error File, select the pick that identifies an entity in error. This will bring up another display listing all the parameters for this entity that are in error, set up as picks.
2. Select a parameter that is in error. The page of the PED that contains the selected parameter will be displayed, with all the ports for the selected parameter bordered in red. All other bad parameters for this entity, if there are any, will appear in the PED, bordered in red.
3. If you want detailed information on a parameter, place the cursor in a red-bordered port and press HELP. This will bring up the Help Display, which provides a definition and often a value range for the parameter. Press CANCEL to return to the prior display.
4. Correct the parameter by either entering the correct value or selecting the correct pick, depending on the parameter type.
5. Press ENTER to process and check the data and parameter selections in the PED. The border color changes from red to green when the parameter is correct.
6. If required, repeat steps 2 through 5 until all the bad parameters for the entity are corrected. Page through the entire PED to make sure you get all the errors.
7. If you want to return to the display that lists all the bad parameters for this entity, press CTL-F2.

8. Press CTL-F10, then CTL-F5 to write the current PED to the IDF, overlaying the original IDF.
9. Press COMND to restore the Command Display.”
10. Execute the Print Entities command with the “Print screen entity” option selected and the Destination pathname entry set to the .EB source file. This will generate a corrected .EB source file, incorporating all the changes you have just made.
11. If you want to verify the changes, use the PRINT FILE command to get a printed copy of the updated .EB file.
12. Execute the Exception Build command again, with the Overwrite option selected.
13. If required, repeat steps 1 through 12 until all the entity errors are corrected.

APPENDIX A - &T POINT TYPES

The following list of point names can be used with &T to build points without a template point in an IDF. These are the point-type names that appear on the Detail Display for each point type. (See 7.2.7.1.)

The following list of point names to be used with the &T directive appears on the left, its description on the right.

A.1 HG POINT TYPES

ANLCMPHG	(Analog Composite)
ANLINHG	(Analog Input)
ANLOUTHG	(Analog Output)
BHASMDHG	(Batch Assign)
BOX	(Box/Slot Configuration)
COUNTHG	(Counter)
CTLCOUNT	(Control Counter)
DIGCMPHG	(Digital Composite)
DIGINHG	(Digital Input)
DIGOUTHG	(Digital Output)
FLAGHG	(Flag)
HIWAY	(Hiway Configuration)
LGCBLKHG	(Logic Block)
LIBRARY	(HG Library)
NUMERCHG	(Numeric)
PRCMODHG	(Process Module)
REGHG	(Regulatory)
TIMERHG	(Timer)

A.2 AM POINT TYPES

	(Batch History-not available)
COUNTAM	(Counter)
CUSTOMAM	(Custom)
FLAGAM	(Flag)
NUMERCAM	(Numeric)
REGAM	(Regulatory)
SWITCHAM	(Switch)
TIMERAM	(Timer)

A.3 AREA DATABASE POINT TYPES

\$ANNDATA (1) - \$ANNDATA (60)	(Annunciator)
\$OTREND	(Area Trend)
\$OBTCHSM	(Batch Summary)
\$OFFLOG (1) - \$OFFLOG (400)	(Free Format Log)
\$OGROUP (1) - \$OGROUP (400)	(Group)
\$DMONBLK (1) - \$DMONBLK (20)	(Hiway Box Annunciation Policy)
\$ANPOLC	(LCN Node Annunciation Policy)
\$OOVERVW	(Overview)
\$OABSTRT	(Pathname Catalog)
\$OPRTRND (1) - \$OPRTRND (100)	(Printed Trend)
\$OPRCJOR (1) - \$OPRCJOR (100)	(Process Journal)
\$OMODGRP (1) - \$OMODGRP (50)	(Process Module Group)
\$RTJNL	(Real Time Journal)
\$OREPORT (1) - \$OREPORT (100)	(Report)
\$OSOEJOR (1) - \$OSOEJOR (20)	(SOE Journal)
\$OLOG (1) - \$OLOG (100)	(Standard Log)
\$OSYSJOR (1) - \$OSYSJOR (100)	(System Journal)
\$OSYSTAT	(System Status)
\$RUNTSTS	(Unit Assignment)
\$OUNITSM (1) - \$OUNITSM (5)	(Unit Summary)
\$OUNTRND (1) - \$OUNTRND (36)	(Unit Trend)

A.4 CG POINT TYPES

ACIDP	(Advanced Control Interface)
ACCRDP	(Calculated Result Interface)

A.5 HM POINT TYPE

HM_HIST	(History Groups)
---------	------------------

A.6 NIM POINT TYPES

ANINNIM	(Analog Input)
ANOUTNIM	(Analog Output)
ARRAY	(Array)
DEVCTL	(Device Control)
DICMPNIM	(Digital Composite)
DIINNIM	(Digital Input)
DIOUNNIM	(Digital Output)
FLAGNIM	(Trend)
LOGICNIM	(Logic)
NIMLIB	(NIM Library Configuration)
NODE	(UCN Node Configuration)
NUMERNIM	(Numeric)
PRMODNIM	(Process Module)
REGCLNIM	(Regulatory Control)
REGPVNIM	(Regulatory PV)
TIMERNIM	(Timer)

APPENDIX B - RESERVED ENTITIES

System-defined reserved entities exist for certain configuration items. Table C-1 lists reserved entity names of interest to the engineer.

Table B-1 — System-Defined Reserved Entities

Entity Name	Content	Where Configured	Notes
\$ANNDATA(n)	Annunciator Policy	Area Database	1
\$ANPOLC	LCN Node Annunciation Policy	Area Database	
\$CHuu(nn)	Continuous History Groups	HM History Groups	1
\$DMONBLK(n)	Hiway Box Annunciation Policy	Area Database	1
\$H(n)LIB(n)	HG Library	Building HG Library	2
\$HIWAYnn	Hiway Configuration	Hiway Configuration (DEB)	2
\$HYnnBnn	Box/Slot Configuration	Box/Slot Configuration (DEB)	2
\$OABSTRT	Pathname Catalog	Area Database	
\$OBTCHSM	Batch Summary Display	Area Database	
\$OFFLOG(n)	Free Format Log	Area Database	1
\$OGROUP(n)	Group Display	Area Database	1
\$OLOG(n)	Standard Log	Area Database	1
\$OMODGRP(n)	Process Module Group Display	Area Database	1
\$OOVERVW	Overview Display	Area Database	
\$OPRCJOR(n)	Process Journal	Area Database	1
\$OPRTRND(n)	Printed Trend	Area Database	1
\$OREPORT	Report	Area Database	1
\$SOEJOR(n)	SOE Journal	Area Database	1
\$OSYSJOR(n)	System Journal	Area Database	1
\$OSYSTAT	System Status Display	Area Database	
\$OTREND	Area Trend Display	Area Database	
\$OUNITSM(n)	Unit Summary Display	Area Database	1
\$OUNTRND(n)	Unit Trend Display	Area Database	1
\$NMnnNnn	UCN Configuration	Network Interface Module (DEB)	2
\$NMnnnnBnn	PM Configuration	Network Interface Module (DEB)	2
\$PRSTSnn	Processor Status Data Point	Not Configured	2
\$RTJNL	Real-Time Journal Assignment	Area Database	
\$RUNTSTS	Unit Assignment Display	Area Database	
NOTE 1: "(n)" indicates that this entity is an array of parameter values. "(n)" designates the value in the array, where "n" is the index number of that value. For \$OGROUP(n), "n" represents the group number (1-120). NOTE 2: "n" represents digits that identify each of several entities of that type. For \$PRSTSnn, nn represents the node number.			

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Honeywell

Honeywell Process Solutions
1860 W. Rose Garden Lane
Phoenix, AZ 85027 USA